2021-2022 EDITION

MATH SURVEY

OFFICIAL PUBLICATION OF THE STUYVESANT MATH DEPARTMENT SINCE 1927

Stuyvesant Math Survey

2021-2022 Edition

The Official Mathematics and Computer Science Publication
of the Stuyvesant High School Math Department since 1927

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Special thanks to Mr. Michael Develin (Class of 1996), Professor Paul Zeitz (Class of 1975), Mr. Patrick Honner, Mr. Thomas Lin, and Dr. Erica Klarreich for their support for this edition.

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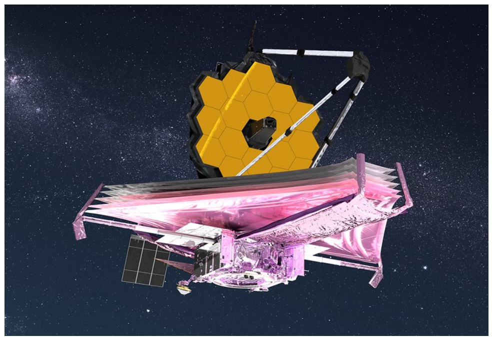
Editor's Note

Josephine Lee



Artist conception of the James Webb Space Telescope

"Webb was worth the wait!"

- John Mather, Webb senior project scientist and Nobel Laureate astrophysicist, NASA Goddard

Did you watch it launch? Did you read about its incredible capabilities? Are you excited to marvel at all the images it will capture?

On the morning of Christmas Day, December 25, 2021, NASA launched the James Webb Space Telescope, more than 30 years after it was first envisioned. According to NASA's website, this 6.5-meter infrared space telescope is the world's largest and most complex, powerful enough to "study every phase of the history of our Universe." Whoa.

Even though I'm not literally or figuratively a rocket scientist and have no background in aeronautics/astronautics, I am in awe of the scope and the scale of this feat of engineering collaboration decades in the making. It took roughly \$10 billion and the perseverance of 20,000 engineers, technicians, computer programmers, and scientists from NASA, the European Space Agency (ESA), and the Canadian Space Agency (CSA) to make this prolonged and arduous journey, but now the dream has become a reality which will expand the boundaries of scientific knowledge for decades to come.

As I look back over my four years at Stuy and my role as Editor-in-Chief of three editions of *Math Survey*, I'm just beginning to realize the value of peering into the past to see farther into the future. I didn't appreciate it at the time, but all the long hours of studying and all the hard work of publishing forced me to tap into an inner reserve of determination that I didn't even know I had. Thanks to the dedication of family, friends, alumni, faculty, and administration, I can now look back at my high school career with a little of that same sense of awe (plus a certain amount of relief) and look forward to college and beyond with anticipation and curiosity.

On behalf of the *Math Survey* staff, I would like to thank Principal Seung Yu and Math and Computer Science Department Assistant Principal Eric Smith for their faithful leadership through a global pandemic that seemed like it would never end. I am also deeply grateful to our faculty advisor, Mr. Gary Rubinstein, whose vision of the future of *Math Survey* was always informed and inspired by its memorable past. To the Stuyvesant math and computer science faculty, both old and new, we students will never forget how tirelessly and sacrificially you invested in our lives and education; thank you, thank you, thank you. And to the resilient staff of *Math Survey*, we did it again, one article, interview, puzzle, and graphic at a time!

To my fellow members of the Class of 2022, graduating from Stuy was worth the wait. Congratulations!

Foreword

Do You Want to Make a "Transformational Impact"? Michael Develin Shares his Advice and his Decades of Academic and Employment Experience

Dr. Michael Develin, Class of 1996, currently works as the Head of Data Science, Merchant Services at Shopify and is best known for his work as a mathematician in combinatorics and discrete geometry. He served as captain of the Math Team while at Stuyvesant but at age 16 entered Harvard University, where he was honored as a Putnam Fellow in the 1997 and 1998 competitions. After just three years, Develin earned his Ph.D. from University of California, Berkeley at age 22, writing his dissertation on Topics in Discrete Geometry under his advisor Bernd Sturmfels. Develin then conducted math research as the recipient of the 2003 American Institute of Mathematics Five-Year Fellowship. Over the course of his academic career, he published 31 research papers in mathematics journals, in fields such as combinatorics, discrete geometry, graph theory, number theory, and topology. Develin's prior employment experience includes quantitative analysis, data science, and analytics at companies such as D.E. Shaw, Facebook, Instagram, and Remitly.

This foreword is based on an interview conducted by Josephine Lee. The content has been edited for clarity.

How Stuyvesant Felt Like Home

I graduated from Stuyvesant in 1996, which is 26 years ago. I had always been interested in mathematics, but it became a reality for me at Stuyvesant. Much of my experience with math involved Math Team. More than a hundred students showed up to work on math problems every day, and realizing that there were not only a few people, but a lot of people pursuing this interest together was eye-opening. I really enjoyed Math Team and the bonding in the culture of math.

Outside of Math Team, I had the rare experience of not only taking higher level math classes, but the option to explore other fields of interest. I remember taking a writing workshop class, which I found much more compelling than reading Shakespeare for seven semesters in a row. But in the end, so much of my Stuy experience came down to the people. Being around others who are nerdy in the way that I was nerdy felt like home. I don't think it would have felt that way at most other high schools.

From the Abstract World to the Real World

I went into graduate school, not because I had an explicit career goal, but because I purely loved working with numbers. While I enjoyed the experience of solving interesting and challenging problems, eventually I realized that working on abstract problems was not the trajectory for the rest of my life. Instead, I was looking for a direct, tangible impact I could have on society.

After I finished my post-doctoral research, I started working at a hedge fund, which is where I learned how to solve problems that were grounded in something concrete. By the time you graduate from college, you have been exposed to a lot of problems that have a right answer, but most of those problems are not connected to the real world. So in those corny word problems, like calculating the time the train leaves, it doesn't matter that it's a train. It's just a situation that you then turn into a math problem. So we emerge from school with that type of mindset and find that the real world does not actually work that way. In the work world, the process is much different; you try to get closer to the truth or build algorithms slightly better than the one before. I grew and learned more of that process after I started working at Facebook and Instagram.

These days, all the doors are open to you if you are fluent in numbers. All the industries have become and are becoming data-driven over time. Back when I was at Stuy, the sports industry was not using analytics like it is today. Now, the need everywhere is for people who can think in a structured way about problems and figure out the objective version of any situation.

What people often do not see is how data is used in tech companies: to give the user a voice. At the end of the day, it really came down to the question of how I can use data, which represents people's voices, to make better decisions.

A Generalist "Aiming for the Stars"

In retrospect, I'm most happy about the fact that I did not take just math classes during high school and college. I think what sets apart people who make an exceptional impact is the ability to excel in areas other than your core competency. There are some roles where you do need to be as good as possible at just math; being a professor is one of those roles. However, don't be afraid to diversify; don't feel like you need to become the world's best programmer. For example, the most impactful software engineers are not just programmers. They are the ones who can also think about the product and the customer, figure out how to architect things, or understand the business problem that leads to founding a company. But to make a truly transformational impact, it is important to develop general communication and comprehension skills and to widen your horizons. You are limiting yourself if you try to become really good at only one thing. When I see people top out in their careers, it's usually not because they fall short as a programmer or analyst, but it's because they can't put the bigger context around the technology, and they can't have impact outside of just the code that they write.

If you are aiming for the stars, it's important to be a bit of a generalist. In that sense, the core skill that I took away from the years of doing math is the ability to set up and frame a problem and figure out the logical problem-solving. You learn some of that in math and computer science, but understanding the broader world is also helpful to solve problems that we actually encounter.

So keep your ears open. Let's say you take a software engineering internship at some point during college. Try to speak not just with the software engineers; pay attention to what others say as well. Those may be the perspectives you will eventually need to understand to potentially make an impact on the world.

Embrace the Imposter Syndrome

It's important to try to learn about things you don't immediately understand. I think a fair number of data folks stay in their comfort zone with data, so it is very easy to just gravitate back to the analysis that's already familiar. And in the meantime, it always feels unfortunate when some writer folks describe the data folks as poor writers.

I'm not a professional caliber writer, but I've practiced a lot to become a better communicator. I'm never going to be the person who can talk to someone and immediately understand them and their life story. But even though I knew I wasn't going to be great at that, I have practiced talking to customers about their problems. It took time to take in their perspectives, understand them thoroughly myself, and then utilize that knowledge to address their problems.

The same process can apply to folks at Stuyvesant. Stuy students have been largely high achievers their entire lives. When you enter a new environment, whether it's high school, college, or a first job, you have the anxiety of being surrounded by really smart people. All of a sudden, you feel that you don't belong there, or you're out of your league. But I think it's important to embrace that Imposter Syndrome. Everyone encounters it to some degree, and there will always be problems you cannot solve. Don't pigeonhole yourself; be willing to embrace it and say, "I should try to improve in this area where I might currently not be that great at right now."

Keep your head up and try to explore; there's so much out there!

Mike Develin
November 2021

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The Power of Infinity: Cantor, Cohen, and Infinite Sets

Daniel Potiesvsky

1 Introduction

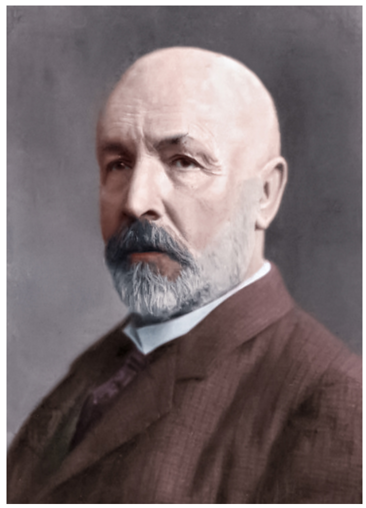


Figure 1: Portrait of Georg Cantor

This article examines some of the interesting and bizarre properties of infinite sets. Thanks to the work of Georg Cantor, a German mathematician of the late 19th and early 20th centuries, set theory began to develop rapidly. His ideas about infinite sets were so shocking and counterintuitive that they were often met with heavy skepticism from the mathematical world. People were outraged at this new and unconventional way of thinking, and even Henri Poincaré called this unusual way to think about sets and to conduct set theory proofs a "disease" in mathematics. In modern math, set theory is considered to be a very important branch, forming the foundation for many mathematical concepts and having many implications outside the world of pure math.

2 Infinite Sets and Bijections

Simply speaking, a set is a collection of elements or objects. Most of Cantor's interesting work was focused on the particular infinite sets that allow us to group numbers: natural numbers, integers, rational numbers, irrational numbers, real numbers, transcendental numbers, and so on. One of Cantor's methods for his exotic assertions would be to develop a **bijection**, or a one-to-one correspondence between two sets, to assert their equivalence. In other words, a bijection creates a relationship between two sets, say X and Y, such that each element from set X is paired with exactly one element from set Y, and vice versa.

For infinite sets, this story is more complex, and Cantor later introduced the concept of a cardinal number—a way to distinguish and to describe the various sizes of infinite sets. Certainly, the size of a finite set is just a number that describes the number of elements in that set. The set $\{1, 3, 5, 7, 9\}$, for instance, is a finite set that has a size of 5. However, for the description of the sizes of various infinite sets, the cardinal numbers are denoted using the Hebrew symbol aleph ($\aleph$) followed by a subscript. In the proofs that follow, it is worth noting that cardinality is defined in terms of bijections. Two sets have the same cardinality if and only if there is a bijection, or a one-to-one correspondence, between them.

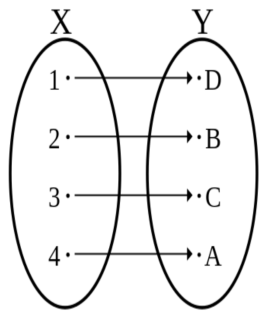


Figure 2: Example of a bijection

3 The Set of Natural Numbers and Integers

Suppose we consider the set of integers ($\mathbb{Z}$) and natural numbers ($\mathbb{N}$). A question arises: Is the set of integers larger than the set of natural numbers? Both of these sets go on forever, but it is clear that the set of natural numbers is a subset of the set of integers. After all, each natural number belongs to the set of integers, but not every integer belongs to the set of natural numbers, suggesting that there must be more integers than natural numbers. Shouldn't the set of integers be a "larger infinity"?

Of course, there is also an argument to the contrary. Suppose we have an infinite set of natural numbers, and we decide to add the negative of each element in the set to it to produce the set of integers. But will adding more elements to an already infinite set change that infinity quantity?

Which of these arguments is right? Although each makes important points, Cantor resolved this with a bijection argument. As one of the most fundamental assertions of infinite set theory, Cantor claimed that the cardinality of the natural number set ($\mathbb{N}$) is equal to that of the set of integers ($\mathbb{Z}$).

To prove his assertion, Cantor established a bijection between two sets, as follows. Arranging all of the integers in a list as 0, 1, -1, 2, -2, 3, -3 ... allows us to count them by creating a function between them and natural numbers. This can be described with the function that has an input of a natural number and an output of a unique integer. In fact, the natural number input is used to produce the position of each integer on the list. For example, the integer 0 is in position 1, 1 in position 2, and -1 in position 3, and so on. Therefore, we obtain:

$$f(1) = 0$$
$$f(n) = \frac{n}{2} \text{ for even values of } n$$
$$f(n) = \frac{-(n-1)}{2} \text{ for odd values of } n$$

Let's check if this function works. For every natural number n we can find a corresponding integer in the list, and cover the entire set of integers. And for every integer we can find a corresponding natural number that is the position of that integer on the list. Thus, we have a bijection.

Note: Sets that are the same size as the set of natural numbers are considered the smallest infinite sets and are denoted with the cardinality of $\aleph_0$. These are also known as **countable sets**.

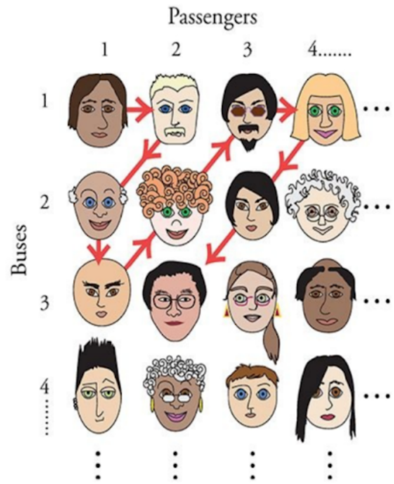
4 The Hilbert Hotel and Rational Numbers

German mathematician David Hilbert, a great admirer of Cantor's work, once told a story of an interesting hotel known now as the Hilbert Hotel. This great hotel was always booked, and yet it always had a vacancy, for there were an infinite number of rooms in the Hilbert Hotel. And if a new guest arrived, all that the manager of the hotel had to do was to move the people from Room 1 to Room 2, Room 2 to Room 3, Room 3 to Room 4, and so on, thus freeing up the first room for the newcomers. Of course, if infinitely many newcomers arrive, the manager can always move the occupants of Room 1 to Room 2, Room 2 to Room 4, Room 3 to Room 6, and so on, thus freeing up all of the odd-numbered rooms at the Hilbert Hotel. Infinitely many odd-numbered rooms for infinitely many demanding newcomers. This manager must be quite pleased with himself.

Now the challenge comes: Infinitely many buses with infinitely many people, all clamoring noisily for a room, ride up to the Hilbert Hotel. Our good-natured manager (although growing less so by the minute) decides to perform the same trick as last time: to vacate infinitely many rooms, all odd, with his "doubling" idea. But will this suffice? We do, after all, have infinitely many rooms, and the number of passengers seems to be of the magnitude of infinity squared...

The trick here is this: If the manager were to start tackling this problem like the previous problem with one bus by trying to assist Bus 1, he would never get through all of the buses. For there are infinitely many passengers in Bus 1, and the rowdy people of Bus 2, for instance, would never see the warm, fluffy beds of the Hilbert Hotel.

The manager takes the problem in stride: He lines up the passengers in an infinite rectangular grid, with passengers and buses, and develops a foolproof algorithm for attending to them all. He begins with Passenger 1 on Bus 1 and gives her the first empty room. The second and third empty rooms go to Passenger 2 on Bus 1 and Passenger 1 on Bus 2, shown on the second diagonal from the corner of the diagram. The manager then tackles the third diagonal of passengers and assists Passenger 1 on Bus 3, Passenger 2 on Bus 2, and Passenger 3 on Bus 1. Notice, in the way that the passengers are processed diagonally, it is ensured that each particular person will be reached in some finite number of steps. If you provide some X number of steps, I can find the particular person on this diagram that corresponds, and if you point out a person, I can identify a unique number of steps! And thus, the saying goes, "There's always room at the Hilbert Hotel."



Mathematically, this is simply a proof that a countable number of countable sets (like an infinity by infinity grid) will be countable, but its implications are massive, particularly with the cardinality of the set of rational numbers. Suppose we were to write out the set of rational numbers like this:

1/1	1/2	1/3	1/4	...
2/1	2/2	2/3	2/4	...
3/1	3/2	3/3	3/4	...
4/1	4/2	4/3	4/4	...
⋮	⋮	⋮	⋮	

Looks familiar? This table encompasses all rational numbers as we move through all of the possible numerators and denominators. Once again, we can move through our table diagonally.

1/1	1/2	1/3	1/4	...
2/1	2/2	2/3	2/4	...
3/1	3/2	3/3	3/4	...
4/1	4/2	4/3	4/4	...
⋮	⋮	⋮	⋮	

Most importantly, we can reach any element in this table in a unique, finite number of steps; but what are those steps, really? The set of steps is equivalent to the set of natural numbers, since these steps are recorded that way: 1 step for 1/1, 2 steps for 2/1, 3 steps for 1/2, 4 steps for 1/3, and so on. This is another bijection! We can establish a one-to-one correspondence between the steps, or the set of natural numbers, and the set of rational numbers. And we know this must be a bijection because a number of steps determines a unique rational number, and a rational number determines a unique number of steps, thus proving the equivalent cardinalities of the sets $\mathbb{N}$ and $\mathbb{Q}$.

5 Real Numbers

How about real numbers $\mathbb{R}$? Could it be true that the number of real numbers and natural numbers is the same, just like the sets $\mathbb{Z}$ or $\mathbb{Q}$? The argument that proved the equivalence between integers and natural number sets was hard to accept because the natural number set is a subset of $\mathbb{Z}$. Can we use the same argument here, despite the fact that there are far more real numbers than integers (or are there)?

This proof by Cantor (one of his finest) shows that the number of real numbers in the interval (0, 1) is uncountable and is done by contradiction.

Suppose we were to create the bijection between natural numbers and real numbers, such that each real number between 0 and 1 appears somewhere in this list.

For instance, our list could start like this:

$$\begin{array}{ll} 1. & .\underline{4}56789\ldots \\ 2. & .9\underline{8}7654\ldots \\ 3. & .67\underline{5}675\ldots \\ 4. & .123\underline{3}56\ldots \\ \vdots & \vdots \end{array}$$

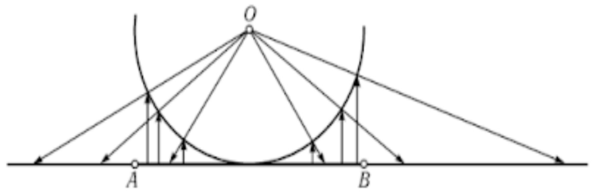
Cantor claimed he could produce a number that was not on the list.

Suppose that we take the first digit from the first number, the second digit from the second number, etc. In other words, take numbers diagonally from the list. The decimal generated from the provided example would be .4853. The next step is to change each digit of this number to any other digit. For example, change the 4 to a 5, the 8 to a 9, the 5 to a 6, and the 3 to a 4. This yields a new number, .5964. This number is our winner. It cannot be the first number on our list, since it has a different first digit from the number there. It is also not the second number, since its second digit is different, etc. If instead we were to make an infinitely long list, implement Cantor's 'diagonal argument,' and change up each digit, we would be making a number that differs from the nth number in the nth decimal place. Ergo, it doesn't appear anywhere on the list, and there is no way that our list can cover all real numbers.

This was explosive in the math world: Cantor's clear proof and identification that the infinite size of the set of real numbers is greater than that of natural numbers. It was then assigned the cardinality $\aleph_1$, greater than $\aleph_0$. Sets with cardinality $\aleph_1$ are also referred to as **uncountable sets**, or continuums.

5.1 Real Numbers in an Interval Versus Real Numbers on the Entire Number Line ($\mathbb{R}$)

We have established that there are more real numbers on an interval (0, 1) than there are natural numbers. But can we compare two sets of real numbers? Specifically, the number of real numbers in a fixed interval, like $(0, 1)$, versus the number of real numbers on the entire number line (two continuums)? Suppose we took the interval $(0, 1)$, picked it up like a string and curved it into a semicircle, tangent to the real number line at its midpoint. We could then place a laser or a flashlight in the center of the semicircle and shine rays of light at it. The laser will hit the string at some point, continue past it, and then hit the number line. And every point on the string will have a corresponding point on the number line. This can be done for any point on the string, establishing the needed one-to-one correspondence, and proving that the cardinality of any two infinite sets of real numbers is equal to $\aleph_1$.



6 Connection Between $\aleph_0$ and $\aleph_1$

So far, we have focused on the countable set ($\aleph_0$) of $\mathbb{N}$, and the uncountable set ($\aleph_1$), or the continuum of real numbers, as very separate entities. But how can we tie them together?

For this, let us return briefly to finite sets and introduce the concept of the **power set**. The power set, $\mathscr{P}$(S), of a set S is defined as the set of all subsets of S. For instance, take a set S(a, b, c). The power set, $\mathscr{P}$(S), is [(), (a), (b), (c), (a, b), (a, c), (b, c), (a, b, c)]. Note that $\mathscr{P}$(S) includes the empty set as well. In this case, there were 3 elements in set S and 8 elements in $\mathscr{P}$(S). Is there some relationship?

It is easy to see that for set $\mathscr{P}$(S) above, the number of elements is

$$\binom{3}{0} = \binom{3}{1} + \binom{3}{2} + \binom{3}{3},$$

since we are choosing all possible numbers of elements from S, which has 3 elements. To generalize: If there is a set with n elements, its power set can be described as containing

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ... + \binom{n}{n}$$

elements. Using the theorem about the sum of the nth row of Pascal's triangle, we can conclude that the power set has 2^n elements. Also of note, the power set must always have more elements than the original set, since $2^n > n$ for all positive n. Therefore, it follows that the power set of the set of natural numbers, $\mathscr{P}(\mathbb{N})$, has $2^{\aleph_0}$ elements, and this power set has more than $\aleph_0$ elements.

Now consider the set of real numbers on the interval $(0, 1)$, but this time, consider it in terms of binary, with each decimal made up entirely of 0s and 1s after the decimal point. The number of digits that any given fraction has is certainly countable, which makes the number of digits in these fractions to be $\aleph_0$. Since there are only two choices for what to put for a given digit, there must be $2^{\aleph_0}$ real numbers between 0 and 1, expressed in binary. Since we can disregard the binary (because every number written in binary can be written equivalently in base 10), we obtain the following: the number of reals between 0 and 1 must be $2^{\aleph_0}$. But we have already proven that the number of reals from between 0 and 1 is equivalent to the number of reals in a continuum $2^{\aleph_0} = \aleph_0$.

So, we can state the following:

$\aleph_0 <$ cardinality $(\mathbb{R})$ = cardinality $((0,1))$ = cardinality$(\mathscr{P}(\mathbb{N})) = 2^{\aleph_0}$.

7 The Continuum Hypothesis

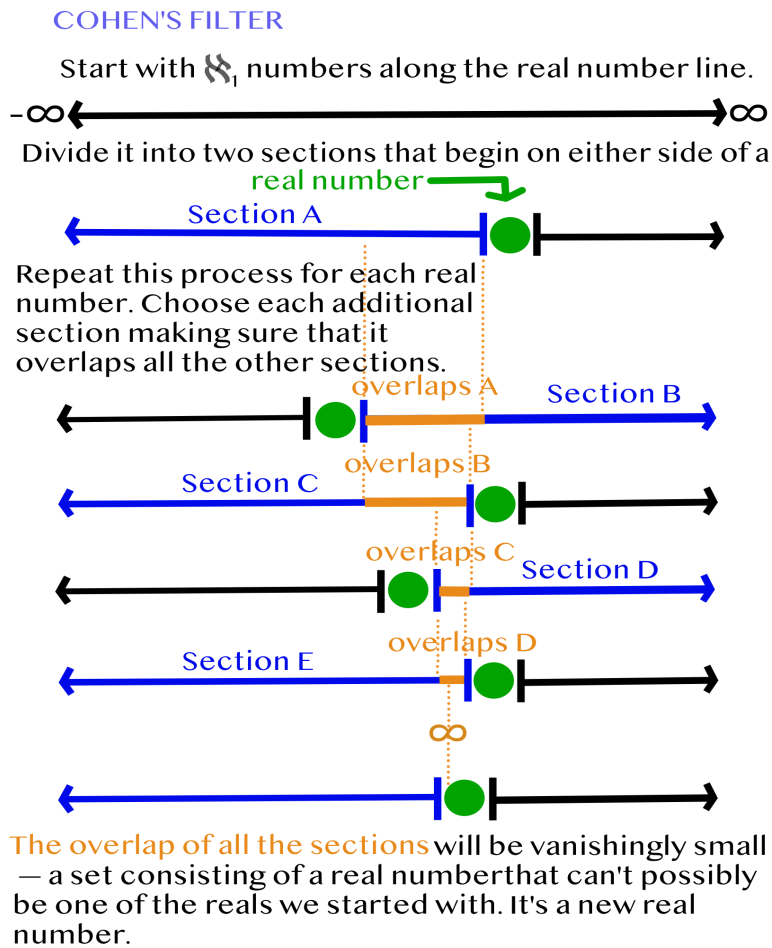
So far, we have described the notion of two kinds of infinities in sets. First are those that are associated with natural numbers, integers, and rational numbers. These are known as countable sets, denoted by $\aleph_0$. The second type is associated with real numbers, also known as uncountable sets, or continuums, denoted by $\aleph_1$. Could there perhaps exist something in between those two types of infinite sets, that is, an infinite set of real numbers that does not have a one-to-one correspondence with natural numbers or a one-to-one correspondence with real numbers?

The continuum hypothesis states that there is no such set of numbers that exists in the realm of cardinality between natural numbers ($\aleph_0$) and real numbers ($\aleph_1$). Cantor could not prove the continuum hypothesis, and the problem remained unsolved in the 20th century. It was considered important particularly by David Hilbert, who placed it first on his famous list of open problems to be tackled in the 20th century.

The trouble with the hypothesis began in 1931, when the Austrian logician Kurt Gödel discovered that any set of axioms that serve as a foundation for mathematics will always be incomplete. There will always be problems that the foundational axioms can't resolve and facts that we cannot prove. As it pertains to the continuum hypothesis, Gödel believed that it is precisely that: a problem that is independent and cannot be solved with the foundational axioms of mathematics and set theory. (There are a total of 10 set theory axioms, known as ZFC, for Zermelo-Fraenkel axioms with the axiom of choice.)

In 1940, Gödel showed that you can't use the ZFC axioms to disprove the continuum hypothesis. And in 1963, through the efforts of American mathematician Paul Cohen, it was shown that the continuum hypothesis can't be proven either, that no axioms by mathematicians could be employed to solve this problem and the hypothesis was entirely independent of set theory and mathematics in general. For this, Cohen won a Fields Medal in 1966.

To do this, Paul Cohen used a technique called **forcing**. He began with a model that included a number line with $\aleph_0$ reals. Since you can "slice" this number line infinitely many times in infinitely many ways, one could slice the line on either side of some real number. This process could then be repeated with all other real numbers. And one could repeat this slicing procedure for each real number. But Cohen used a filter, so that this infinite slicing takes place within an interval that gets narrower and narrower. This begins with a slice that cuts the number line into two sections, and you pick one of them to work with, called Section A. The subsequent slice then comes from a point that is in Section A, and also creates a Section B. The third point for slicing can be found in the intersection of Sections A and B.



As the overlap of all of these sections continues, the remaining set gets smaller and smaller, until the overlap is nothing more than a single real number. But we also know that each real number where we performed the slicing was excluded from the sections that we produced. Therefore, the real number that was obtained at the end was not a real number that this slicing started with: It must be a new real number, which allows us to expand the set of with cardinality of $\aleph_1$ to include any number of reals, like $\aleph_2$ or $\aleph_{1.78}$.

But this seems to raise more questions than it answers. It produces the strange effect that any real number can be invented by any arbitrary forcing procedure, which seems undoubtedly confusing. Does an entity exist because it was invented by forcing?

8 Modern Discoveries and the Battle with the Continuum Hypothesis

These issues with forcing were resolved by creating another set of axioms for it. One important rule is called Martin's maximum, which states that any object that is created by forcing indeed exists as a mathematical object if it satisfies certain conditions that show that it behaves consistently within the set of reals. Interestingly enough, despite the flexibility that this allows, it still seems to suggest that the set of reals moves up only by 1: to $\aleph_2$ and nothing greater.

Later in the 1990s, a mathematician named William Hugh Woodin introduced an axiom called (*) [star] that also tries to show that the continuum has a cardinality of $\aleph_2$. (*) involves a model universe of sets that satisfies the nine ZFC axioms plus the axiom of determinacy. A particularly important part of (*) is the statement that for all sets of $\aleph_1$ reals, there exist reals not in those sets, which is a clear negation of the continuum hypothesis.

The problem is that the foundational axioms with which Martin's maximum and (*) were constructed were the axioms of choice and determinacy, respectively, which contradict each other.

However, a new 2018 work by mathematician David Asperó shows that any extension of Martin's maximum, denoted Martin's maximum++, implies the correctness of (*). In order to do this, he used a procedure called L-forcing, and breaking it up into a sequence of recursive forcing procedures that satisfied the conditions of Martin's maximum.

Thus, with the newfound connection between Martin's maximum++ and (*), the case against the continuum hypothesis continues to build, as does the case for a continuum the size of $\aleph_2$. A final wrinkle: When he created (*), Woodin also stated some of its stronger counterparts, called (*)+ and (*)++, which apply to the power set of the set of reals. It's known that, in various models, (*)+ contradicts Martin's maximum. In a new proof, Woodin demonstrated that (*)+ and (*)++ are equivalent, meaning that (*)++ contradicts Martin's maximum as well. And because Martin's maximum seems to contradict (*)+ and (*)++, it raises doubts as to whether these theorems actually contradict the continuum hypothesis in the first place. Even Woodin himself admitted that he has given up hope on (*), and has rediscovered belief in the hypothesis. The battle for the truth about the continuum hypothesis continues to rage on.

9 Conclusion

This article examined the foundations of infinite set theory, including the sets of natural numbers, integers, rational numbers, and real numbers. This includes a comparison of the cardinalities of these infinite sets and the battle over the continuum hypothesis that continues to this day. The complexity of this field lies perhaps in its unconventional and abstract way of thinking. In that sense, this branch of mathematics shows us just how powerful math can be and what it can do: to understand, elucidate, and manipulate the power of infinity.

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Exploring Catalan Numbers and Their Strikingly Distinct Applications

Krish Bhandari

1 Introduction

The Catalan numbers are a set of nonnegative integers used in various combinatorics problems. Though named after the French-Belgian mathematician Eugène Charles Catalan, the sequence was earlier discovered by Leonhard Euler while exploring the triangulation of polygons. Indeed, the Catalan numbers can be used to count the number of triangulations of a polygon.

1.0.1 Example

How many triangulations are there of a polygon? To understand this problem, we define a triangulation.

1.0.2 Definition

A triangulation is a dissection of a polygon into triangles with all vertices being those of the original polygon, such that none of the sides intersect.

First, verify that every polygon actually has a triangulation.

1.0.3 Claim

Every polygon has a triangulation.

1.0.4 Proof

We proceed by strong induction on n, the number of vertices in the polygon.

Base Case: $n = 3$. This case is trivial as the polygon is a triangle itself.

Inductive Step: For $n > 3$, assume that the theorem holds for all $k < n$. Now, let P be a polygon with n vertices. P must have a diagonal, so we can draw one which splits P into two adjacent polygons, M and N. M must have at least one vertex that is not shared with N. Similarly,

N must have at least one vertex that is not shared with M. Thus, both M and N have fewer than n vertices. By our inductive hypothesis, both have triangulations, meaning that P also has a triangulation.

Now, let's delve into the combinatorial problem of how many triangulations there actually are of an n-gon. Let us begin with a recursive relation. Consider an arbitrary polygon Z with $n+2$ sides. By Claim 1.0.3, Z has a triangulation. Denote the number of ways to triangulate it as C_n. Now, fix a side of Z called AB. In any triangulation, AB forms a triangle with another vertex of Z; let it be C. $\triangle ABC$ is in the triangulation. This triangle cuts the original polygon into three polygons: $\triangle ABC$, a polygon F with side AC, and a polygon G with side BC.

Without loss of generality, let the polygon F have $k+2$ sides and k triangles in its triangulation. Then the polygon G will have $n-k+1$ sides and $n-k-1$ triangles in its triangulation, where k can change from 0 to $n-1$. Thus, there are

$$\sum_{k=0}^{n-1} C_k C_{n-1-k}$$

ways to triangulate the originally defined polygon. This is the recurrence that defines the Catalan numbers.

Now, consider another problem.

1.0.5 Example

Given an even number of people seated at a round table, how many ways can simultaneous handshakes occur without anyone's arms crossing?

1.0.6 Claim

This satisfies the Catalan recurrence.

1.0.7 Proof

Let the people be numbered

$$1, 2, 3, ..., 2n$$

around the table. If person 1 shakes hands with an odd numbered person, then there will be an odd number of people in-between the pair. Given that no crossing handshakes are allowed, among the odd number of people, there will always be a lone person without a pair. Thus, person 1 must shake hands with an even numbered person.

Now, we can split the problem into cases and quantify them using C_n. If person 1 shakes hands with person 2, then the remaining people have C_{n-1} ways to shake hands. If person 1 shakes hands with person 4, then person 2 can shake hands with person 3 in C_1 ways, and the other people can shake hands together in C_{n-2} ways. What do you notice now? This clearly satisfies the same recurrence $\sum_{k=0}^{n-1} C_k C_{n-1-k}$.

1.0.8 Theorem

The Catalan recurrence can be succinctly defined by the non-recursive relation

$$C(x) = \sum_{n=0}^{\infty} \binom{2n}{n} \frac{x^n}{n+1}.$$

1.0.9 Proof

The generating function for the Catalan numbers is defined by the power series

$$C(x) = \sum_{n=0}^{\infty} C_n x^n.$$

However, we obtain a different generating function if we expand the Catalan recurrence relation:

$$C(x) = 1 + x \cdot C(x)^2.$$

Rewriting as $xC(x)^2 - C(x) + 1 = 0$ and solving for $C(x)$ using the quadratic formula, we get

$$\frac{1 \pm \sqrt{1-4x}}{2x}.$$

Regarding the $\pm$, we must take the $-$ in order to satisfy $C_0 = \lim_{x \to 0} C(x) = 1$. Since the power series is centered at 0, its coefficients are equal to the Catalan numbers, and thus the square root term $\sqrt{1-4x}$ can be rewritten as the power series

$$\sum_{n-1}^{\infty} \frac{1}{2(2n-1)} \binom{2n}{n} x^{n-1}.$$

A simple substitution yields

$$C(x) = \sum_{n=0}^{\infty} \binom{2n}{n} \frac{x^n}{n+1},$$

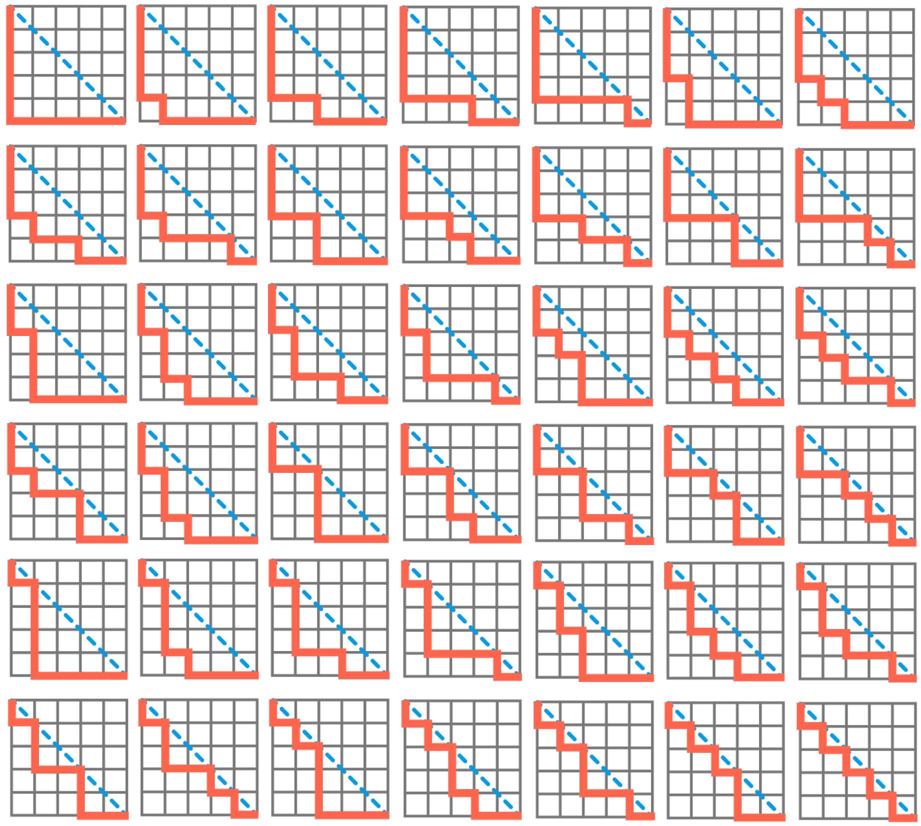
so we are done.

1.0.10 Exercise

Verify the statement above using induction.

1.0.11 Example

A well-known problem involving the non-recursive definition of the Catalan sequence is block walking. We aim to find a closed formula for the number of paths of length $2n$ from one corner of an $n \times n$ square to the opposite one without crossing the diagonal connecting them (solutions for $n = 5$ shown below).



First, we can calculate the total number of paths (ignoring the restriction for now). There must be n vertical steps and n horizontal steps, so out of the $2n$ steps, choosing any n of them to be vertical determines a unique path. Therefore, there are $\binom{2n}{n}$ total paths.

Then, define a "bad" path as one that crosses the main diagonal and touches the next diagonal up/down. If we reflect the invalid portion of the path, there are now $n+1$ vertical steps and $n-1$ horizontal steps. Thus the number of bad paths can be expressed as

$$\binom{n-1+n+1}{n-1} = \binom{2n}{n-1} = \binom{2n}{n+1}.$$

Moreover, the number of desired paths is

$$\binom{2n}{n} - \binom{2n}{n+1}.$$

Using induction to verify that expression this is equivalent to all the other Catalan formulas is left as an exercise to the reader.

1.0.12 Exercise

Consider a bijection of this to handshaking; correspond each person shaking hands with an even number but not an odd number to walking along the black line but not the red line.

1.0.13 Remark

The idea of taking a bijection considered in the exercise above can be applied to all Catalan-type recurrence relations.

1.1 Problems

1. Compute C_6 and C_7 using both the recursion and the explicit formula.
2. How many ways are there to arrange n open brackets "[" and n closed brackets "]" such that when read left to right, the number of closed brackets is less than or equal to the number of open brackets?

2 The Hausdorff Moment Problem

We begin with a definition.

2.0.1 Definition

$d-$dimensional Catalan numbers $C_d(n)$ are defined by

$$\prod_{p=0}^{d-1} \frac{p!}{(n+p)!}(dn)! \qquad [1]$$

where $2 \leq d \in \mathbb{Z}$, and $0 \leq n \in \mathbb{Z}$.

2.1 Lemma

The integral representation of the Catalan sequence is

$$C_n = \frac{1}{2\pi}\int_0^4 x^n \sqrt{\frac{4-x}{x}}dx.$$

The following section will provide a walk-through of the extensive proof of the key theorem below:

2.1.1 Theorem

These $C_d(n)$'s are the nth Hausdorff power moments of positive functions $W_d(x)$ for $x \in [0, d^d]$.

Catalan numbers that can be associated with a dimension d allow for $d-$dimensional extensions of the theorems for the two-dimensional Catalan numbers explored in the Introduction. Note that by $[1]$, we have that $C_d(0) = 1$ for all d. We seek the solutions of the Hausdorff Moment Problem

$$\int_0^{R(d)} x^n W_d(x)dx = C_d(n),$$

where $R(d)$ will be the upper edge of $W_d(x)'s$ support:

$$R(d) = \lim_{n\to\infty}((C_d(n))^{\frac{1}{n}}) = d^d.$$

2.1.2 Claim

The $W_d(x)$ that we seek in the Hausdorff Moment Problem is positive.

2.1.3 Proof

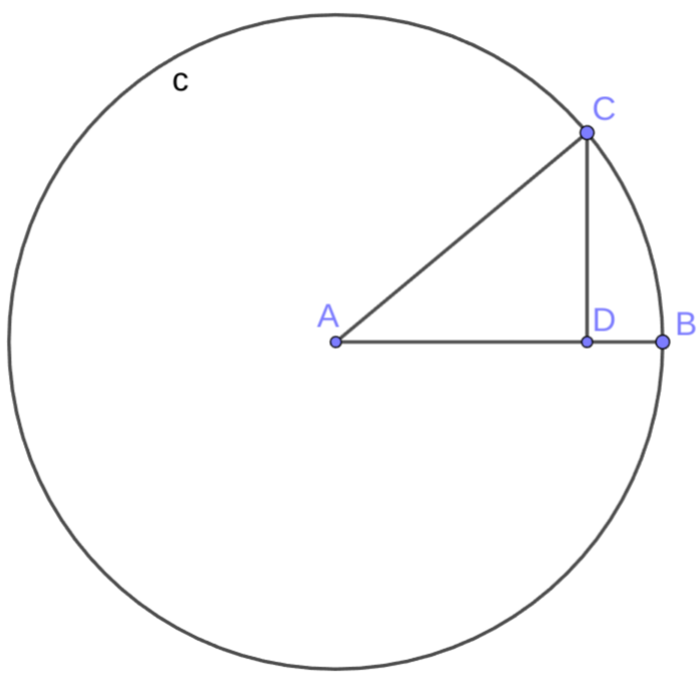
By the Gauss-Legendere multiplication formula for the gamma function on [1] and for complex z such that $z = n + 1$, we have that

$$C_d(z-1) = (2\pi)^{1-d/2} \cdot d^{\frac{1-2d}{2}} \cdot \prod_{k=0}^{d-1} k! \cdot (d^d)^z \cdot \prod_{j=0}^{d-1} \frac{\Gamma(n+j+1)/d}{\Gamma(z+j)}$$

is the Mellin transform of $W_d(x)$. From here, we note that j has a Mellin transform that indicates its proportionality to the standard beta distribution over $[0,1]$. Looking at the $C_d(z-1)$ as the product of d individual terms, we can see that $W_d(x)$ is continuous and thus positive, because it is a $d-$fold Mellin transformation of positive, continuous functions over $[0,1]$. Then, we note that the Hausdorff Moment Problem is dependent on $\frac{x}{d^d}$, a $d-$fold Mellin convolution. Executing this Mellin convolution is done by identifying $W_d(x)$ by the Meijer $G-$function, and then converting it to hypergeometric form. This gives us $R(d) = d^d$.

From here, computing the coefficients $c_j(d)$ using the integral representation of the Catalan numbers along with a density argument gives that the $W_d(x)$ give solutions, as desired.

3 A Surprising Geometric Result



Consider circle c with radius $r = AB = AC$. Let the length of CD be 1. Then,

$$AD = \sqrt{r^2 - 1}$$

and

$$DB = r - \sqrt{r^2 - 1}.$$

Seeking to find the length of DB, we can try plugging in some small values of r. For example, $r = 5$ gives $DB = 0.101020514433643803...$, and if you look closely, you'll spot the numbers $1, 1, 2, 5, 14$, and a number approaching 42 hiding in the decimal expansion. These are the first six Catalan numbers! But wait, there's more... Plugging in $r = 50,000,000,000,000,000$, we get:

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8586045349200000000000000459508043246217423640000000000000176733862787006701400000000000
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0119592996693088040000000002257117854077248073253720000000008740328711533173390046320
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7138054608572000000197826165775616065362377445600000076847856705143163852308161560000029
8691669457726259501424175120000116157871455782434250553845880000045195971802795347144760
9509424000175941461660881887099247987597200068524569278448734975496584643120026700952856
7748519042452209126640104088460289122304033498318812080040594499512757698573064344336711
2158385096459612004268677277903889661821279585848556504870808472163384139737743045626825
7114585462733214295850558771979787935384946380161847916987581665947900904271354709404185
7315091966887248989424386591326815845603127349939157058491473033725021956517463358552614
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0489365077377841208237632349300034187831110332113753629151880925616680544303788118287607
2200796260983191303633039113283862083948663403387926809884575211361279087552567774401153
2458459690151854647358184343717526489945428384602949088090035002473362719316328770046679
0289661893883204824632412975061349361581563016776834735140517915662521840347558019470788
8052034668674198730310039511802864246328916867275604157266738359730366580434768767311107...

Between the zeros lie Catalan numbers galore! Why does this work?

Think about generating functions. When we use the Taylor series expansion of the Catalan recurrence relation, $\dfrac{1-\sqrt{1-4x}}{2x}$, we can let $r = 5 \cdot 10^{n-1}$. Plugging into our expression $r - \sqrt{r^2 - 1}$ and expanding with the Taylor series will imply the problem; the rest is left as an exercise to the reader.

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The Graph Theory Behind Games: The Seven Bridges of Konigsberg and the Game of Hex

Shaon Anwar

1 The Problem of the Seven Bridges of Konigsberg

1.1 Introduction

During the 18th century, the city of Konigsberg (currently Kaliningrad, Russia) was divided into four districts by the Pregel River. Seven bridges connected the four districts (as shown below). Legend had it that residents would stroll throughout the city and eventually challenged each other to find a way to traverse all seven bridges only one time each.

In other words, the problem was to design a walking tour of Konigsberg (through all its four districts) by crossing all seven bridges exactly once.

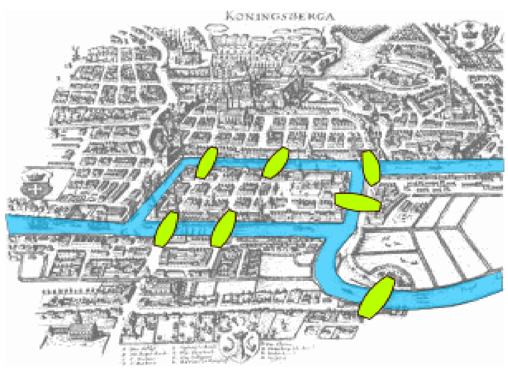


Figure 1: Map of Konigsberg in Euler's time

The puzzle reached Swiss mathematician Leonhard Euler who found it trivial, but also intriguing. In 1741, he published a solution to the problem. Since he could not find a sufficient explanation using branches of math such as Algebra or Geometry, Euler solved it using an abstract process that led to a new branch of mathematics called "Graph Theory," where he applied a new mathematic model called a graph.

1.2 Euler's Approach and Solution

According to Euler, physical maps did not matter. Rather, the important element to consider was which land masses were connected by bridges. He transformed the problem into a network of vertices and connecting edges, known today as a **connected graph**, with each vertex representing a land mass and each edge a connecting bridge.

For the purpose of our explanation, we establish some definitions:

- **Graph:** A set of vertices which are connected by edges.
- **Vertex:** A node on a graph.
- **Edge:** A link connecting a pair of vertices.
- **Degree of a vertex:** The number of edges incident to a vertex.
- **Walk:** A finite alternating list of vertices and edges.
- **Trail:** A walk with no repeated edges.
- **Path:** A walk with no repeated vertices.
- **Circuit:** A closed walk (starts and ends at the same vertex) that contains at least one edge and does not contain a repeated edge.

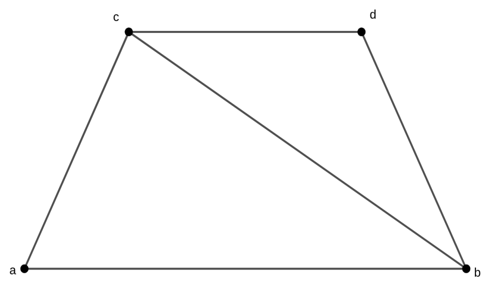


Figure 2: A graph with 4 vertices and 5 edges

For example, in the figure above, we have:

$\deg(a) = 2$
$\deg(b) = 3$
$\deg(c) = 3$
$\deg(d) = 2$

Now, returning to Euler's discourse...

Euler denoted each district with capital letters: A, B, C, and D. He also observed that Konigsberg had one district that was connected by five bridges, while the other three regions were connected by three bridges (as shown in the figure below).

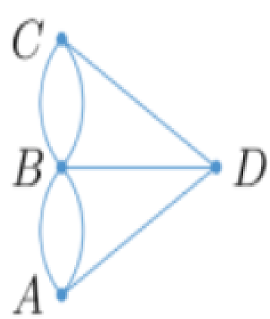


Figure 3: Euler's graph theory representation of the map of Konigsberg

The graph above could be used to translate the Konigsberg problem into the following question:

Is it possible to start at A, B, C or D, traverse each edge exactly once, and end at the starting point?

Finding such a possibility would be the equivalent of finding the solution to the puzzle.

We can see that when we start from the vertex A and go to B, C, and D, we always have to use one edge to get in and another edge to get out from each vertex. That signifies that we must have an even number of edges leading into and out of the vertices B, C and D. However, in our scenario, $\deg(C) = \deg(D) = 3$, and $\deg(B) = 5$. Hence, given the odd degrees of B, C and D, there is no such route that starts and ends at A and leads across all bridges only once. For example, as one of my possible trip routes, I can begin my travel at A and walk to B, then C and next I can go back to B and proceed back to A (my journey's starting point). However, since I need to visit D as well, I am forced to use the third "outbound" edge of A. Now, I cannot return to A because there is no "inbound" edge I could possible use that leads me to A and has not been traveled already. Similarly, considering vertices other than A as a starting and ending point, we can arrive at the same conclusion. (Readers can verify this by attempting to trace the graph without lifting a pen and redrawing on line.)

Thus, it is impossible to find a closed path that satisfies the required condition of the Konigsberg puzzle.

In honor of Euler, such a closed path, with no isolated vertices and where each edge was passed exactly once, became known as an **Euler circuit**.

1.3 Euler's Theorem

Generating one rule to solve puzzles with arbitrary number of bridges and land masses led to the following theorem:

1.3.1 Statement

Let the graph $G(V, E)$ be an undirected graph (in that an edge can be traversed in both directions) with no isolated vertices.

G has an Euler circuit if and only if:

1. G is connected.

2. Every vertex has an even degree.

1.3.2 Proof

First, we show that if G has an Euler circuit, then for all $v \in V$, the trail T $\{v_0, v_1, ...v_{k-1}, v_k\}$ with $v_k = v_0$, G is connected.

Trail T visits each vertex of G at least once (it is possible to revisit a vertex), and since every vertex appears somewhere along the route as a part of sequence of adjacent vertices and edges of

G, we can deduce that G is connected. Since the route passes through any given vertex v of G as the process of entering and departing by using distinct edges that lead to and from the vertex, we can see that the degree of each vertex v is even.

Conversely, let's consider a connected graph G with every vertex having an even degree (assuming the vertex set of G is nonempty). To show that G contains an Euler circuit, we proceed by induction on the number of edges n in G.

In the base case, there are two edges, and since the degree is even, then these edges must be between the same vertices. Therefore, an Euler circuit is immediately represented.

Now, we show the statement holds for fewer than n edges.

Let's consider a connected graph G with vertices of even degrees, and v_0 as our starting vertex of G. Now, let's pick any sequence of adjacent edges and vertices, without repeated edges containing v_0, beginning and ending at v. Let's call the resulting circuit C.

If we see that C includes every edge of G, then we are done.

If this is not the case, we proceed to traverse C, and we keep removing its edges until we get to the vertex where there are no more outcoming edges, and this vertex must be v_0. (We are stuck here because by assumption there are no vertices with an odd degree that would allow us to depart.)

In the process of removing edges, we end up with a subgraph of G, which we can denote by G'. As we traversed C, we have removed an even number of edges from each vertex, so G' has vertices of an even degree. Though it may not be connected, each component subgraph of G' is connected, and each vertex of these components has an even degree.

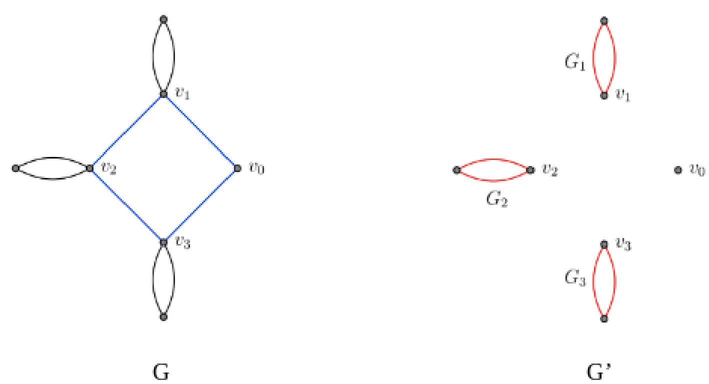


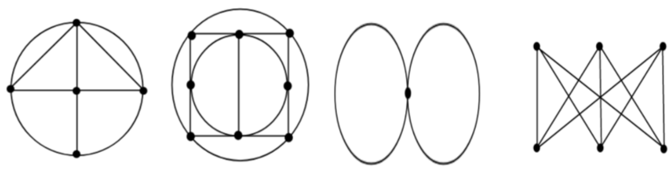
Figure 4: The graph G before and after removing edges

In addition, each of these Euler circuits has at least one vertex in common with G'. Next, we may choose any sequence of adjacent vertices and edges of G'. For example, starting at v_0 and then arriving at v_1, moving along the Euler circuit of G_1, and then moving to v_2 and completing the Euler circuit of G_2, and then to v_3 and so on, until we get to the starting vertex because the degree of v_0 is an even integer and hence, there must be another edge that brings us back to v_0. That way, we can reconstruct an Euler circuit for G, noting that our algorithm must terminate because G has finite number of edges.

1.4 Exercises in Finding a Euler's Circuit

[Solutions to all exercises included at the end.]

1. Which graph from the following collection represents an Eulerian circuit?



2. An architect wants to add additional bridges to the blueprint of Konigsberg city. How many should she draw to make an Eulerian circuit?

3. Leonhard wants to visit the Western states of the United States. He wants to start and finish in the same state crossing each border exactly once. Is this possible?

1.5 Concluding Thoughts

Euler's intellect and vision to begin with a specific and perhaps banal problem but derive from it a new theory through a generalization process is indeed remarkable and inspiring. Today, graph theory has become an important branch of mathematics because web pages connected by links, social media pages, family charts, and the lattices formed by atoms are all modeled by graphs.

2 The Mathematics Behind the Game of Hex

"Problems worthy

of attack

prove their worth

by hitting back." (Piet Hein)

2.1 Introduction

Piet Hein, a Danish mathematician and writer, and John Nash, an American mathematician and economist, invented the game of Hex independently in 1942 and 1948 respectively. Initially, in Copenhagen's Niel Bohr's Institute of Theoretical Physics, the game was called Polygon because of the diamond-shaped board used by players. In Princeton University, where Nash was an undergraduate student, the game was known as "John" (because the hexagonal pattern of tiles found in the bathroom created a suitable area to play). Ultimately, however, in 1952, the game was renamed Hex when a Salem game board company Parker Brothers, Inc. (mostly recognized for Monopoly) started its circulation. In 1953, Claude Shannon and E. F. Moore created a first Hex-playing analog machine at the Bell Telephone Laboratories. The general public was introduced to the game by Martin Gardner in his "Mathematical Games" column in *Scientific American* in 1959.

2.2 Let's Play!

The premise of the game is relatively simple. The board is composed of an 11×11 hexagonal grid, and two players alternate turns by claiming hexagons on the board. One player "owns" two opposite sides of the board, and the other "owns" the remaining two sides. (In our picture below, the colors are blue and red.) The objective of the game is to make a continuous path of "owned" hexagons from one side to the other.

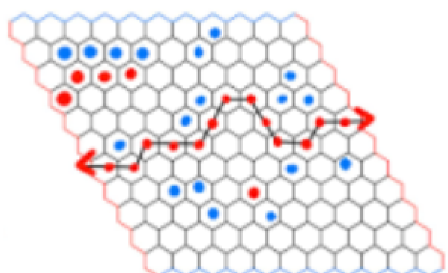


Figure 5: A sample game of Hex

2.3 The Hex Theorem

One particularly interesting component of Hex is that the game never ends in a draw; a player wins if and only if the other loses.

This Hex Theorem can be proved by interpreting the board as a planar graph, with vertices as corners of every hexagonal cell as shown in below figure.

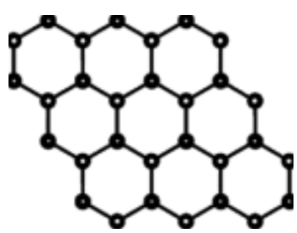


Figure 6: The Hex board as a planar graph

To prove the theorem, we begin with a graph lemma:

2.3.1 Lemma

Any finite graph with vertices of degree at most 2 is a union of disjoint components, and each component is either:

- an isolated vertex
- a simple cycle (defined as a closed path where middle vertices are not visited twice)
- a simple path (no vertices are repeated).

2.3.2 Proof

We proceed by induction.

Given a graph G with N vertices, where each vertex has degree of at most 2 and the maximum number of edges N, we can denote a graph with k edges as G_k.

For G_0, our base case, all the nodes are isolated, as we have N vertices and 0 edges.

When a graph has $n+1$ edges, we randomly choose an edge (s, v) to remove. The vertices s and v now have a degree at most 1, since they had a degree at most 2 initially.

Therefore, s and v cannot be on any cycles. By assumption, g_n is the union of disjoint isolated nodes, simple cycles, and simple paths.

When we add (s, v) back into the graph, the components that were disjoint from s and v in g_n remain the same by the addition of (s, v), and the vertices s and v are now either on the same simple path or cycle.

Therefore, G_{n+1} is also the union of disjoint isolated vertices, simple cycles, and simple paths.

Thus, the lemma is true for all g_k with $0 \leq k \leq N$.

2.3.3 Proof of the Hex Theorem

Now, let's substitute the two colors in the game with Xs and Os, and add four vertices u_1, u_2, u_3 and u_4 at the four corners of the main graph and connected to the graph by four edges e_1, e_2, e_3, and e_4.

Every tile will be marked with either an X or O. Also, opposite sides will be changed to O-O' and X-X' respectively, as shown below.

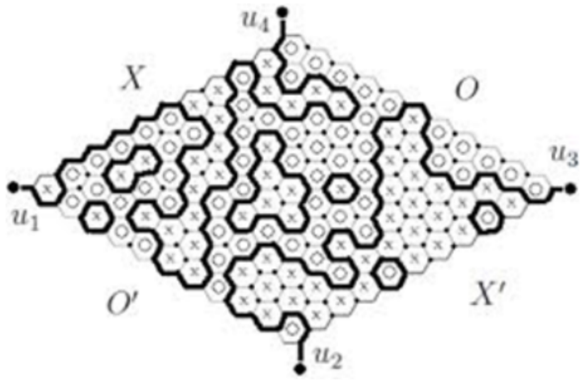


Figure 7: Adjusted Hex board with Xs and Os

Following our changes in representation, we can restate the Hex Theorem in the following way:

If every tile of the Hex board is marked either X or O, then there is either an X-path connecting regions X and X' or an O-path connecting regions O and O'.

In order to prove the Hex Theorem, it suffices to show that there is a simple path connecting any two corner vertices.

Step 1: Let's construct a subgraph $G'(V, E')$ containing the same vertices and the subset of the edges that lie between X and O tile faces.

Step 2: The corner vertices u_1, u_2, u_3, and u_4 have degree 1.

Step 3: If a vertex is surrounded by two same patterned tiles and one differently patterned tile, then it has two edges. Thus, each vertex has degree of 0 or 2.

Step 4: By proof of the lemma above, if a graph has vertices with degree at most 2, the graph is a union of disjoint components: isolated vertices, simple cycles, or simple paths.

Step 5: Thus, there exist two paths connecting two of the corner endpoint vertices. Hence, there is a winning path for one of the players.

2.4 Zermelo and the Theory of Games

Since we have proven that Hex cannot end in a draw, one of the players must have a winning strategy. In other words, independently of how the opponent plays, with a finite number of moves, and the absence of chance, the player's strategy will result in a win. This forms the basis of Zermelo's Theory, which is often cited as the first mathematical analysis of graph theory.

In his paper "On an Application of Set Theory to the Theory of the Game of Chess" (1913), German mathematician Ernst Zermelo analyzed chess as an example of such finite, two-player, zero-sum games (where one player's gain corresponds with another player's loss) to figure out the optimal strategies for both players that decide both players' payoff. It is important to note that modern game theory was marked by John Von Neumann and Oskar Morgenstern's proof and characterization of mixed equilibria in zero-sum games. John Nash subsequently generalized the analysis beyond zero-sum games to equilibrium in games that presented more complex situations, where people interact strategically and purposefully to maximize their advantage.

To show why Zermelo's theory is true, let's assume that the first player does not have a winning strategy and initiates a move. In this case, the second player prevents the first from winning by moving strategically in response. This preemptive set of moves guarantees that the second player cannot lose because she is presented with at least a drawing strategy. Now, since the game cannot end in draw, this drawing strategy is equivalent to forcing a win. Hence, she has a winning strategy.

In the game of Hex, the two players take turns, make a strategic finite number of moves without the element of chance, and cannot tie. Thus, Zermelo's theory posits that one of the two players must have a winning strategy.

2.5 The Strategy-Stealing Argument

Additionally, John Nash determined another interesting component of the game: the first player has always a winning strategy. (My younger sister mercilessly exploited this assertion when playing against me.) In 1949, Nash proved the strategy-stealing argument using the *reductio ad absurdum*, which can succinctly be explained as follows:

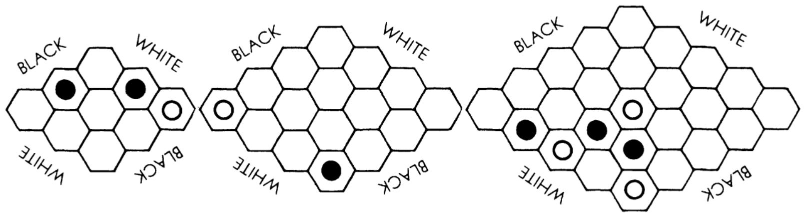
1. Assume that the second player has a winning strategy.
2. The first player makes a move by marking a cell anywhere on the board.
3. After the second player's move, the first player purposefully ignores his already claimed cell and pretends that the opponent's move is the initial one. This extra claimed cell on the board does not interfere with the player's game and serves only as an advantage.
4. In effect, the first player "steals" the second player's winning strategy, as he now masquerades as the second player.
5. This is a contradiction to our starting assumption that the second player has a winning strategy.
6. Thus, discarding the assumption, we can conclude that there is a winning strategy for the first player (remembering that a draw is impossible).

2.6 State of Play

Since the first player has a strong advantage in Hex, players are allowed to implement a swap rule, which allows for the second player to take the first player's initial move and claim his cell. This rule is often utilized during Hex tournament at Computer Olympiad organized by the International Committee of Mathematical Games (ICMG) since 2000.

2.7 Conclusion

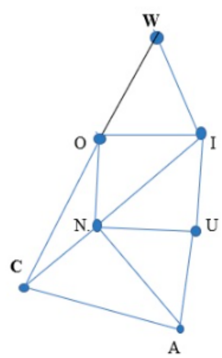
There are numerous online games available for the readers who would like to simply try Hex. For those readers who would like a challenge, Martin Gardner offered three Hex problems devised by Hein with the objective of finding first move that would ensure the win for "white."



Hex is a beautiful game with a simple set of rules yet full of critical tactical and strategic intricacies. With a variety of openings and 121 hexagons, creating 2.4×10^{56} legal positions (exceeding that of chess!), as Piet Hein fittingly put it in his grook, Hex indeed can become a problem that is worthy of attack.

Solutions to Exercises in Finding a Euler's Circuit

1. The third graph represents an Euler circuit because it is the only graph that is connected with vertices with an even degree.
2. The architect can draw two bridges in order to make an Euler circuit (for example, in Figure 1: another bridge that connects landmasses C and D, and one that connects landmasses A and B)
3. It is impossible.



The graph contains 4 vertices with an odd degree.

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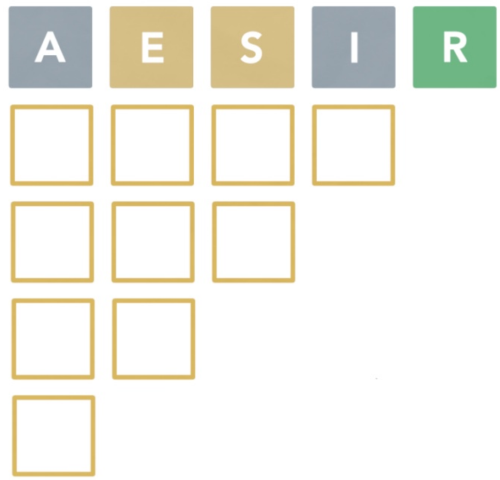
Hacking Wordle: A CS Perspective on the Viral Word Game

Jeremy Ku-Benjet



1 Introduction: What Is It?

Does Josh Wardle's Wordle need any introduction? The simple word game has spread like wildfire, attracting millions of players across the world at its peak in early 2022. Every day, the website chooses a random five letter word that the player tries to solve for in as few guesses as possible. After every guess of a letter, the color of the five background tiles changes to grey, yellow, or green, indicating how close they were to the chosen word. It is grey if the letter is not in the solution word, yellow if the letter is in the solution word but in the wrong position, and green if the letter is present and in the right place. The player can keep guessing until they have either guessed the chosen word or used all six tries.

2 What is the Best Strategy?

With luck, any strategy can be considered the best strategy. If the starting word suggested by the strategy happens to be the solution word, then the strategy is technically proven to be effective.

3 The Worst Case

However, the lucky case, or optimal outcome for a strategy, is not helpful when evaluating how well one will perform when it is actually used, as it is almost always an edge case that occurs with very low probability. More insightful measures include of how many guesses the average game of Wordle will take, or how many guesses the longest game of Wordle could take. For players never wanting to lose, the worst case, where they use up all their chances to guess the word, is most important.

Wordle has a list of roughly 10,000 valid guess words and a separate list of roughly 2,300 possible solution words, all of which are guessable. The guess words tend to be less familiar while the solution words are more common. When players guess a possible solution word, they get feedback that allows them to eliminate some of the possible solution words. Players win only when they guess the solution word. In the worst case, players would have to eliminate all but a single word with previous guesses to win.

A succinct way to represent this system is by associating each word with a "bucket" of other words. The bucket has all of the words which are still possible after that word is guessed. For example, if the possible solution words had been narrowed to "power," "mower," and "mover," and the guessed word was "tower," the associated bucket would be the list of two words "power" and "mower." The game of Wordle can then be described as having a starting bucket of all the possible solution words, and then replacing that starting bucket with buckets associated with guesses until a bucket associated with a guess has only one word in it. That one word is the solution word.

4 An Upper Bound on Number of Guesses

Intuitively, a good strategy should eliminate many words with each guess, with the goal being able to choose guesses with small associated buckets. One such strategy is to greedily choose the guess which, in the worst case, would have the smallest bucket. For each possible remaining solution word, assume for a moment it is the current solution. Then find the bucket associated with each guess. This creates a list of possible buckets for each possible guess, each bucket assuming a different solution word. The heuristic for trying to minimize the worst case based on the above intuition is to eliminate as many words as possible with each guess, so the strategy should be to guess the word whose maximum bucket over all the possible solution words is minimum.

Simulating this strategy with a computer gives "aesir" as a starting word, which surprisingly matches the intuition of guessing words with common vowels first. Testing every possible solution word shows that the longest running game takes five guesses. This greedy strategy will make sure one never loses a game of Wordle!

This puts an upper bound on the amount of moves needed to guess a solution word in the worst case: five. But as the greedy solution is only a heuristic for a good strategy, it says nothing about a lower bound for the worst case number of needed guesses, which is definitely a larger computation.

5 A Lower Bound on Number of Guesses

A game of Wordle can further be represented by a path through a tree. The root of the tree is a bucket containing all the possible solution words. The edges represent guesses and connect nodes representing buckets before and after a certain guess. The size of the tree grows exponentially as longer paths of guesses are considered, quickly becoming too hard to manage.

Fortunately, considering long paths is irrelevant, since the point of searching this tree is to figure out if it is possible to solve every Wordle puzzle in four or fewer guesses. It is already known to be possible in five through the greedy strategy. Therefore, any path with more than four edges, a game where the player guesses more than four times, can be ignored.

The search space is still large, though now not so large that it cannot be searched with brute force. A recursive search, using the greedy strategy above as a heuristic to find solutions to words more quickly, will show that solving every Wordle puzzle with only four guesses is not possible; a lower bound for the worst case number of guesses is five.

This shows the greedy strategy is optimal for minimizing guesses in the worst case, but it is not the only one. Many strategies can achieve optimal performance, some being more possible for use by humans than others. Even a simple list of starter words can be effective, though at that point the problem becomes a test of human skill more than anything else.

6 A Quick Note on Greedy Adversarial Wordle

The strategies above are for the player to make the game as short as possible when Wordle happens to choose the word that forces the game to run as long as possible. But this brings about the question of what an adversarial game, a game trying to force the player to make as many guesses as possible, would look like.

An adversarial game which forces five guesses from the greedy strategy to be confident in a solution can always have the solution word "badly." Always having the same word won't work against every strategy though, since the strategy of always guessing "badly" quickly becomes very ineffective.

Instead, making a game which adapts to guesses is more versatile. The game will appear to have a set solution word, but in actuality it changes the solution word to prolong the game as the player guesses.

An example is a game which works like the greedy guessing strategy. It will have a bucket of possible solution words which it can choose from. When a player guesses, the bucket is split, and the part of the bucket containing the remaining possible solution words depends on the game's response of greys, yellows, and greens. Analogous to how the greedy guessing strategy would choose the smallest bucket, the game would choose the response causing the current bucket to be as large as possible, helping to narrow to the least set of possible solutions.

While this creates a hard game to beat as a human, it is not optimal in prolonging guesses. The best it can do is force a four-word solution, for example, the sequence of guesses "aiery," "canst," "dolma," or "offal." As the game described above is deterministic, the sequence of words will always win.

The reason players cannot win in three guesses is because for them to be able to win in three guesses, they would have to narrow the word down to a single word by the time of their second guess. In response to each guess, the adversarial game divides the current bucket of words into at most $3^5 = 243$ smaller buckets; one bucket for each possible response. Each of these buckets contains distinct words, so for there to only be one word in every bucket after the second guess, the maximum bucket size after the first guess would have to be at most 243 words. All guesses which sufficiently narrow the possible solutions do not lead to a three-word win. They are found through simply trying every possible pair of words after the first guess.

7 Conclusion

While the worst case is interesting, the most useful question for playing the game is: Which algorithm minimizes the average number of guesses? Strategies for optimizing the average case are endless, and writing up those strategies is similarly endless, so they won't be found here, though it is a worthwhile problem to think and read about. It might even be fun to take a crack at the daily Wordle today. Good luck!

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Banach-Tarski and the Paradox of Infinite Cloning

Lamisa Aziz

1 Introduction

What do you think of when you hear the word "infinity"? The symbol ∞ or something that never ends likely come to mind. By definition, infinity is a value greater than any assignable quantity or countable number. In close relation to infinity is cloning, or producing an identical replica of the individual organism or object. Now, consider the following situation: Imagine you and your friend become hungry during a hike, so you decide to split an apple into two. At this point, you think your share of the apple is $\frac{1}{2}$ of the original, but your friend disagrees. She says that theoretically, both pieces of the apples are exactly the same size as the original full-sized apple. What does she mean by this?

This scenario represents the argument called the Paradox of Infinite Cloning, an idea promulgated by mathematicians Stefan Banach and Alfred Tarski in 1924. In essence, this paradox claims that concept of infinite points makes it possible to turn one whole apple into two whole apples. The paradox itself proves that according to the fundamental rules of mathematics, a solid three-dimensional ball can be split into parts that can combine again to form two exact identical copies of the original. This paradox arose largely due to the complex concept of infinity as mentioned earlier. People often mistake infinity to be a real number, and as a result get confused by its unique properties. For instance, you can add or subtract finite numbers to infinity, and the outcome will always be infinity. The difference between a countable and uncountable infinity makes the natural numbers a smaller infinity compared to the real numbers, and this is called a difference in cardinality. In 1891, Georg Cantor proved that the infinite number of points on a line has the same cardinality as the infinite number of points for a volume of a shape.

Cantor's discovery inspired Banach and Tarski's thinking when they realized that they could use an uncountable infinite set of points to turn one sphere into two, or in other words, perform infinite cloning. The Banach-Tarski Paradox is a theorem which states: Given a solid ball in 3D space, there exists a decomposition of the ball into a finite number of disjoint subsets, which can then be put back together in a different way to yield two identical copies of the original ball. The theorem is a paradox because it contradicts the basic principles of geometry. The idea that an object can be doubled by dividing and transforming it without adding any new points seems impossible and supernatural.

Fun Fact: A stronger form of the theorem implies that given any two reasonable solid objects, the cut pieces of either one can be reassembled into the other. This is often stated informally as "a pea can be chopped up and reassembled into the Sun" and called the "pea and the Sun paradox."

2 A Four-Step Proof

There are four basic steps to prove this theorem.

2.1 Step 1

The first step is to find a paradoxical decomposition of the free group in two generators. The free group with two generators a and b consists of all finite strings that can be formed from the four symbols a, a^{-1}, b, and b^{-1} such that no a appears directly next to a^{-1} and no b appears directly next to b^{-1}. The group F_2 is a group that is "paradoxically decomposed," and let $S(a)$ be the set of all non-forbidden strings that start with a. The resulting equation would be:

$$F_2 = e \cup S(a) \cup S(a^{-1}) \cup S(b) \cup S(b^{-1}) = F_2 = aS(a^{-1}) \cup S(a).$$

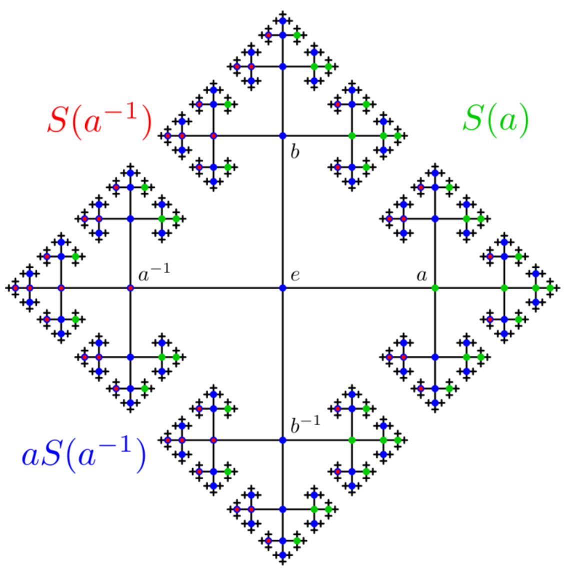


Figure 1: Cayley Graph of F_2

2.2 Step 2

The second step is to find a group of rotations in a 3D space isomorphic to the free group in two generators. In order to find a free group of rotations of 3D space that behaves just like the free group F_2, two orthogonal axes are taken. Another arithmetic proof of the existence of free groups in some special orthogonal groups using integral quaternions lead to paradoxical decomposition of the rotation group.

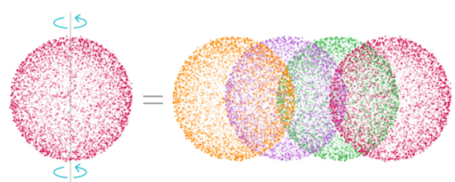
2.3 Step 3

The third step is to use the paradoxical decomposition of that group and the axiom of choice to produce a paradoxical decomposition of the hollow unit sphere. The unit sphere S^2 is separated into orbits by the action of our group H: two points belong to the same orbit if there is a rotation in H which moves the first point into the second. The paradoxical decomposition of H yields a paradoxical decomposition of S^2 into four pieces: A_1, A_2, A_3, A_4.

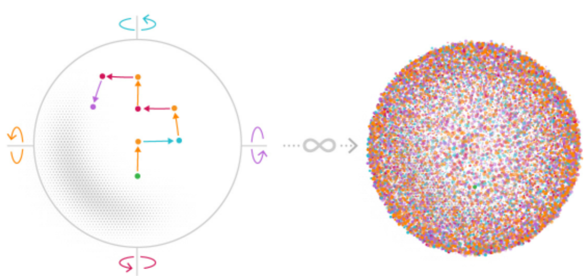
$$A_1 = S(a)M \cup M \cup B$$
$$A_2 = S(a^{-1})M/B$$
$$A_3 = S(b)M$$
$$A_4 = S(b^{-1})M$$

2.4 Step 4

The final step of the proof is to extend this decomposition of the sphere to a decomposition of the solid unit ball. As a result, this connects every point on S^2 with a half-open segment to the origin; the paradoxical decomposition of S^2 then yields the difference between paradoxical decomposition of the solid unit ball and the point at the ball's center. To apply this problem, for example, think of the group that contains all points derived from a final rotation to the East. Rotate this group to the West, and this instantly negates all those final East rotations, transforming the group into the collection of points that immediately preceded the formation of the original sets. The group now contains points that finished on North, South, and East rotations. In other words, the rotated piece contains both the new content and its old self. The nature of infinity makes this seeming increase possible.



Rotating the **East group** to the **West** duplicates the **South**, the **North**, the **starting points**, and the **East group** itself.



3 The Von Neumann Paradox

The Banach-Tarski paradox also corresponds to the Von Neumann paradox, which explains the impossibility of a distinction between planar and high-dimensional cases of the decomposition of a disk of Banach-Tarski using Euclidean congruence. Jon Von Neumann's paradox includes the idea that, unlike the 3D rotation group, the $E(2)$ group of Euclidean motions is indeed solvable, so it rules out the paradoxical decomposition of non-negligible sets. Von Neumann goes on to question, "Can such a paradoxical decomposition be constructed if one allows a larger group of equivalences?" Since any two squares in a plane can become equivalent, a paradoxical decomposition of a square could be counter-intuitive for the same reasoning as the Banach-Tarski decomposition of a ball.

Using the Banach-Tarski paradox can actually strengthen the paradox for a square by claiming that any two bounded subsets of the Euclidean plane with non-empty interiors are equidecomposable with respect to the area-preserving affine maps.

The Von Neumann paradox has made progress in recent years. For example, in 2017, it was found that although the paradox concerns the Euclidean plane, there are other classical groups where paradoxes are possible; in 2019, the paradox was found to use finitely many pieces in the duplication. In return, the class of groups that was isolated by Von Neumann in the study of the Banach-Tarski paradox has been helpful in other areas of mathematics.

4 Conclusion

Today, mathematicians consider the Banach-Tarski paradox as a demonstration of the richness of mathematics. It offers a law-abiding example of how mathematics can stray from physical intuition without contradicting itself. As American mathematician Philip J. Davis once reflected, "One of the endlessly alluring aspects of mathematics is that its thorniest paradoxes have a way of blooming into beautiful theories."

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The Facility Location Problem

Yuhao "Ben" Pan

1 The Facility Location Problem

Given:

- A set of facilities, F, and a set of clients, C.
- Each connection between a facility i and a client j has an associated cost, c_{ij}.
- Each facility i has a setup cost of f_i.
- Defined on a metric space, i.e., the costs follow the triangle inequality (we will get to that in section 5.2).

Goals:

- Choose a subset of facilities and open them.
- Assign/connect each client to **exactly one** facility.
- Minimize the overall cost (facility + connection).

Consider the following scenario: Imagine you are a shipping company like FedEx or UPS. You are looking to expand into New York State and build warehouses (facilities) to store goods. You have some major cities to serve, like New York, Buffalo, Syracuse, Albany, etc. You also have some potential locations in mind. How can you choose some of these locations to open warehouses while minimizing the total distance between the cities and their assigned warehouses so that goods can be delivered faster? This is a real life example of what is called the facility location problem.

2 Introduction to Linear Programming

Before we get into the details of the problem, we will introduce the concept of linear programming. Suppose we are given a maximization problem to be formulated in linear programming in the following way:

Maximize

$$c_1x_1 + c_2x_2 + \cdots + c_nx_n,$$

subject to the following m constraints:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\leq b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\leq b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &\leq b_m \\ x_1, x_2, \cdots, x_n &\geq 0. \end{aligned}$$

We can think of c_i as "cost coefficients"and x_i as decision variables. We will also define the **objective function** as the expression we are trying to maximize, namely $c_1x_1 + c_2x_2 + \cdots + c_nx_n$.

2.1 Dual LP

Suppose we solve a linear program, like the one above, and we want an alternate method to verify that our solution is correct. How can we do that? The answer lies in the **dual** of a linear program.

The dual of a linear program is simply another equivalent linear program based off of the original LP, or the **primal** LP. By solving the dual LP, we obtain an optimal solution that should be the same as the optimal solution given by solving the primal LP. In other words, solving the dual LP verifies the solution of the corresponding primal LP.

2.2 Derivation of Dual from Primal

Let us use the standard primal LP above.

First, we multiply each constraint by a constant y_i:

$$\begin{aligned} y_1a_{11}x_1 + y_1a_{12}x_2 + \cdots + y_1a_{1n}x_n &\leq y_1b_1 \\ y_2a_{21}x_1 + y_2a_{22}x_2 + \cdots + y_2a_{2n}x_n &\leq y_2b_2 \\ &\vdots \\ y_ma_{m1}x_1 + y_ma_{m2}x_2 + \cdots + y_ma_{mn}x_n &\leq y_mb_m. \end{aligned}$$

Then, we add all constraints and combine like terms:

$$\begin{aligned} &(y_1a_{11} + y_2a_{21} + \cdots + y_ma_{m1})x_1 + \\ &(y_1a_{12} + y_2a_{22} + \cdots + y_ma_{m2})x_2 + \\ &\qquad\vdots \\ &(y_1a_{1n} + y_2a_{2n} + \cdots + y_ma_{mn})x_n \leq y_1b_1 + y_2b_2 + \cdots + y_mb_m. \end{aligned}$$

Since we have an upper bound of some linear function of x $(y_1b_1 + y_2b_2 + \cdots + y_mb_m)$, the logical way of formulating an alternative linear program is to minimize the upper bound so that it is as close as possible to the maximum of the objective function.

Therefore, in the dual LP, we will minimize $y_1b_1 + y_2b_2 + \cdots + y_mb_m$.

Just like the primal LP, the dual has to satisfy a set of constraints. Since we want the upper bound to satisfy the maximum of the objective function, we want the coefficients of the new constraints to be greater than the cost coefficients (otherwise our upper bound is invalid):

$$\begin{aligned} y_1a_{11} + y_2a_{21} + \cdots + y_ma_{m1} &\geq c_1 \\ y_1a_{12} + y_2a_{22} + \cdots + y_ma_{m2} &\geq c_2 \\ &\vdots \\ y_1a_{1n} + y_2a_{2n} + \cdots + y_ma_{mn} &\geq c_n. \end{aligned}$$

This completes our dual LP.

2.3 Matrix Form of Primal and Dual

We can write the LPs above using matrices:

Let $C = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$, $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$, $Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$.

Primal:
Maximize $C^T X$, subject to the constraints:

$$\begin{aligned} AX &\leq B \\ X &\geq 0. \end{aligned}$$

Dual:
Minimize $B^T Y$, subject to the constraints:

$$\begin{aligned} A^T Y &\geq C \\ Y &\geq 0. \end{aligned}$$

2.4 How to Solve Linear Programs: The Simplex Method

There are many algorithms to solve linear programs, of which the most well-known is the Simplex Method. We won't go into the details of the algorithm, but we will explain the intuition behind it. Linear programs are essentially a system of linear inequalities, and it is a well-known result in mathematics that a system of linear inequalities defines a convex polyhedron. Intuitively, you would expect the extremum of the objective function to occur at a vertex of the polyhedron, so the Simplex Method first chooses a vertex of the polyhedron.

In a maximization problem, we will proceed as follows. During each iteration of the algorithm, we scan the neighbors of the current vertex to determine which vertex, when moved to, will increase the value of the objective function at the fastest rate. Then we move to that vertex and repeat. The algorithm terminates when, during the scanning step, moving to any neighbor will result in a decrease of the value of the objective function, which means the current vertex is the maximum solution.

The Simplex Method will only work for maximization problems. However, for minimization problems, we can simply apply the Simplex Method to their duals.

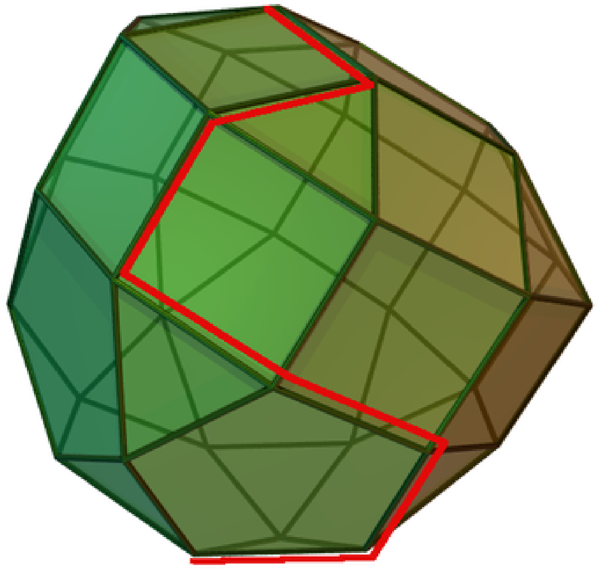

Figure 1: The path produced by our "scan"

3 Theorems of Primal/Dual

To better understand the relationship between the primal and dual, we will introduce some theorems that will also be useful later.

3.1 The Dual of the Dual is the Primal

To form the dual of the dual, we simply perform the same process by which we convert the primal to the dual, i.e., flipping of signs and matrix variables. Flipping signs and matrix variables twice would result in the original signs and variables, so the dual of the dual is simply the primal.

3.2 Strong Duality Theorem

3.2.1 Theorem

At optimality, the primal objective function value equals the dual objective function value.

3.3 Weak Duality Theorem

3.3.1 Theorem

Let $(x_1, x_2, \cdots, x_n)$ and $(y_1, y_2, \cdots, y_m)$ be feasible solutions to the primal (maximization) and dual (minimization), respectively. Then,

$$\sum_{j=1}^{n} c_j x_j \leq \sum_{i=1}^{m} b_i y_i.$$

In other words, any primal feasible solution is less than or equal to any dual feasible solution.

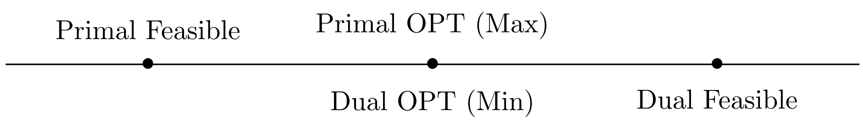

Figure 2: The relationships between feasible and optimum solutions in a maximization problem

3.3.2 Proof

$$\begin{aligned}
\sum_{j=1}^{n} c_j x_j &\le \sum_{j=1}^{n}\sum_{i=1}^{m}(a_{ij}y_i)x_j \\
&= \sum_{i=1}^{m} y_i \sum_{j=1}^{n} a_{ij}x_j \\
&\le \sum_{i=1}^{m} y_i b_i .
\end{aligned}$$

3.4 Complementary Slackness Condition

3.4.1 Theorem

Primal version: Either $x_j^* = 0$ or $\sum_{i=1}^{m} a_{ij}y_i = c_j$.
Dual version: Either $y_i^* = 0$ or $\sum_{j=1}^{n} a_{ij}x_j = b_i$.

3.4.2 Remark

Essentially, CSC describes the “slackness” of the constraints between a primal or dual LP. A constraint is “slack” if it is an inequality and “tight” if it is an equality. A variable is “slack” if it is positive and “tight” if it is 0. CSC claims that a primal variable and its corresponding constraint in the dual (or vice versa) cannot have slack at the same time. Otherwise, the solution to the LP is not optimal (“slack”) and needs to be “tightened.”

4 The Facility Location Problem, Revisited with LP

4.1 LP-formulation

Let $x_{ij} = 1$ if facility i is connected to client j, 0 otherwise.
Let $y_i = 1$ if facility i is open, 0 otherwise.
The facility location problem can then be written as a linear program:

Minimize

$$\sum_{i\in F} f_i y_i + \sum_{i\in F, j\in C} c_{ij}x_{ij},$$

subject to the following constraints:

$$\begin{aligned} x_{ij} &\leq y_i,\ \forall i,j \\ \sum_{i \in F} x_{ij} &\geq 1,\ \forall j \\ x_{ij} &\in \{0,1\} \subseteq \mathbb{Z} \\ y_i &\in \{0,1\} \subseteq \mathbb{Z}. \end{aligned}$$

Note that this linear program is a bit different from the standard linear programs. There are **integrality** constraints that must be satisfied. This is known as an **integer linear program**. In general, integer linear programs are known to be "unsolvable" in a reasonable amount of time.

4.1.1 Remark for readers who are familiar with time complexity and P vs NP

There are many algorithms that can solve a standard linear program within polynomial time. (The number of "steps" in the algorithm can be upper-bounded by a polynomial function in which the exponents are constants. In simpler terms, polynomial time = reasonable time.) However, there is no algorithm that can solve an **integer linear program** within polynomial time because you can reduce the vertex cover problem to an ILP, but the vertex cover problem is NP-hard, so integer linear programs are NP-hard as well.

Therefore, we must convert the integer linear program above to a standard linear program without integrality constraints, a process known as **LP-relaxation**. Removing the integrality constraints, we have the following linear program:

Minimize

$$\sum_{i \in F} f_i y_i + \sum_{i \in F, j \in C} c_{ij} x_{ij},$$

subject to the following constraints:

$$y_i \geq x_{ij}\ \forall i,j \tag{Primal 1}$$

$$\sum_{i \in F} x_{ij} \geq 1,\ \forall j \tag{Primal 2}$$

$$x_{ij} \geq 0.$$

However, LP-relaxation will sacrifice the accuracy of the resulting optimum solution.

4.1.2 Theorem

Given an ILP, the LP after relaxation will produce an optimum solution that is less than or equal to the optimum solution of the original ILP.

4.1.3 Remark

Note that the optimum solution of the ILP is also a feasible solution of the relaxed LP, since it satisfies all constraints of the relaxed LP. (The converse is not necessarily true.) In a minimization problem, the optimum solution is less than or equal to all feasible solutions.

4.2 Dual LP

In section 2.2, we discussed how to derive the dual from the primal. We can apply the same procedure to our primal LP above. First, multiply each constraint by some variables α and β which will become our dual variables, corresponding to x, y respectively.

$$y_i(\beta_{ij}) \geq (\beta_{ij})x_{ij} \quad \forall i, j \tag{1}$$

$$(\alpha_j)\sum_{i \in F} x_{ij} \geq \alpha_j \quad \forall j. \tag{2}$$

Comparing the coefficients of y_i, we have:

$$\sum_j (\beta_{ij})y_i \leq f_i y_i \quad \forall i$$

$$\sum_j (\beta_{ij}) \leq f_i \quad \forall i.$$

Now we compare the coefficients of x_{ij}. First we can rearrange (1) and (2):

$$-(\beta_{ij})(x_{ij}) \geq -(\beta_{ij})(y_i), \text{and}$$

$$(\alpha_j)\sum_{i \in F} x_{ij} \geq \alpha_j.$$

Next, combine the left-hand side of both inequalities and compare their coefficients of x_{ij} with c_{ij}:

$$\alpha_j(\sum_{i \in F} x_{ij}) - \beta_{ij}x_{ij} \leq c_{ij}x_{ij}, \quad \forall i, j$$

$$\alpha_j x_{ij} - \beta_{ij}x_{ij} \leq c_{ij}x_{ij}$$

$$\alpha_j - \beta_{ij} \leq c_{ij}$$

Thus, the dual is as follows:

Maximize

$$\sum_j \alpha_j,$$

subject to the following constraints:

$$\sum_j \beta_{ij} \leq f_i \quad \forall i \tag{Dual 1}$$

$$\alpha_j - \beta_{ij} \leq c_{ij} \quad \forall i, j \tag{Dual 2}$$

$$\alpha_j, \beta_{ij} \geq 0 \quad \forall i, j.$$

5 Introduction to Approximation Algorithms

Approximation algorithms are a class of algorithms that find feasible solutions to "unsolvable" (NP-hard) optimization problems in a reasonable amount of time. (The technical term is **polynomial time**, which means the running time function can be upper bounded using the Big-O notation by a polynomial expression.)

To measure the effectiveness of an approximation algorithm, we will use a metric called **performance guarantee**. For example, in a minimization problem, an approximation algorithm that produces a feasible solution at most 4 times the optimum solution has an approximation guarantee of 4. As a shorthand, we will call this algorithm a **4-approximation algorithm**.

5.1 A 4-Approximation Algorithm for the Facility Location Problem

Now we have both the primal and dual. We will define some important terms:

- Let (x^*, y^*) and (α^*, β^*) be the optimum solution to the primal and the dual, respectively.
- Let Primal OPT, Dual OPT be the value of the objective function of the relaxed LP evaluated by (x^*, y^*) and (α^*, β^*), respectively.
- Note that by the Strong Duality Theorem, Primal OPT = Dual OPT.
- Let OPT be the optimum solution to the ILP. In other words, OPT is the desired (and correct) solution to the original problem, without LP-relaxation.
- Note that by Theorem 4.1.2, Primal OPT = Dual OPT $\leq$ OPT.

5.1.1 Definition

$N_j = \{i \in F | x^*_{ij} > 0\}$ represents a neighborhood of client j.

The antecedent $x^*_{ij} > 0$ means that client j gets fractional service from facility i in our optimum solution.

5.1.2 Lemma

If $x^*_{ij} > 0$, then $c_{ij} \leq \alpha^*_j$.

5.1.3 Proof

Primal CSC states that either $x^*_{ij} = 0$ or $\alpha^*_j - \beta^*_{ij} = c_{ij}$.
Since $x^*_{ij} > 0$, $\alpha^*_j - \beta^*_{ij} = c_{ij}$.
Since $\beta^*_{ij} \geq 0, c_{ij} \leq \alpha^*_j$.

5.1.4 The 4-Approximation Algorithm

Our algorithm can be broken down into the following steps:

Step 1: Solve the primal and dual LP optimally to yield $(x^*, y^*), (\alpha^*, \beta^*)$.

Step 2: Pick a client with the smallest α^* value.

Step 3: Choose the cheapest facility to open in its neighborhood and assign the client to the facility.

Step 4: Remove all clients whose neighborhoods overlap with the client chosen in step 3.

Step 5: Repeat steps 2 through 4 until there are no more clients left.

Step 6: All clients that were removed in step 4 are assigned to the "nearest" open facility.

5.2 Analysis: Bounding the Connection Costs $\sum c_{ij}$

The most interesting step in this algorithm is the removal of all clients whose neighborhoods overlap with the neighborhood of the chosen client.

In other words, for the client j with the smallest α^*, we remove all j' if $N_j \cap N_{j'} \neq \emptyset$.

We want to be able to bound the connection cost between the removed client j' and some facility i opened by client j, using an intermediate facility i' "shared" by both j and j'.

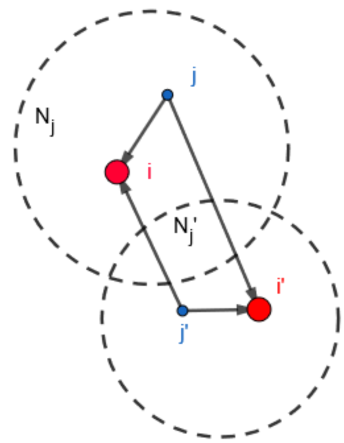

Figure 3: Overlapping neighborhoods

More formally, let j' be a client removed in step 4 of our algorithm.

Let j be the nearest open facility of j'. N_j must overlap with $N_{j'}$ because otherwise $N_{j'}$ will have no neighborhood "shared" with any open facility, but that would also mean that j' is not removed, which is a contradiction.

Let i be the facility opened when client j was chosen in step 2, and let i' be a facility such that $i' \in N_j \cap N_{j'}$.

Then we can write:

$$\begin{aligned} c_{ij'} &\leq c_{ij} + c_{i'j} + c_{i'j'} \text{ by the triangle inequality} \\ &\leq \alpha_j^* + \alpha_j^* + \alpha_{j'}^* \text{ by Lemma 5.1.2} \\ &\leq \alpha_{j'}^* + \alpha_{j'}^* + \alpha_{j'}^* \text{ by step 2 of our algorithm} \\ &\leq 3\alpha_{j'}^*. \end{aligned}$$

5.2.1 Theorem

The connection cost of removed clients can be upper bounded by $3\alpha_j^* = 3\text{Dual OPT}$.

Let's consider the connection cost of clients that were not removed, i.e., vertices that opened some facility in step 3. This is fairly straightforward since such a client is assigned to a facility that is in its neighborhood.

Therefore, we can apply Lemma 5.1.2 to upper bound the connection cost by α_j^*.

5.2.2 Theorem

The connection cost of non-removed clients can be upper bounded by $\alpha_j^* = \text{Dual OPT}$.

Combining Theorems 5.2.1 and 5.2.2, we can conclude that:

5.2.3 Theorem

The total connection cost can be upper bounded by 3 Dual OPT.

5.3 Analysis: Bounding the Facility Costs

5.3.1 Lemma

$$\sum_{i \in N_j} y_i^* \geq 1.$$

5.3.2 Proof

From Primal 2, we have

$$\sum_{i \in F} x_{ij}^* \geq 1.$$

We know that $N_j = \{i | x_{ij}^* > 0\}$, so $\forall i \in F \setminus N_j$,

$$x_{ij}^* = 0.$$

Thus,

$$\sum_{i \in N_j} x_{ij}^* = \sum_{i \in F} x_{ij}^* \geq 1.$$

From Primal 1, $y_i \geq x_{ij} \quad \forall i, j$, so

$$\sum_{i \in N_j} y_i^* \geq \sum_{i \in N_j} x_{ij}^* \geq 1.$$

Let $i_0(j)$ be the facility with the minimum opening cost in N_j, and let $f_{i_0(j)} = \min\ f_i, \forall i \in N_j$. Then, we have:

5.3.3 Lemma

$$f_{i_0(j)} \leq \sum_{i \in N_j} f_i y_i^*.$$

5.3.4 Proof

By Lemma 5.3.1,

$$\sum_{i \in N_j} y_i^* \geq 1.$$

Thus,

$$\begin{aligned}
f_{i_0(j)} &\leq f_{i_0(j)}(\sum_{i \in N_j} y_i^*) \\
&= \sum_{i \in N_j} f_{i_0(j)} y_i^* \\
&\leq \sum_{i \in N_j} f_i y_i^* \text{ since } f_{i_0(j)} \leq f_i, \forall i.
\end{aligned}$$

5.3.5 Bounding the Facility Cost

Let D be the set of open facilities. Now we can bound the total facility cost:

$$\begin{aligned}
\text{Total facility cost} &= \sum_{j \in D} f_{i_0(j)} \\
&\leq \sum_{j \in D} \sum_{i \in N_j} f_i y_i^* \text{ by Lemma 5.3.1} \\
&= \sum_{k \in F} f_k y_k^* \\
&\leq \sum_{k \in F} f_k y_k^* + \sum_{i \in F, j \in C} c_{ij} x_{ij} \\
&= \text{Primal OPT.}
\end{aligned}$$

5.3.6 Theorem

The total facility cost can be upper bounded by Primal OPT; combining Theorems 5.3.3 and 5.2.3, we have:

$$\begin{aligned}
\text{Total Cost} &= \text{Connection Cost} + \text{Facility Cost} \\
&\leq 3 \text{ Dual OPT} + \text{Primal OPT} \\
&\leq 4 \text{ Primal OPT} \\
&\leq 4 \text{ OPT by Theorem 4.1.2.}
\end{aligned}$$

6 Conclusion

Our algorithm yields a solution that is guaranteed to be at most 4 times the optimum solution, making it a "4-approximation algorithm." The gist of the algorithm is filtering (removing clients in step 4) and LP-rounding (opposite of LP-relaxation, represented by the assignment of removed clients to open facilities). The algorithm was created by Shmoys, Tardos, and Aardal in 1997. There are other algorithms developed recently that overperform the algorithm presented in this article, up to an approximation ratio of 1.52!

The facility location problem also has many variants. The variant presented in this article is called the "Uncapacitated Facility Location Problem," where each facility has no restrictions on how many clients it can serve. There is a capacitated version in which each facility has a capacity and each client has a demand. The problem can be extended to the k-center problem and more generally, cluster analysis. The k-center problem is the same as the facility location problem with one caveat: you want to minimize the maximum distance, not the total distance, between facilities and clients. Both the facility location problem and the k-center problem belong to a set of problems known as cluster analysis, in which one is asked to group objects such that objects in the same group are the most similar. In the case of the facility location problem, the groups are simply the set of all clients connected to a facility which are as close ("similar") to the facility as possible.

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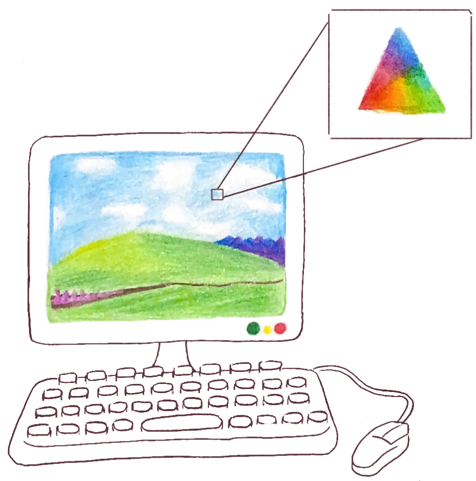
Barycentric Coordinates: Trading a Point of Origin for a Triangle

Aditya Pahuja



1 Introduction

Mathematics is, in many ways, a challenge of finding the right perspective from which to view a problem. In combinatorics, this may translate into counting objects in a certain order, while number theorists might interpret it as finding a useful factorization of an algebraic expression, for instance. Geometry is full of such examples. There are subtle instances that simply involve changing the way you look at the interactions between the parts of a diagram; one of my favorites, despite its simplicity, is translating the power of a point to the language of similar triangles.

Then, there are more drastic examples, such as assigning an ordered pair to each point in the plane, as in Cartesian coordinates or polar coordinates, or assigning a unique complex number to each point, as in the complex plane. Each coordinate system has its own specialties: Cartesian coordinates are well equipped for understanding configurations involving numerous lines and their intersections, while polar coordinates make natural the language of rotation around some central point. Complex numbers are particularly interesting, for they are a strange amalgam of the previous two systems.

In this article, we explore another one of these coordinate systems: barycentric coordinates, which are specially equipped to work with ratios between areas and lengths relative to a central triangle.

2 A Brief Primer of Barycentric Coordinates

Barycentric coordinates are centered around a *reference triangle*, which we will refer to as $\triangle ABC$ throughout this article. Essentially, they fix the coordinates of the vertices of a specific triangle, and build around them such that all other vertices are expressed in terms of this central triangle.

In 3D space, let $\triangle ABC$ be positioned such that no three of points O, A, B, and C are collinear, where O is the origin of the barycentric plane. Then, any vector P whose endpoint is in the plane of $\triangle ABC$ can be expressed as

$$\vec{P} = \alpha\vec{A} + \beta\vec{B} + \gamma\vec{C}$$

with $\alpha + \beta + \gamma = 1$. This ordered triple (α, β, γ) is defined as the coordinates of P.

2.1 Coordinates as Area Ratios

Geometrically, we can interpret the coordinates as area ratios, as shown in the following theorem.

2.1.1 Theorem

The barycentric coordinates of a point P are

$$\left(\frac{[PBC]}{[ABC]}, \frac{[PCA]}{[ABC]}, \frac{[PAB]}{[ABC]}\right),$$

where $[\lambda]$ denotes the signed area[1] of the geometric figure λ, and where we take P with ratio to the reference triangle ABC.

2.1.2 Proof

We can apply a linear transformation such that $\vec{A}$, $\vec{B}$, and $\vec{C}$ are the vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ respectively. Then, let P have barycentric coordinates (α, β, γ); $\vec{P}$ is by definition (α, β, γ). This means the determinant

$$\begin{vmatrix} \alpha & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \alpha$$

gives the ratio of the volumes of tetrahedra $OPBC$ and $OABC$, where O is the origin. Since the bases of the tetrahedra are coplanar, the tetrahedra's heights are equivalent, so this ratio is equivalent to the ratio of the signed areas of $\triangle PBC$ and $\triangle ABC$, as claimed by the statement of the theorem. An analogous argument tells us that β and γ are $\frac{[PCA]}{[ABC]}$ and $\frac{[PAB]}{[ABC]}$ respectively.

2.1.3 Implications

As a result of the previous theorem, the following characteristics of barycentric coordinates are made apparent.

1. The coordinates of P are unique; that is, if two points P_1 and P_2 both have the same coordinates, then they are the same point.

[1] Taking ABC to always have positive area, if λ is oriented the same way as ABC, then its signed area is its absolute area (i.e., it is positive); otherwise, the signed area is the negative of its absolute area.

2. P lies inside $\triangle ABC$ if and only if α, β, and γ are all positive.

3. P lies on an edge of $\triangle ABC$ if and only if at least one of the coordinates is 0.

4. P is outside $\triangle ABC$ if and only if at least one of α, β, γ is negative.

5. A, B, and C have coordinates $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, respectively.

While the coordinates in their current form are useful in their own right, it is sometimes helpful to entirely get rid of the constraint $\alpha + \beta + \gamma = 1$: given an arbitrary triple of numbers (α, β, γ) with $\alpha + \beta + \gamma \neq 0$, we can divide each number by $\alpha + \beta + \gamma$ to get a triple of numbers whose sum is one. In particular, this means we can simply consider the ratio of $\alpha : \beta : \gamma$ to be the point P. When the coordinates are written this way, we use the notation $(\alpha : \beta : \gamma)$ to indicate that we are only looking at the ratio of the coordinates. Coordinates of this form are called *dehomogenized coordinates*.

An application of these is shown in the following theorem:

2.1.4 Theorem

Let D be a point on side BC of $\triangle ABC$ such that $BD : DC = d : 1 - d$.[2] Then, the set of points on line AD (excluding A) are of the form $(t : 1 - d : d)$, where t is a real number.

2.1.5 Proof

Let P be a point on line AD with $P \neq A$. Then, since $BD : DC = d : 1 - d$, the ratio of $[PCA] : [PAB]$ is $1 - d : d$. The area of $\triangle PBC$ is entirely arbitrary here, so the coordinates of P are $(t : 1 - d : d)$ where t can take on any real value. This fails to include the coordinates of A because $[PCA]$ and $[PAB]$ have zero area when $P = A$, so $t : 1 - d : d = 1 : 0 : 0$, which is possible only if $1 - d = d = 0$, but this is itself impossible.

In particular, dehomogenized coordinates can help us determine the characteristics of collinearity, as seen above.

2.1.6 Exercise

Prove Ceva's theorem; namely, given points D, E, F on sides BC, CA, AB of $\triangle ABC$, prove that AD, BE, and CF concur if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1,$$

where lengths are directed.

2.1.7 Exercise

What are the coordinates of the incenter, centroid, circumcenter, and orthocenter in terms of the side lengths $a = BC$, $b = CA$, and $c = AB$ of ABC?

[2]Here, lengths are directed, meaning that given collinear points A, B, C, $AB + BC = AC$ is true regardless of the permutation of the points. Specifically, lengths are negated as necessary to satisfy this relation.

2.2 Properties of Lines

Let $\triangle PQR$ be a triangle that lies in the plane of $\triangle ABC$, with P, Q, R having barycentric coordinates (p_1, p_2, p_3), (q_1, q_2, q_3), (r_1, r_2, r_3). Then, we can determine ratio of the signed areas of $\triangle PQR$ and $\triangle ABC$ with the determinant

$$\begin{vmatrix} p_1 & p_2 & p_3 \\ q_1 & q_2 & q_3 \\ r_1 & r_2 & r_3 \end{vmatrix}.$$

P, Q, and R are collinear if and only if the area of $\triangle PQR$ is 0, which means they are collinear if and only if the aforementioned determinant is 0. This allows us to obtain a general equation of a line.

2.2.1 Theorem

The equation of a line in barycentric coordinates is of the form

$$ux + vy + wz = 0,$$

where (x, y, z) is some point on the line and u, v, w are some constants.

2.2.2 Proof

Let $S = (s_1 : s_2 : s_3)$ and $T = (t_1 : t_2 : t_3)$ be arbitrary points in the plane of $\triangle ABC$. Then, any point $P = (x, y, z)$ on line ST satisfies

$$\begin{vmatrix} x & y & z \\ s_1 & s_2 & s_3 \\ t_1 & t_2 & t_3 \end{vmatrix} = 0.$$

Evaluating this determinant by minors yields

$$(s_2t_3 - t_2s_3)x + (s_3t_1 - t_3s_1)y + (s_1t_2 - t_1s_2)z = 0.$$

Then, define the constants u, v, and w to be the coefficients of x, y, and z respectively, which tells us that the set of points on a line is defined by the equation

$$ux + vy + wz = 0$$

as desired.

2.2.3 Implications

The power of this relatively simple form of a line is that it is easy to find where lines intersect: the intersection points come from solutions to the system

$$\begin{aligned} u_1x + v_1y + w_1z &= 0 \\ u_2x + v_2y + w_2z &= 0 \\ x + y + z &= 1. \end{aligned}$$

If the system has no solutions, then the lines must of course be parallel. However, there is something more subtle here; since the system has no solutions, we have

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ 1 & 1 & 1 \end{vmatrix} = 0.$$

But this implies that

$$\begin{aligned} u_1x + v_1y + w_1z &= 0 \\ u_2x + v_2y + w_2z &= 0 \\ x + y + z &= 0 \end{aligned}$$

has a nontrivial solution!

This apparently nonsensical triple of coordinates allows for the introduction of the concept of *points at infinity.* The point at infinity of any line is the nontrivial triple of coordinates satisfying the equation of the line whose coordinates sum to zero; this now means that every pair of lines intersects at a unique point, including parallel lines. This makes the following theorem more powerful, as we now have a way to interpret coordinates whose sum is zero.

2.2.4 Theorem

Three lines $u_ix + v_iy + w_iz = 0$ for $i = 1, 2, 3$ are either concurrent or all parallel if and only if

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} = 0.$$

2.2.5 Proof

To find the intersection point of the three lines, we have a system of equations in x, y, z:

$$\begin{aligned} u_1x + v_1y + w_1z &= 0 \\ u_2x + v_2y + w_2z &= 0 \\ u_3x + v_3y + w_3z &= 0 \end{aligned}$$

Obviously, the system has the trivial solution $(x, y, z) = (0, 0, 0)$, but it has nontrivial solutions if and only if the matrix representing this system of equations has determinant zero; that is, the existence of a common intersection point on the three lines is equivalent to

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} = 0.$$

Now, supposing the determinant is indeed zero, solutions to this system must satisfy either $x+y+z = 0$ or $x + y + z \neq 0$. If $x + y + z \neq 0$, then the lines intersect at a point $(x : y : z)$ in the plane, meaning that they concur. On the other hand, if $x + y + z = 0$, the lines intersect at a point at infinity, so they are all parallel.

3 Applications of Barycentric Coordinates

Now that we are familiar with the most essential properties of barycentric coordinates, it helps to look at some applications to see how they relate to various areas of math.

3.1 Olympiad Geometry

One such application is their use in math olympiads. To demonstrate this, let's look at a problem from the 2016 Indian National Mathematical Olympiad:

3.1.1 Problem

(INMO 2016/1) Let ABC be a triangle in which $AB = AC$. Suppose the orthocenter of the triangle lies on the incircle. Find the ratio AB/BC.

3.1.2 Solution

First, what suggests that this problem can be solved with barycentric coordinates? The fact that we're asked to find a ratio of side lengths of the triangle is a good indicator, since the triangle's side lengths appear in the coordinates of both the incenter and the orthocenter.

However, what about the circle? We don't know how to deal with circles! Fortunately, the fact that $\triangle ABC$ is isosceles implies that the incenter I lies on the altitude from A to BC, and the point where the incircle touches BC is the foot of that altitude; we will call that point M. Thus, we can deduce that the foot of the altitude from A is diametrically opposite the orthocenter, which we will label H. In other words, I is the midpoint of HM; this relation allows us to entirely ignore the circle for the rest of the problem!

Additionally, some useful notation before continuing: define

$$S_A = \frac{b^2 + c^2 - a^2}{2} = ab\cos C,$$

where a, b, c are the lengths of the sides of $\triangle ABC$ opposite A, B, C respectively, and define S_B and S_C analogously. This relation allows us to simplify otherwise messy expressions, such as that of the orthocenter, which we can write as

$$H = (S_B S_C : S_C S_A : S_A S_B).$$

Now, continuing the problem: since we are only interested in the ratio of $AB : BC$, let $BC = 1$ and $AB = AC = a$, so that a is the ratio we want to find. I has coordinates $(1 : a : a)$ and M, being the midpoint of BC, has coordinates $(0 : 1 : 1)$. We also get that

$$S_A = a^2 - \frac{1}{2} \text{ and } S_B = S_C = \frac{1}{2},$$

so the coordinates of H are

$$(1 : 2a^2 - 1 : 2a^2 - 1).$$

However, we can also write the coordinates of H using the incenter. Since IM is half of HM, the area of $\triangle BHC$ is twice that of $\triangle IHC$. Hence,

$$\frac{[HBC]}{[ABC]} = \frac{2[IBC]}{[ABC]} = \frac{2}{2a+1}.$$

By the symmetry in the isosceles triangle,

$$[HCA] = [HAB],$$

which implies that

$$\frac{[HAB] + [HBC] + [HCA]}{[ABC]} = \frac{2}{2a+1} + \frac{2[HCA]}{[ABC]} = 1.$$

Therefore,

$$\frac{[HCA]}{[ABC]} = \frac{[HAB]}{[ABC]} = a - \frac{1}{2},$$

meaning $H = (4 : 2a - 1 : 2a - 1) = (1 : 2a^2 - 1 : 2a^2 - 1)$. Since these triples of coordinates designate the same point,

$$\frac{4}{2a-1} = \frac{1}{2a^2 - 1}.$$

This yields the quadratic equation $8a^2 - 2a - 3 = 0$, whose only positive solution is $a = \boxed{\frac{3}{4}}$.

In general, olympiad geometry problems that involve lots of lines and ratios of areas or segments are likely to be solvable with barycentric coordinates, while problems involving circles are typically tough to work with.

3.2 Computer Graphics

Instead of defining barycentric coordinates on a plane in three-dimensional space, it is possible to define them in two dimensions. This makes it easier to convert between barycentric coordinates and two-dimensional systems, which computers can more readily interpret due to their screens being rectangles. Each point on the computer screen can then be assigned barycentric coordinates relative to some triangle. The coordinates of points inside the triangle (which are nicely filtered out by the signs of the coordinates) can be converted to a value between 0 and 255, so that each point is a triple of numbers from 0 to 255, which gives it an RGB color. This result in a smooth gradient between red, green, and blue.

There have also been attempts to generalize barycentric coordinates to polygons with more vertices, some of which have been used for color interpolation, image warping, and shape deformation. However, a closed-form construction of barycentric coordinates has still not been found.

4 Connections to Other Coordinate Systems

Barycentric coordinates are closely related to trilinear coordinates, which define points by signed distance to the edges of the triangle. For some point P and $\triangle ABC$, the area of $\triangle PAB$ is half the distance from P to AB times the length of AB, so a triple of trilinear coordinates $(x : y : z)$ is equivalent to $(ax : by : cz)$ in barycentric coordinates.

Using our three-dimensional formulation of barycentric coordinates, some linear algebra can be used to convert barycentrics to regular three-dimensional coordinates. The two-dimensional definition that was previously mentioned can similarly be used to convert them to Cartesian coordinates as well.

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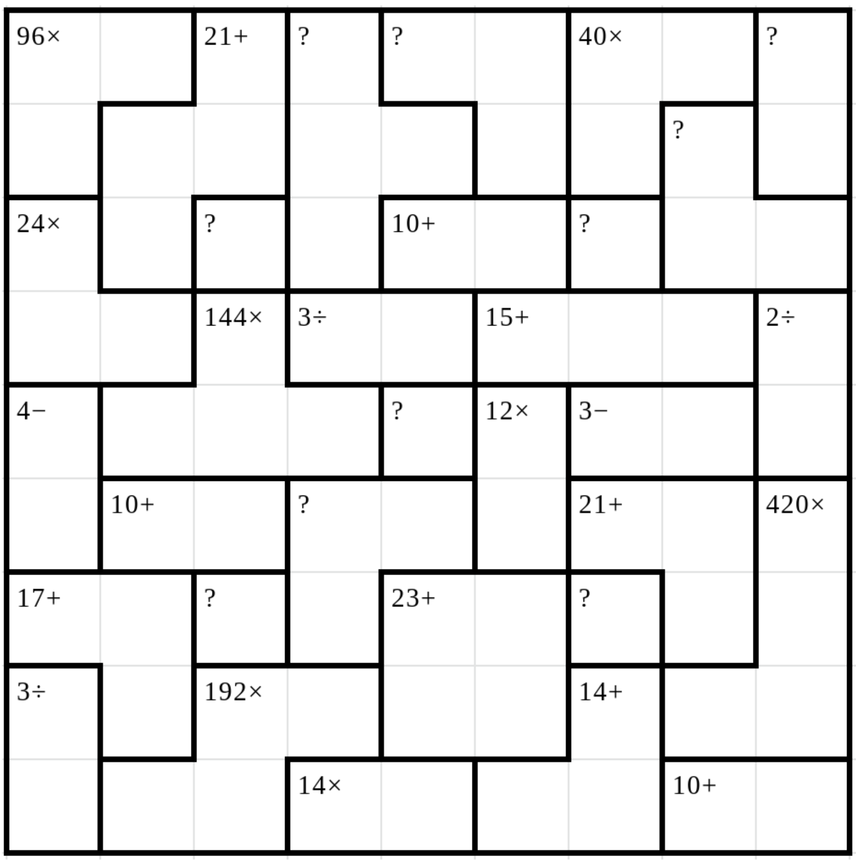
Mayo, Please Stop Making a Mess

Edward Wu

You just finished making your favorite sandwich and headed over to your printer to pick up your freshly printed KenKen. You lay your trusty mechanical pencil (because they're better than pens) and eraser down on the table. You pick up your sandwich and take a bite, when *Splat!* huge drops of mayo fling themselves all over your puzzle. After you wipe off the mayo globs, you realize that some of the clues are still completely covered. Normally, you would be freaking out, but your experiences with the last three messed up KenKens has made you a KenKen master, and you think you can still do it. Also, since you have been a dedicated environmentalist for many years, you don't really see a reason to print another because that's a waste of paper and ink. See if you can finish it! (P.S. You can.)



Solution to Mayo, Please Stop Making a Mess

Edward Wu

2	6	3	9	8	7	4	5	1
8	9	4	6	5	1	2	7	3
6	5	7	3	1	9	8	4	2
4	1	9	2	6	5	7	3	8
5	2	1	8	7	3	6	9	4
1	8	2	5	3	4	9	6	7
7	3	5	4	9	2	1	8	6
9	7	6	1	4	8	3	2	5
3	4	8	7	2	6	5	1	9

Chromatic Contention

Aditya Pahuja

In Chromatopia, there are two species of chromatopes, *chromatopods* and *chromataceans*, each of which are one of n possible colors. As an annual ritual, the Council of Colors arranges chromatopes in a line. However, because chromatopods and chromataceans are sworn enemies, chromatopes of the same color but different species are hostile towards each other and refuse to stand next to one another. In particular, if a chromatopod is next to at least one identically-colored chromatacean, the chromatopod and one of the adjacent chromataceans both leave the line, and vice versa. As per the sacred ritual, the Council has arranged them according to the following rules:

1. There are no chromatopods adjacent to identically-colored chromataceans.

2. If all chromatopes of a color are removed from the line, then all the other creatures will eventually leave the line, as the absence of other creatures causes them to be adjacent to a hostile creature.

For example, when $n = 2$, a valid line is a_1 a_2 b_1 b_2, since removing either a_1 and b_1 or a_2 and b_2 will result in the remaining two being of the same color but different species, causing them to leave the line, resulting in an empty line.

What does such a line look like for $n = 3$? What about $n = 4$? How can this be generalized to arbitrary values of n?

Solution to Chromatic Contention

Aditya Pahuja

What does such a line look like for $n = 3$ and $n = 4$?

Let a_k and b_k denote chromatopods and chromataceans, respectively, such that a chromatopod a_i and chromatacean a_j are of the same color if and only if $i = j$.

When $n = 3$, a valid line is

$$a_1\ a_2\ b_1\ b_2\ a_3\ a_2\ a_1\ b_2\ b_1\ b_3$$

To show this works, consider what happens when all chromatopes with color 3 are removed (that is, remove a_3 and b_3). The resulting line is

$$a_1\ a_2\ b_1\ b_2\ a_2\ a_1\ b_2\ b_1,$$

but then b_2 and a_2 refuse to stand next to each other, so they leave the line, resulting in

$$a_1\ a_2\ b_1\ a_1\ b_2\ b_1,$$

and so on, eventually reducing the line to nothing; a similar process can be applied to colors 1 and 2 to show that they work as well.

An example of a line that obeys the rules of the ritual for $n = 4$ is

$$a_1\ a_2\ b_1\ b_2\ a_3\ a_2\ a_1\ b_2\ b_1\ b_3\ a_4\ a_1\ a_2\ b_1\ b_2\ a_3\ a_2\ a_1\ b_2\ b_1\ b_3\ b_4.$$

What about arbitrary values of n?

Let S_i be a sequence satisfying the rules for $n = i$. Then, the sequence

$$S_i\ a_{i+1}\ S_i'\ b_{i+1}$$

also abides by the ritual's constraints, where S_i' is the sequence obtained by reversing S_i, and replacing all the a_k with b_k and vice versa. For example, a possible line S_2 is $a_1\ a_2\ b_1\ b_2$, in which case S_2' would be

$$a_2\ a_1\ b_2\ b_1.$$

This is very reminiscent of commutators, as a_k and b_k are effectively inverses in this scenario. To see why

$$S_i\ a_{i+1}\ S_i'\ b_{i+1}$$

is a valid line, consider what happens if all creatures with a color corresponding to a value less than or equal to i. S_i entirely disappears by definition, and so does S_i' since chromatopods and chromataceans are interchangeable. This leaves behind $a_{i+1}\ b_{i+1}$ which goes to nothing, as desired. Alternatively, if the creatures of color $i + 1$ are removed, then S_i and S_i' are inverse sequences, so they cancel each other out.

It is worth noting that these are not the shortest forms of valid lines, but valid lines in general do take the form of commutators. For example, a shortened variation of the $n = 4$ case is

$$a_1\ a_2\ b_1\ b_2\ a_3\ a_4\ b_3\ b_4\ a_2\ a_1\ b_2\ b_1\ a_4\ a_3\ b_4\ b_3,$$

which is actually a commutator of two commutators!

XOR-bitant!

Aditya Pahuja

The *exclusive or* (XOR) function is a binary operation which, when given two true or false values, outputs "true" if and only if the two values are distinct. Replacing "false" and "true" with 0 and 1 respectively, the XOR operation, represented by the operator $\oplus$, can be defined as

$$a \oplus b = a + b \pmod 2$$

where a and b take on some value from $\{0, 1\}$.

We can also extend this in a way that encompasses inputs beyond just zeros and ones to get the *bitwise XOR* operation ($\overline{\oplus}$): Given two binary strings, for each pair of corresponding digits, use the XOR operator on the pair of numbers to get the respective digit in the final result, using leading zeros as necessary. For example,

$$\begin{aligned} &101001_2 \\ \overline{\oplus}\ &010111_2 \\ =\ &111110_2. \end{aligned}$$

In the spirit of XOR, let's describe a binary operation $\oplus^m$, which acts on numbers a and b between 0 and $m-1$:

$$a \oplus^m b = a + b \pmod m$$

This operation can be extended to arbitrary inputs base m in a similar way as with XOR; take corresponding digits a and b, and use $a \oplus^m b$ as the digit in the resulting base-m number. In a similar fashion to the bitwise XOR operator, let's call this operation $\overline{\oplus}^m$. To demonstrate this operation,

$$\begin{aligned} &64523 \\ \overline{\oplus}^9\ &01422 \\ =\ &65045. \end{aligned}$$

1. *Show that there is a unique number e base m such that*

$$a \overline{\oplus}^m e = e \overline{\oplus}^m a = a.$$

2. *Is there always a number a' such that $a \overline{\oplus}^m a' = e$?*

3. One day, a mathematician accidentally writes $\oplus^m$ as $\otimes^m$. Then, her friend, reading her work, sees $a \otimes^m b$ and interprets it to mean

$$a \times b \pmod m$$

when a and b are between 0 and $m-1$, with the analogous extension to arbitrary base m numbers denoted by $\overline{\otimes}^m$.

Does $\overline{\otimes}^m$ satisfy the same properties as $\overline{\oplus}^m$? If not, are there any restrictions we can impose on a, b, and/or m for which these properties are satisfied?

4. *Do $\overline{\oplus}^m$ and $\overline{\otimes}^m$ satisfy the distributive property? That is, does*

$$a \overline{\otimes}^m (b \overline{\oplus}^m c) = (a \overline{\otimes}^m b) \overline{\oplus}^m (a \overline{\otimes}^m c)?$$

Solution to XOR-bitant!

Aditya Pahuja

1. *Show that there is a unique number e base m such that*

$$a\overline{\oplus}^m e = e\overline{\oplus}^m a = a.$$

First, we see that $e = 0$ clearly satisfies this property. To show that this value is unique, suppose that some nonzero number f also satisfies this property. Then, there is some digit k whose value is between 1 and $m-1$. Let the corresponding digit of a be ℓ; then, $k \oplus \ell \not\equiv \ell$, which means the result of $a\overline{\oplus}^m f$ differs from a in at least one digit, so there is no such f.

2. *Is there always a number a' such that $a\overline{\oplus}^m a' = e$?*

Given some number a in base m, we can determine a value of a' satisfying the above equation as follows: The nth digit of a' is the additive inverse modulo m of the nth digit of a. For example, if $a = 239871_{10}$, then the value of a' satisfying

$$a\overline{\oplus}^{10} a' = e$$

would be 871239_{10}. This value of a' has the property mentioned in the problem because the corresponding digits must combine to 0 after applying $\oplus^m$, so a' exists for every a.

3. *Does $\overline{\otimes}^m$ satisfy the same properties as $\overline{\oplus}^m$? If not, are there any restrictions we can impose on a, b, and/or m for which these properties are satisfied?*

No, this operation doesn't have the two properties mentioned above. This is because there can be several values of e for which

$$a\overline{\otimes}^m e = a.$$

For example, let $a = 1234_7$. Then,

$$1234_7\overline{\otimes}^7 1111_7 = 1234_7\overline{\otimes}^7 11111_7 = 1234,$$

so both 1111_7 and 11111_7 have this property. In fact, any string of 1s with at least as many digits as a will return a upon combination by $\overline{\otimes}^m$, since the excess leading 1s will just get cancelled out by the leading zeros in a.

If we take e to be the string of 1s with the same number of digits as a, however, then there will be an a' for which $a\overline{\oplus}^m a' = e$ when $gcd(f_n(a), m) = 1$, where $f_n(k)$ is the nth digit of k and n is any positive integer less than or equal to the number of digits in a. This is because, if $a\overline{\oplus}^m a' = e$, then $f_n(a) \cdot f_n(a') \pmod{m} = 1$, which means $f_n(a')$ is the modular inverse of $f_n(a)$, which exists if and only if $gcd(f_n(a), m) = 1$.

4. *Do $\overline{\oplus}^m$ and $\overline{\otimes}^m$ satisfy the distributive property? That is, does*

$$a\overline{\otimes}^m (b\overline{\oplus}^m c) = (a\overline{\otimes}^m b)\overline{\oplus}^m (a\overline{\otimes}^m c)?$$

The answer is yes. Let $f_n(k)$ be the nth digit of k. Then,

$$\begin{aligned}
f_n(a \overline{\otimes}^m (b \overline{\oplus}^m c)) &= f_n(a) \cdot f_n(b \overline{\oplus}^m c) \pmod{m} \\
&= f_n(a) \cdot (f_n(b) + f_n(c)) \pmod{m} \\
&= (f_n(a) \cdot f_n(b)) + (f_n(a) \cdot f_n(c)) \pmod{m} \\
&= f_n((a \overline{\otimes}^m b) \overline{\oplus}^m (a \overline{\otimes}^m c)),
\end{aligned}$$

which means the nth digit of $a \overline{\otimes}^m (b \overline{\oplus}^m c)$ is equivalent to the nth digit of $(a \overline{\otimes}^m b) \overline{\oplus}^m (a \overline{\otimes}^m c)$ regardless of the value of n. Hence, the two numbers are equivalent, which implies that $\overline{\otimes}^m$ is distributive over $\overline{\oplus}^m$.

Pun-ny Math

Amanda Louie

1 Introduction

Here is the key that you will need to use to solve the following problem set. You will solve the given problems in order to answer the questions (that are in words). You will need to rearrange the letters (keep in mind that some letters will repeat) you get while solving the individual problems in order to arrive at your final answer.

A = 0	F = 5	K = 10	P = 15	U = 20	Z = 25
B = 1	G = 6	L = 11	Q = 16	V = 21	Note: All the problems are given in ascending order of the resulting numbers
C = 2	H = 7	M = 12	R = 17	W = 22	
D = 3	I = 8	N = 13	S = 18	X = 23	
E = 4	J = 9	O = 14	T = 19	Y = 24	

2 Problems

1. Question: What do you get when you multiply a famous place in New York City by itself?
   - Answer: - - - - -   - - - - -

     (a) (Part i's answer) - [(Part b's answer) + (Part e's answer)] =
     (b) (Letter used twice): The only number whose principal square root is half the number itself =
     (c) $2^4 - 2^3 =$
     (d) (f's answer)$-5 =$
     (e) $(5! - 4!) \div 6 =$
     (f) $\sqrt{300 - 11} =$
     (g) (Letter used twice): $6 + 324 \div 6 - 42 =$
     (h) Find the positive difference between 2116 and the year this year's entering freshman class (2021-2022) will graduate high school. Now flip the resulting number's digits (for example, the number 48 would result in 84) =
     (i) $(2021 \div 47) - (2020 \div 202) - (2019 \div 673) - (2018 \div 1009)$
     $- (2017 \div 2017) - (2016 \div 288) =$

2. Question: What do you call brothers who love math?
   - Answer: - - - - - - - -

     (a) (g's answer)-[(d's answer)+(e's answer)] =
     (b) $|1^2 - 2^2| \div 3 =$
     (c) $(6 - 5) \times 4 =$
     (d) $7 + 8 - 9 =$
     (e) $10 - 11 + 12 =$

(f) $13 + |14 - 15| =$
(g) $16 - 17 + 18 =$
(h) $19 + 20 - 21 =$

3. Question: What did the triangle say to the circle?

   - Answer: _ _ _'_ _ _ _ _ _ _ _ _ _ _

   (a) (Letter used twice): (f's answer)-(c's answer) =
   (b) $1 + 2^2 + 3 =$
   (c) First Digit $= |4 - 5|$, Second Digit $= |6 - 7|$
   (d) $8 - 9 + 10 - 11 + 12 - 13 + 14 - 15 + 16 - 17 + 18 =$
   (e) (Letter used twice): $18 - ||19^2 - 20^2| - |21^2 - 22^2|| =$
   (f) $|23 - 24| \times (25 + 26)$, flip the resulting number's digits =
   (g) $(2930 - 2728 - 32) \div 10 =$
   (h) (Letter used twice): $31 + 33 - 35 - 11 =$
   (i) $38 \div (36 - 34) =$
   (j) $40 \div (44 - 42) =$
   (k) $|42 - 46|! =$

4. Question: What do you call a political party in favor of agriculture?

   - Answer: _ _ _-_ _ _ _ _ _ _ _

   (a) (g's answer)-[(b's answer)+(e's answer)] =
   (b) This number is the only even prime number =
   (c) (Letter used twice): This number has a prime factorization of (b's answer)(7) =
   (d) This number is the smallest two digit positive multiple of 5 that has only odd factors =
   (e) (Letter used twice): This is the second largest prime under 20 =
   (f) This number has a prime factorization of (b's answer)(e's answer-c's answer)^2 =
   (g) (Letter used twice) This is the largest prime under 20 =

5. Question: What do you call a male mathematician who spent all summer on the beach?

   - Answer: _ _ _ _ _ _ _ _

   (a) (Letter used twice): (e's answer)-[(c's answer)+(d's answer)] =
   (b) The function of $p \heartsuit q$ is defined as $2p + 3q$. If $x \heartsuit 36 = 188$, find $x - 36 =$
   (c) The function of h❖k is defined as $h^2 - k^2$. If x❖$21 = 184$, find $|x| - 19 =$
   (d) (Letter used twice): The function of t☆u is defined as $-3(t + u)$. If 5☆$x = 9$, find $5 - x =$
   (e) (Letter used twice): The function of $y \sqcap z$ is defined as $2(y - 2)^2$. If $x \sqcap 20 = 722$ and $y < z$, find $20 - x =$

Solution to Pun-ny Math

Amanda Louie

1 Key/Code

A = 0	F = 5	K = 10	P = 15	U = 20	Z = 25
B = 1	G = 6	L = 11	Q = 16	V = 21	Note: All the problems are given in ascending order of the resulting numbers
C = 2	H = 7	M = 12	R = 17	W = 22	
D = 3	I = 8	N = 13	S = 18	X = 23	
E = 4	J = 9	O = 14	T = 19	Y = 24	

Although this solution outlines one of the ways you could have solved these problems, there are many other paths you could have taken to arrive at the same answer.

2 Answers

1. Question: What do you get when you multiply a famous place in New York City by itself?
   - Answer: **TIMES SQUARE**
     (a) (Part i's answer) - [(Part b's answer) + (Part e's answer)] $= 20 - (4 + 16) = \mathbf{0}$
     (b) (Letter used twice): The only number whose principal square root is half the number itself $= \mathbf{4}$
     (c) $2^4 - 2^3 = 16 - 8 = \mathbf{8}$
     (d) (f's answer)$-5 = 17 - 5 = \mathbf{12}$
     (e) $(5! - 4!) \div 6 = [(5 \times 4 \times 3 \times 2 \times 1) - (4 \times 3 \times 2 \times 1)] \div 6 = (120 - 16) \div 6 = 96 \div 6 = \mathbf{16}$
     (f) $\sqrt{300 - 11} = \sqrt{289} = \mathbf{17}$
     (g) (Letter used twice): $6 + 324 \div 6 - 42 = 6 + 54 - 42 = 60 - 42 = \mathbf{18}$
     (h) Find the positive difference between 2116 and the year this year's entering freshman class (2021-2022) will graduate high school. Now flip the resulting number's digits (for example, the number 48 would result in 84) $= 2116 - 2025 = 91$ which becomes **19**
     (i) $(2021 \div 47) - (2020 \div 202) - (2019 \div 673) - (2018 \div 1009)$
     $- (2017 \div 2017) - (2016 \div 288) = 43 - 10 - 3 - 2 - 1 - 7 = 43 - 23 = \mathbf{20}$
   - Letters: A, E, E, I, M, Q, R, S, S, T, U. Unscramble and you should get **TIMES SQUARE**

2. Question: What do you call brothers that love math?
   - Answer: **ALGEBROS**
     (a) (g's answer)-[(d's answer)+(e's answer)] $= 17 - (6 + 11) = \mathbf{0}$
     (b) $|1^2 - 2^2| \div 3 = |1 - 4| \div 3 = 3 \div 3 = \mathbf{1}$
     (c) $(6 - 5) \times 4 = 1 \times 4 = \mathbf{4}$
     (d) $7 + 8 - 9 = 15 - 9 = \mathbf{6}$
     (e) $10 - 11 + 12 = 12 - 1 = \mathbf{11}$
     (f) $13 + |14 - 15| = 13 + 1 = \mathbf{14}$

   (g) $16 - 17 + 18 = 18 - 1 = \mathbf{17}$
   (h) $19 + 20 - 21 = 39 - 21 = \mathbf{18}$

- Letters: A, B, E, G, L, O, R, S. Unscramble and you should get **ALGEBROS**

3. Question: What did the triangle say to the circle?

- Answer: **YOU'RE POINTLESS**
   (a) (Letter used twice): (f's answer)-(c's answer) $= 20 - 18 + (13 - 11) = 2 - 2 = \mathbf{0}$
   (b) $1 + 2^2 + 3 = 1 + 4 + 3 = \mathbf{8}$
   (c) First Digit $= |4 - 5|$, Second Digit $= |6 - 7|$, **11**
   (d) $8 - 9 + 10 - 11 + 12 - 13 + 14 - 15 + 16 - 17 + 18 = 8 + 1 + 1 + 1 + 1 + 1 = \mathbf{13}$
   (e) (Letter used twice): $18 - ||19^2 - 20^2| - |21^2 - 22^2|| = 18 - |361 - 400| - |441 - 484|| = 18 - |39 - 43| = 18 - 4 = \mathbf{14}$
   (f) $|23 - 24| \times (25 + 26)$, flip the resulting number's digits $= 1 \times 51 = 51$ which becomes **15**
   (g) $(2930 - 2728 - 32) \div 10 = (202 - 32) \div 10 = 170 \div 10 = \mathbf{17}$
   (h) (Letter used twice): $31 + 33 - 35 - 11 = 64 - 35 - 11 = 29 - 11 = \mathbf{18}$
   (i) $38 \div (36 - 34) = 38 \div 2 = \mathbf{19}$
   (j) $40 \div (44 - 42) = 40 \div 2 = \mathbf{20}$
   (k) $|42 - 46|! = 4! = 4 \times 3 \times 2 \times 1 = \mathbf{24}$

- Letters: Letters: E, E, I, L, N, O, O, P, R, S, S, T, U, Y. Unscramble and you should get **YOU'RE POINTLESS**

4. Question: What do you call a political party in favor of agriculture?

- Answer: **PRO-TRACTORS**
   (a) (g's answer)-[(b's answer)+(e's answer)] $= 19 - (2 + 17) = \mathbf{0}$
   (b) This number is the only even prime number = **2**
   (c) (Letter used twice): This number has a prime factorization of (b's answer)(7) $= 2 \times 7 = \mathbf{14}$
   (d) This number is the smallest two digit positive multiple of 5 that has only odd factors = **15**
      – 15: 1, 3, 5, 15 (all numbers are odd)
   (e) (Letter used twice): This is the second largest prime under 20 = **17**
   (f) This number has a prime factorization of (b's answer)(e's answer-c's answer)$^2 = 2 \times (17 - 14)^2 = 2 \times 3^2 = 2 \times 9 = \mathbf{18}$
   (g) (Letter used twice) This is the largest prime under 20 = **19**

- Letters: A, C, O, O, P, R, R, S, T, T. Unscramble and you should get **PRO-TRACTORS**

5. Question: What do you call a male mathematician who spent all summer on the beach?

- Answer: **A TAN GENT**
   (a) (Letter used twice): (e's answer)-[(c's answer)+(d's answer)] $= 19 - (6 + 13) = \mathbf{0}$
   (b) The function of p$\heartsuit$q is defined as 2p + 3q. If x$\heartsuit$36 = 188, find x-36 $= 40 - 36 = \mathbf{4}$
   (c) The function of h❖k is defined as $h^2 - k^2$. If x❖21 = 184, find $|x| - 19 = 25 - 19 = \mathbf{6}$
   (d) (Letter used twice): The function of t☆u is defined as $-3(t + u)$. If 5☆x = 9, find 5 - x = $5 - (-8) = \mathbf{13}$
   (e) (Letter used twice): The function of y$\sqcap$z is defined as $2(z - y)^2$. If x$\sqcap$20 = 722 and $y < z$, find $20 - x = 20 - 1 = \mathbf{19}$

- Letters: A, A , E, G, N, N, T, T. Unscramble and you should get **A TAN GENT**

KenKen (12 × 12)

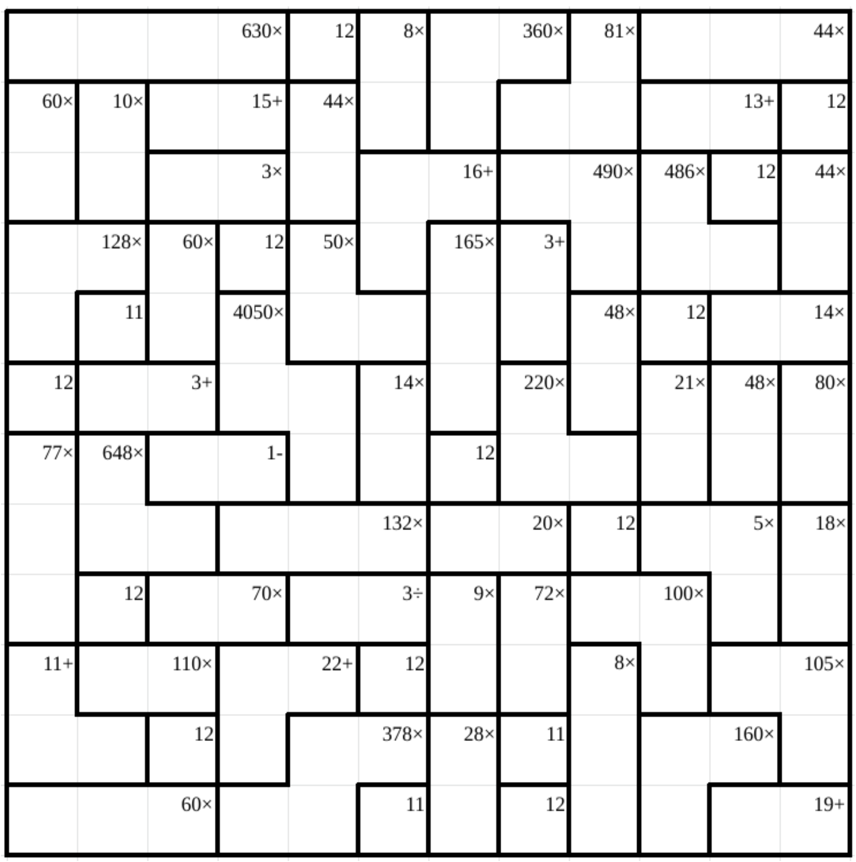
Amanda Louie

Preface

Standard KenKen rules apply:

1. All rows and columns must have exactly one number (from 1 to 12) in each respective row/column. No repeats!

2. The numbers in each square must satisfy the enclosed area they are put in (i.e., if a 5 and 6 are put in an area that reads 30×, this means these two numbers will multiply to 30. You cannot, for example, have the numbers 4 and 6 in this enclosed area because they do not multiply to 30.)



Solution to KenKen (12 × 12)

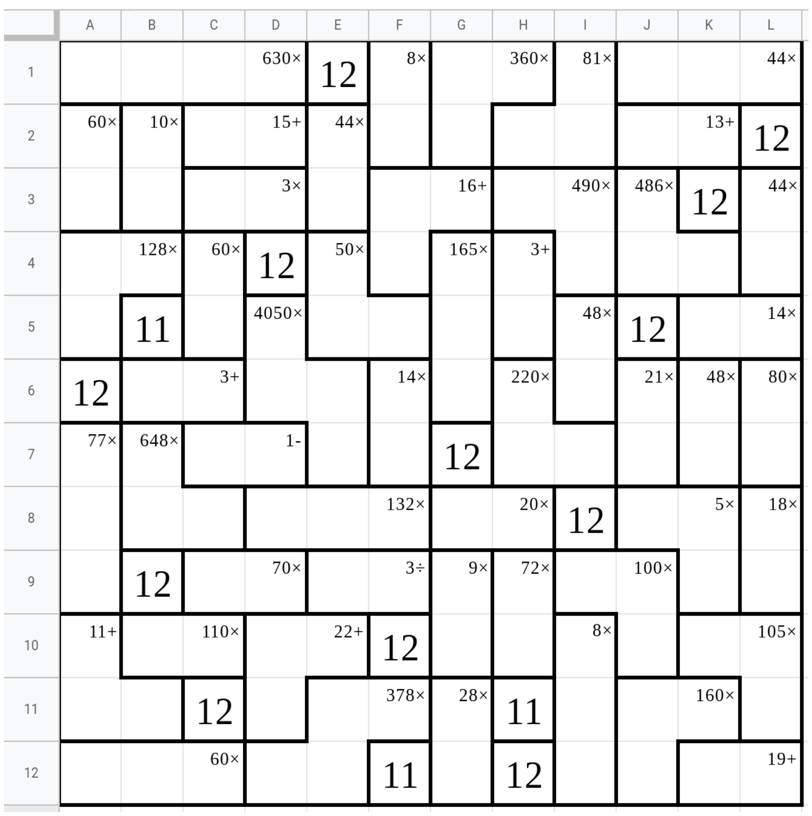
Amanda Louie

Preface

This solution presents one of the many ways to solve this KenKen. Per usual, standard KenKen rules apply, and in the following steps visual aid will be provided in order to help depict where certain numbers go within the KenKen. The KenKen itself will be centered inside a grid labeled horizontally (A-L) and vertically (1-12) in order to indicate specifically which square (referred to as a cell throughout this solution) a number will be placed in. The naming system for these squares will work like a chess board: Each letter will refer to the given column the square is in and each number will refer to the given row the square is in, so where the given column and row intersect is where you will find your square.

1 Fill in the givens

We can start by filling in the single cells that already have a given number, and we get this grid:



You will notice that all of the 12s have been given, in addition to some 11s. We will keep this in mind later when solving the rest of the puzzle as we can no longer count 12 as a possibility in any future cell.

2 Determine possibilities that are definite for operations on two cells

- Set: Cells F1 and F2 which multiply to 8. The possible solutions are 1×8 and 2×4. We will determine which of these is the correct solution later.
- Set: Cells A2 and A3 which multiply to 60. The only possible solutions are 6×10 and 5×12. However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 6×10.
- Set: Cells B2 and B3 which multiply to 10. The only possible solutions are 1×10 and 2×5. We will determine which of these is the correct solution later.
- Set: Cells C3 and D3 which multiply to 3. The only possible solution is 1×3.
- Set: Cells E2 and E3 which multiply to 44. The only possible solution is 4×11.
- Set: Cells L3 and L4 which multiply to 44. The only possible solution is 4×11.
- Set: Cells C4 and C5 which multiply to 60. The only possible solutions are 6×10 and 5×12. However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 6×10.
- Set: Cells H4 and H5 which add to 3. The only possible solution is $1+2$.
- Set: Cells I5 and I6 which multiply to 48. The only possible solutions are 4×12 and 6×8. However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 6×8.
- Set: Cells K5 and L5 which multiply to 14. The only possible solution is 2×7.
- Set: Cells B6 and C6 which add to 3. The only possible solution is $1+2$.
- Set: Cells F6 and F7 which multiply to 14. The only possible solution is 2×7.
- Set: Cells J6 and J7 which multiply to 21. The only possible solution is 3×7.
- Set: Cells K6 and K7 which multiply to 48. The only possible solutions are 4×12 and 6×8. However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 6×8.
- Set: Cells L6 and L7 which multiply to 80. The only possible solution is 8×10.
- Set: Cells G8 and H8 which multiply to 20. The only possible solutions are 2×10 or 4×5. We will determine which of these is the correct solution later.
- Set: Cells L8 and L9 which multiply to 18. The only possible solutions are 2×9 and 3×6. We will determine which of these is the correct solution later.

- Set: Cells C9 and D9 which multiply to 70. The only possible solution is 7×10.
- Set: Cells E9 and F9 which divide to 3. The only possible solutions are $12 \div 4$, $9 \div 3$, $6 \div 2$, or $3 \div 1$. We know that all the 12s in this puzzle have already been put in place, so this leaves us with the last three solutions as possible combinations.
- Set: Cells G9 and G10 which multiply to 9. The two possible solutions are 9×1 or 3×3. However, 3×3 would not work because we cannot have more than one 3 in a singular column, so we know that these two cells will contain 9 and 1.
- Set: Cells H9 and H10 which multiply to 72. The only possible solutions are 6×12 and 8×9. However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 8×9.
- Set: Cells B10 and C10 which multiply to 110. The only possible solution is 10×11.
- Set: Cells G11 and G12 which multiply to 28. The only possible solution is 4×7.
- Set: Cells K12 and L12 which add to 19. The only possible solutions are $7+12$, $8+11$, or $9+10$. We know that we already have an 11 and 12 in this row, so the only possibility is $9+10$.

The rest of the cells with sets of two have more possibilities, so we can leave them for later.

3 Determine possibilities that are definite for operations on three cells

- Set: Cells G1, G2, and H1 which all multiply to 360. The prime factorization of 81 is $3 \times 3 \times 3 \times 3$. The only possible combinations from this are $4 \times 9 \times 10$, $5 \times 8 \times 9$, or $6 \times 6 \times 10$. We will determine which of these is the correct solution later.
- Set: Cells I1, I2, and H2 which multiply to 81. The prime factorization of 81 is $3 \times 3 \times 3 \times 3$. The only two ways that we can group these 3s are $3 \times 3 \times 9$ (I1=3, I2=9, H2=3) or $1 \times 9 \times 9$ (I1=9, I2=1, H2=9). We will determine which of these is the correct solution later.
- Set: Cells J1, K1, and L1 which all multiply to 44. The prime factorization of 44 is $2 \times 2 \times 11$. The only way we can group these numbers so that there is no repetition is $1 \times 4 \times 11$.
- Set: Cells H3, I3, and I4 which all multiply to 490. The prime factorization of 490 is $2 \times 5 \times 7 \times 7$, so in order to fit this into 3 cells, we must have $10 \times 7 \times 7$. (None of the other solutions of $14 \times 5 \times 7$, $35 \times 2 \times 7$, or $49 \times 2 \times 5$ have all three numbers between 1 and 12.) In addition, since we know that we cannot repeat numbers within columns or rows, we must place a 7 in H3, a 10 in I3, and a 7 in I4.
- Set: Cells J3, J4, and K4 which all multiply to 486. The prime factorization of 486 is 2×3^5. The only way we can group these numbers such that all three are between 1 and 12 is $6 \times 9 \times 9$. In addition, since we cannot repeat numbers within columns or rows, we must place a 9 in J3, a 6 in J4, and a 9 in K4.
- Set: Cells A4, A5, and B4 which all multiply to 128. The prime factorization of 128 is 2^7 or $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$. The only two ways that we can group these 2s to create three numbers is $2 \times 8 \times 8$ and $4 \times 4 \times 8$. We will determine which is the correct solution later.

- Set: Cells E4, E5, and F5 which all multiply to 50. The prime factorization of 50 is $2{\times}5{\times}5$. The only ways we can group these numbers are $2{\times}5{\times}5$ or $1{\times}5{\times}10$. We will determine which is the correct solution later.
- Set: Cells G4, G5, and G6 which all multiply to 165. The prime factorization of 165 is $3{\times}5{\times}11$, which is the only possible solution.
- Set: Cells H6, H7, and I7 which all multiply to 220. The prime factorization of 220 is $2{\times}2{\times}5{\times}11$. The only possible ways to group these numbers are $2{\times}10{\times}11$ or $4{\times}5{\times}11$. We will determine which is the correct solution later.
- Set: Cells A7, A8, and A9 which all multiply to 77. The prime factorization of 77 is $7{\times}11$, so the only possible solution is $1{\times}7{\times}11$.
- Set: Cells B7, B8, and C8 which all multiply to 648. The prime factorization of 648 is $2^3 \times 3^4$ and the only way we can group these numbers such that they are between 1 and 11 (not 12 since every row and column already has a 12) is $8{\times}9{\times}9$. In addition, since we know that we cannot repeat numbers within columns or rows, we must place a 9 in B7, an 8 in B8, and a 9 in C8.
- Set: Cells D8, E8, and F8 which all multiply to 132. The prime factorization of 132 is $2{\times}2{\times}3{\times}11$, and the only ways we can group these numbers such that they are between 1 and 11 (not 12 since every row and column already has a 12) are $2{\times}6{\times}11$ or $3{\times}4{\times}11$. We will determine which is the correct solution later.
- Set: Cells J8, K8, and K9 which all multiply to 5. Since 5 is prime, the only solution is $5{\times}1{\times}1$. In addition, since we cannot repeat numbers in columns or rows, there must be a 1 in J8, a 5 in K8, and a 1 in K9.
  - We also can conclude that there must be a 9 in G9 and a 1 in G10 (see Step 2 for reference) because there cannot be another 1 in Row 9.
- Set: Cells I9, J9, and J10 which all multiply to 100. The prime factorization of 100 is $2{\times}2{\times}5{\times}5$, and the only possible groupings of these numbers are $1{\times}10{\times}10$, $2{\times}5{\times}10$, or $4{\times}5{\times}5$. We will determine which is the correct solution later.
- Set: Cells I10, I11, and I12 which all multiply to 8. The prime factorization of 8 is 2^3. However, we know that we cannot repeat numbers within columns, so the only possible grouping of these 2s is $1{\times}2{\times}4$.
- Set: Cells K10, L10, and L11 which all multiply to 105. The prime factorization of 105 is $3{\times}5{\times}7$, and this is the only possible grouping of these numbers, since they are all prime.
- Set: Cells A12, B12, and C12 which all multiply to 60. The prime factorization of 60 is $2{\times}2{\times}3{\times}5$, and the only possible groupings of these numbers are $2{\times}6{\times}5$, $2{\times}3{\times}10$ or $3{\times}4{\times}5$. We will determine which is the correct solution later.
- Set: Cells J11, J12, and K11 which all multiply to 160. The prime factorization of 160 is $2^5 \times 5$, and the only possible ways to group these numbers are $2{\times}8{\times}10$, $4{\times}4{\times}10$, $4{\times}5{\times}8$. We will determine which is the correct solution later.

The rest of the cells with sets of three have more possibilities, so we can leave them for later.

4 Determine possibilities that are definite for operations on four cells

- Set: Cells A1, B1, C1, and D1 which all multiply to 630. The prime factorization of 630 is $2\times3\times3\times5\times7$, so the only possible solutions are $1\times7\times9\times10$, $2\times5\times7\times9$, and $3\times5\times6\times7$.
- Set: Cells D5, D6, E6, and E7 which all multiply to 4050. The prime factorization of 4050 is $2\times3\times3\times3\times3\times5\times5$. We have to group this into $5\times9\times9\times10$, or else we have too many numbers or numbers that are too large.
- Set: Cells D12, E11, E12, and F11 which all multiply to 378. The prime factorization of 378 is $2\times3\times3\times3\times7$. The only possible groupings of these numbers are $1\times6\times7\times9$ or $2\times3\times7\times9$. We will determine which is the correct solution later.

After Steps 2, 3, and 4, you should have the following grid:

	A	B	C	D	E	F	G	H	I	J	K	L
1	2×5×7×9	3×5×6×7	1×7×9×10	630×	12	8×	4×9×10 5×8×9 6×6×10	360×	81×	1×4×11		44×
2	60×	10×		15+	44×	1×8 or 2×4			1×1×9 3×3×9		13+	12
3	6×10	1×10 or 2×5	1×3	3×	4×11		16+	7	10	9	12	44×
4	2×8×8 4×4×8	128×	60×	12	50×		165×	3+	7	6	9	4×11
5		11	6×10	4050×	1×5×10 2×5×5		3×5×11	1+2	48×	12	2×7	14×
6	12	1+2	3+	5×9×9×10		14×		220×	6×8	21×	48×	80×
7	77×	9		1-		2×7	12	2×10×11 4×5×11		3×7	6×8	8×10
8	1×7×11	8	9	2×6×11 3×4×11		132×	2×10 or 4×5	20×	12	1	5	18×
9		12	7×10	70×	9÷3 6÷2 3÷1	3÷	9	72×	1×10×10 2×5×10 4×5×5	100×	1	2×9 or 3×6
10	11+	10×11	110×		22+	12	1	8×9	8×		3×5×7	105×
11			12			378×	28×	11	1×2×4	2×8×10 4×4×10 4×5×8	160×	
12	2×5×6 2×3×10 3×4×5		60×	1×6×7×9 2×3×7×9		11	4×7	12			9+10	19+

5 Take the definite solutions to cells to eliminate possible solutions for other cells, as well as fill some cells based on the same definite solutions

- We can eliminate the solution $1\times7\times9\times10$ for Cells A1, B1, C1, and D1. In the same row (Cells J1, K1, L1) there is already going to be a 1 within one of the cells as a definite solution, so the 1 for this row cannot come from Cells A1, B1, C1, and D1.

- We can eliminate the solutions $4 \times 9 \times 10$ and $5 \times 8 \times 9$ for cells G1, G2, and H1. In Columns G and H there are already going to be definite 9s in each column (G9 and H9 or H10). From here, we are left with the solution $6 \times 6 \times 10$, and we know that we cannot repeat numbers in columns or rows. There must be a 10 in G1, a 6 in G2, and a 6 in H1.
  - From here we can also eliminate $3 \times 5 \times 6 \times 7$ as a solution from Cells A1, B1, C1, and D1 because there is already a 6 in the first row. This leaves us with only the solution of $2 \times 5 \times 7 \times 9$ to fill these cells.
- We must put a 1 in L1. Columns J and K already have 1s in them (J8 and K9), so L1 is the only cell to place the 1 for the solution of J1, K1, and L1.
- We must place a 10 in A2 and a 6 in A3. Row 2 already has a 6 in it (G2), so we have to place the 6 of the solution for A2 and A3 in A3. This leaves us with only A2 to place the 10.
- We can eliminate the solution 1×10 for Cells B2 and B3. Both Rows 2 and 3 already have 10s in them (A2 and I3), so 10 for these rows is not going to come from B2 or B3. This leaves us with the solution of 2×5 for Cells B2 and B3.
  - From here we also know that we must place a 1 in B6 and a 2 in C6. Column B already has a 2 in it (B2 or B3), so we have to place the 2 of the solution for B6 and C6 in C6. This leaves us with only B6 to place the 1.
- We must place a 10 in C4 and a 6 in C5. Row 4 already has a 6 in it (J4), so we have to place the 6 of the solution for C4 and C5 in C5. This leaves us with only C4 to place the 10.
- We must place a 2 in H4 and a 1 in H5. Row 5 already has a 2 in it (K5 or L5), so we have to place the 2 of the solution for H4 and H5 in H4. This leaves us with only H5 to place the 10.
  - From here, we can also eliminate $2 \times 8 \times 8$ as a solution for Cells A4, A5, and B4 because in this solution we would have to place a 2 in Row 4, however there is already a 2 in Row 4 (H4). This leaves us with the solution $4 \times 4 \times 8$, and we know that we cannot repeat numbers in columns or rows. There must be an 8 in A4, a 4 in A5, and a 4 in B4.
    * From here, we also know that we must place a 4 in L3 and an 11 in L4. Row 4 already has a 4 in it (B4), so we have to place the 4 of the solution for L3 and L4 in L3. This leaves us with only L4 to place the 11.
    * From here, we also know that we must place a 4 in E2 and an 11 in E3. Row 3 already has a 4 in it (L3), so we have to place the 4 of the solution for E2 and E3 in E2. This leaves us with only E3 to place the 11.
  - From here, we can also eliminate $2 \times 5 \times 5$ as a solution for cells E4, E5, and F5, because there is already going to be a 2 in Rows 4 and 5 (H4 and K5 or L5). This leaves us with the solution $1 \times 5 \times 10$. Since there already is a 10 in Row 4 and a 1 in Row 5, we must put the 10 in Row 5 (E5 or F5) and the 1 in Row 4 (E4).
  - From here, we can also eliminate $1 \times 1 \times 9$ as a solution for Cells H2, I1, and I2 because there is already going to be a 1 in Column H (H5), the column in which a 1 would have gone in if we chose the solution set $1 \times 1 \times 9$. This leaves us with the solution $3 \times 3 \times 9$. We know that we cannot repeat numbers in columns or rows. There must be a 3 in H2, a 3 in I1, and a 9 in I2.

- We must place an 8 in I5 and a 6 in I6. Row 5 already has a 6 in it (C5), so we have to place the 6 of the solution for I5 and I6 in I6. This leaves us with only I5 to place the 8.
- We must place an 8 in K6 and a 6 in K7. Row 6 already has a 6 in it (I6), so we have to place the 6 of the solution for K6 and K7 in K7. This leaves us with only K6 to place the 8.
- We must place a 10 in L6 and an 8 in L7. Row 6 already has an 8 in it (K6), so we have to place the 8 of the solution for L6 and L7 in L7. This leaves us with only L6 to place the 10.
  - From here, we also know that we must place a 10 in E7. The set of Cells D5, D6, E6, and E7 which all multiply to 4050 have the solution $5 \times 9 \times 9 \times 10$, and we know the the 10 cannot be placed in Rows 5 or 6 because there are already 10s in these rows (E5/F5 and L6). We can also place the 5 and 9s as we know that we cannot repeat numbers in columns or rows. There must be a 9 in D5, a 5 in D6, and a 9 in E6.
    * From here, we must place a 5 in E5 and a 10 in F5. Column E already has a 10 in it (E7), so we have to place the 10 of the solution for E5 and F5 in F5. This leaves us with only E5 to place the 5.
    * From here we can also place an 11 in G6 since the solution for Cells G4, G5, and G6 must be $3 \times 5 \times 11$, and there is already a 3 and 5 in Row 6 (J6 for 3 and D6 for 5), so the only number that can be placed in G6 from this solution set is 11. We must also place a 5 in G4 and a 3 in G5. Row 5 already has a 5 in it (E5), so we have to place the 5 of the solution for G4 and G5 in G4. This leaves us with only G5 to place the 3.
- We can eliminate the solution $2 \times 10 \times 11$ for cells H6, H7, and I7. Both Columns H and I already have 2s in them (H4 and I10/I11/I12). This means that the 2 for these columns is not going to come from H6 or H7 or I7. We are left with the solution $4 \times 5 \times 11$, and we can put the 11 in I7 since Column H already has an 11 in it (H11). We must place a 4 in H6 and a 5 in H7. Row 6 already has a 5 in it (D6), so we have to place the 5 of the solution for H6 and H7 in H7. This leaves us with only H6 to place the 4.
- We must place a 7 in F6 and a 2 in F7. Row 6 already has a 2 in it (C6), so we have to place the 2 of the solution for F6 and F7 in F7. This leaves us with only F6 to place the 7.
  - From here, we also know that we must place a 3 in J6 and a 7 in J7. Row 6 already has a 7 in it (F6), so we have to place the 7 of the solution for J6 and J7 in J7. This leaves us with only J6 to place the 3.
  - From here, we can also eliminate the solution 2×4 for cells F1 and F2. Column F already has a 2 in it (F7), so the 2 for this column cannot come from F1 or F2. This leaves us with the solution of 1×8 for cells F1 and F2. We also know that we must place the 1 in F2 and the 8 in F1. Row 1 already has a 1 in it (L1), so we have to place the 1 of the solution for F1 and F2 in F2. This leaves us with only F1 to place the 8.
- We must place a 1 in A7 since the solution for Cells A7, A8, and A9 must be $1 \times 7 \times 11$, and there is already a 7 and 11 in Row 7 (J7 for 7 and I7 for 11), so the only number that can be placed in A7 from this solution set is 1. We must also place a 7 in A8 and an 11 in A9. Row 9 already has a 7 in it (C9/D9), so we have to place the 7 of the solution for A8 and A9 in A8. This leaves us with only A9 to place the 11.

- We can eliminate the solution 4×5 for Cells G8 and H8. Row 8 already has a 5 in it (K8), so 5 for these rows cannot come from G8 or H8. This leaves us with the solution of 2×10 for Cells G8 and H8. We also know that we must place the 2 in G8 and the 10 in H8. Column H already has a 2 in it (H4), so we have to place the 2 of the solution for G8 and H8 in G8. This leaves us with only H8 to place the 10.
  - From here, we can also eliminate 2×6×11 as a solution for Cells D8, E8, and F8, because there is already going to be a 2 in Row 8 (G8). This leaves us with the solution 3×4×11. Since there already is an 11 in Column F (F12) and Column E (E2 or E3), we must put the 11 in D8. There already is a 4 in Column E (E2 or E3), so we must put the 4 in F8. This leaves us with only E8 to place the 3.
  - From here we can also eliminate 2×9 as a solution for cells L8 and L9 because there is already going to be a 2 and 9 in Row 8 (G8 for 2 and C8 for 9). This leaves us with the solution 3×6. Since there already is a 3 in Row 8 (D8/E8/F8), we have to place the 3 of the solution for L8 and L9 in L9. This leaves us with only L8 to place the 6.
- We must place a 10 in D9 and a 7 in C9. Column C already has a 10 in it (C4), so we have to place the 10 of the solution for C9 and D9 in D9. This leaves us with only C9 to place the 7.
- We can eliminate the solutions 9÷3 and 3÷1 for Cells E9 and F9. Row 9 already has a 9 in it (G9), a 3 in it (L9), and a 1 in it (K9), so the 1, 3, and 9 for this row cannot come from E9 or F9. This leaves us with the solution of 6÷2 for Cells E9 and F9. We must place 6 in F9 and 2 in E9. Column F already has a 2 in it (F7), so we have to place the 2 of the solution for E9 and F9 in E9. This leaves us with only F9 to place the 6.
- We must place a 9 in H10 and an 8 in H9. Row 9 already has a 9 in it (G9), so we have to place the 9 of the solution for H9 and H10 in H10. This leaves us with only H9 to place the 8.
- We can eliminate the solutions 1×10×10 and 2×5×10 for cells I9, J9, and J10. Row 9 already has a 10 in it (D9), a 1 in it (K9), and a 2 in it (E9), so the 1, 2, and 10 for this row cannot come from I9 or J9. This leaves us with the solution of 4×5×5 for cells I9, J9, and J10. Since we cannot repeat numbers in columns or rows, there must be a 5 in I9, a 4 in J9, and a 5 in J10.
  - From here, we also know that we must place an 11 in J1 and a 4 in K1. Column J already has a 4 in it (J9), so we have to place the 4 of the solution for J1 and K1 in K1. This leaves us with only J1 to place the 11.
- We must place a 10 in B10 and an 11 in C10. Column B already has an 11 in it (B5), and Column C already has a 10 in it (C4), so we have to place the 10 and 11 in the opposite columns of the existing 10 and 11s, and yielding a 10 in B10 and an 11 in C10.
- We must place a 3 in K10, a 5 in L11, and a 7 L10. Column L already has 3 in it, so the 3 has to go into Column K and the only cell in this set it can occupy is K10. There is also already a 5 in Row 10, so the 5 has to go into Row 11, and the only cell in this set it can occupy is L11. Finally, this leaves us with only cell L10 to place the 7.
  - From here, we also know that we must place 7 in K5 and a 2 in L5. Column L already has a 7 in it (L10), so we have to place the 7 of the solution for K5 and L5 in K5. This leaves us with only L5 to place the 2.

- We must place a 10 in K12 and a 9 in L12. Column L already has a 10 in it (L6), so we have to place the 10 in K12. This leaves us with only L12 to place the 9.
  - From here, we can also eliminate $2\times3\times10$ as a solution for Cells A12, B12, and C12 because there is already going to be a 10 in Row 12 (K12). This leaves us with the solutions $2\times5\times6$ and $3\times4\times5$.
  - From here, we can also place a 9 in F11 because the possible solution for Cells D12, E11, E12, and F11 all have to contain a 9 within them; however, Row 12 already has a 9 in it (L12), and Column E already has a 9 in it (E6), so the only place for the 9 of this set to go in is F11. We can also eliminate the solution $2\times3\times7\times9$ because if we used this solution, we would have to place the 2 in D12 (Column E already has a 2 in E9) but also the 3 in D12 (Column E also already has a 3 in E8), and it is not possible to put two numbers within one cell. This leaves us with the solution $1\times6\times7\times9$. We can also place a 1 in D12 because Column E already has a 1 in it, so the only cell within the set that we can place the 1 in is D12.
    - From here, we also know that we must place 1 in C3 and a 3 in D3. Column D already has a 1 in it (D12), so we have to place the 1 of the solution for C3 and D3 in C3. This leaves us with only D3 to place the 3.
- We can eliminate the solutions $4\times4\times10$ and $4\times5\times8$ for cells J11, J12, and K11. Columns J and K already have 4s in them (J9 and K1), so the 4 for these columns cannot come from J11 or J12 or K11. This leaves us with the solution of $2\times8\times10$ for Cells J11, J12, and K11. We can put the 8 in J12 because we know that within the possible solutions to cells in Row 12, J12 is the only place where we can get an 8 from. From here, we can also place 10 in J11 because Column K already has a 10 in it (K12) and this leaves us with only Cell K11 to place the 2.

After Step 5, you should obtain the following grid:

	A	B	C	D	E	F	G	H	I	J	K	L
1	2×5×7×9			630×	12	8	10	6	3	11	4	1
2	10	10×		15+	4	1	6	3	9		13+	12
3	6	2×5	1	3	11		16+	7	10	9	12	4
4	8	4	10	12	1		5	2	7	6	9	11
5	4	11	6	9	5	10	3	1	8	12	7	2
6	12	1	2	5	9	7	11	4	6	3	8	10
7	1	9		1-	10	2	12	5	11	7	6	8
8	7	8	9	11	3	4	2	10	12	1	5	6
9	11	12	7	10	2	6	9	8	1	5	1	3
10	11+	10	11	6	8	12	1	9	8×	1	3	7
11			12	8	6,7	9	28×	11	1×2×4	10	2	5
12	2×5×6 3×4×5		60×	1	6,7	11	4×7	12		8	10	9

6 Fill in the remaining blanks to finish the puzzle

- Starting with Cells A1, B1, C1, and D1, we know that the solution for this set is going to be 2×5×7×9. We know that we must place 7 in B1 because Column B already has a 2, 5, and 9 (B2, B3, B7). Next, we must place 9 in A1 because Columns B, C, and D already have a 9 in them (B7, C8, D5). Finally, we must place 5 in C1 and 2 in D1. Column C already has a 2 in it (C6), so we have to place the 2 of the solution for C1 and D1 in D1. This leaves us with only C1 to place the 5.
- In Column G, we notice that the only missing number is 8, so we must place an 8 in G3. From here, we can also notice that the only missing numbers in column F are 3 and 5. We must place a 5 in F3 and a 3 in F4. Row 3 already has a 3 in it (C3 or D3), so we have to place the 3 of the solution for F3 and F4 in F4. This leaves us with only F3 to place the 5. We now know that the solution set for Cells F3, F4, and G3 is 3+5+8.
  - From here, we also know that we must place a 5 in B2 and a 2 in B3. Row 3 already has a 5 in it (F3), so we have to place the 5 of the solution for B2 and B3 in B2. This leaves us with only B3 to place the 2.
  - From here, we can notice that the missing numbers in Row 2 are 2, 8, 7, and 11 and the missing numbers in Columns J and K are 2 and 11, respectively. This means that we

must place an 8 in C2, a 7 in D2, a 2 in J2, and an 11 in K2. We know that the missing numbers in Columns J and K will be filled in, so the only numbers left in Row 2 will be 7 and 8. However, there is already a 7 in Column C (C9), so we can't place the 7 in C2, and we have to place it in D2. This leaves us with only cell C2 to place the 8.

- In Row 7 we can notice that the only missing numbers are 3 and 4. Therefore, know that the solution for Cells C7 and D7 will be 4-3. We must place the 3 in C7 and the 4 in D7. Column D already has a 3 in it (D3), so we have to place the 3 of the solution for C7 and D7 in C7. This leaves us with only D7 to place the 4.

- We must place a 1 in I11. Rows 10 and 12 already have 1s in them (G10 and D12), so we have to place the 1 in Row 11.

- In Columns D and E, we notice that the 8s for these columns are missing. The only set of cells where the 8s can come from are Cells D10, D11, and E10 which add up to 22. We know that we cannot repeat numbers in columns or rows. We must place the 8s in separate rows, so we will place an 8 in D11 and an 8 in E10. We know that Cells D10, D11, and E10 will all have to add up to 22, and the only number that will allow these cells to add up to 22 is 6, so we must place a 6 in D10.

- In Column C, we notice that the only missing number is 4, so we must place a 4 in C12. This means that the solution set for Cells A12, B12, and C12 is $3 \times 4 \times 5$. We must place the 5 in A12 and the 3 in B12. Column B already has a 5 in it (B2), so we have to place the 5 of the solution for A12 and B12 in A12. This leaves us with only B12 to place the 3.

- In Column B, we notice that the only missing number is 6, so we must place a 6 in B11. The remaining numbers in the solution set for Cells A10, A11, and B11 have to be 2 and 3 (the only remaining numbers in Column A). We must place the 2 in A10 and the 3 in A11. Row 10 already has a 3 in it (K10), so we have to place the 3 of the solution for A10 and A11 in A11. This leaves us with only A10 to place the 2.

- We must place a 6 in E11 and a 7 in E12. Row 11 already has a 6 in it (B11), so we have to place the 6 of the solution for E11 and E12 in E11. This leaves us with only E12 to place the 7.

- We must place a 4 in G11 and a 7 in G12. Row 11 already has a 7 in it (E11), so we have to place the 7 of the solution for G11 and G12 in G12. This leaves us with only G11 to place the 4.

- We must place a 4 in I10 and a 2 in I12. Row 10 already has a 2 in it (A10) and Row 12 already has a 4 in it (C12), so we have to place the 4 and 2 in the opposite columns of the existing 4 and 2s, and we will end up with a 4 in I10 and a 2 in I12.

After Step 6, you should get this as your final solution:

	A	B	C	D	E	F	G	H	I	J	K	L
1	9	7	5	2	12	8	10	6	3	11	4	1
2	10	5	8	7	4	1	6	3	9	2	11	12
3	6	2	1	3	11	5	8	7	10	9	12	4
4	8	4	10	12	1	3	5	2	7	6	9	11
5	4	11	6	9	5	10	3	1	8	12	7	2
6	12	1	2	5	9	7	11	4	6	3	8	10
7	1	9	3	4	10	2	12	5	11	7	6	8
8	7	8	9	11	3	4	2	10	12	1	5	6
9	11	12	7	10	2	6	9	8	1	5	1	3
10	2	10	11	6	8	12	1	9	4	1	3	7
11	3	6	12	8	7	9	4	11	1	10	2	5
12	11	3	4	1	6	11	7	12	2	8	10	9

Code Wheel

Nathan Kuo

Introduction

Code wheels have an inner wheel and an outer wheel. Numbers on the outside wheel correspond with letters on the inside wheel. By turning the wheel, a shift is created, assigning new numbers to these letters. Try to figure out these encoded messages!

The Puzzles

Puzzle # 1

Wklv vhqwhqfh lv vkliwhg eb wkuhh. Fdhvdu fiskhuv zhuh vxssrvhgob xvhg eb Mxolxv Fdhvdu. Wklv phwkrg lv hdvlob ghfiskhuhg eb euxwh iruflqj

Puzzle # 2

Olyl'z huvaoly zptpshy spulhy zopma, iba dpao h zopma vm zlclu.

Puzzle # 3

Skv dxplz githo atm uxjyn tcvi 13 ebwr qtfn.

Hint: This sentence gets shifted by a linear function $y = mx + b$.
Undo the linear shift, and you get the original sentence.

Puzzle # 4

Gngjyg vmcjvp xg e Pcvqpk gvrycg.

Puzzle # 5

64 25 144 144 225 400 64 25 324 25

Solution to Code Wheel

Nathan Kuo

Puzzle # 1

Code: Wklv vhqwhqfh lv vkliwhg eb wkuhh. Fdhvdu flskhuv zhuh vxssrvhgob xvhg eb Mxolxv Fdhvdu. Wklv phwkrg lv hdvlob ghflskhuhg eb euxwh iruflqj

Answer: This sentence is shifted by three. Caesar ciphers were supposedly used by Julius Caesar. This method is easily deciphered by brute forcing.

Explanation: While you can brute force this problem by shifting each letter until you find actual words, there's an easier way to do this. Knowing that "E" is the most used letter in the alphabet, followed closely by "T," "A," "I," and "O," if you notice the most used letters in the prompt, you can figure out how much the sentence has been shifted by. There's a lot of "H"s used in the prompt, which could possibly be "E." The letters are three spaces apart, so try shifting the entire sentence back by three. If this did not work, repeat the process with a different letter.

Puzzle # 2

Code: Olyl'z huvaoly zptpshy spulhy zopma, iba dpao h zopma vm zlclu.

Answer: Here's another similar linear shift, but with a shift of seven.

Explanation: This problem uses the same steps used to solve the first puzzle. Find the most used letters in the sentence and substitute another common letter to replace them.

Puzzle # 3

Code: Skv dxplz githo atm uxjyn tcvi 13 ebwr qtfn.

> Hint: This sentence gets shifted by a linear function $y = mx + b$.
> Undo the linear shift, and you get the original sentence.

Answer: The quick brown fox jumps over 13 lazy dogs.

Explanation: Assigning letters to 1-26 and using the equation for a line ax+b will give you the shift. To undo it, use the modulo function (ax+b) mod 26. Working backwards will give you a = 5 and b = 1.

Puzzle # 4

Code: Gngjyg vmcjvp xg e Pcvqpk gvrycg.

Answer: Caesar cipher in a Caesar cipher.

Explanation: If you assembled a table of Caesar ciphers, with letters getting shifted in increments of one, you can assign letters according to a certain chosen keyword. For this puzzle, the keyword is “encrypt.” If this were a regular linear shift cipher, the repetition of characters would easily lend itself into being deciphered.

Puzzle # 5

Code: 64 25 144 144 225 400 64 25 324 25

Answer: hello there

Explanation: This one is a bit weird to catch on, but realize that these numbers are perfect squares and are associated with numbers. Take the square root of these numbers and assign them to letters until the sequence makes sense. Other variations and mathematical operations such as addition, subtraction, or division could be used to further complicate this cipher.

Questions and Answers with Professor Emeritus Paul A. Zeitz, Class of 1975

Paul A. Zeitz, Class of 1975, is a Professor Emeritus of Mathematics at the University of San Francisco, where he also served as Chair of the Mathematics Department. He is also the author of the classic text, The Art and Craft of Problem Solving (1998).

At Stuyvesant, Zeitz served as captain of the Math Team. At age 16, he qualified for the very first U.S. team to participate in the International Mathematical Olympiad (IMO) in 1974, where he won a bronze medal. He later contributed to writing national math contest problems, including USAMO. He would eventually go on to coach multiple U.S. IMO teams, including the first ever perfect-scoring 1994 "Dream Team."

He received his bachelor's degree from Harvard University in 1981, having won a Westinghouse (now Regeneron) Science Talent Search scholarship. He earned his Ph.D. from the University of California, Berkeley in 1992.

In 1999, Zeitz co-founded the Bay Area Mathematical Olympiad and co-founded the San Francisco Math Circle in 2005. In 2009, he produced a series of video lectures for The Teaching Company on problem solving. The Mathematical Association of America awarded Zeitz with the Deborah and Franklin Haimo Awards for Distinguished College or University Teaching of Mathematics in 2003.

Currently, Zeitz serves as Chair of the Board of Directors of The Proof School, located in downtown San Francisco. Starting in 2013, Zeitz envisioned and then co-founded this mathematics-focused middle school and high school which opened its doors in 2015 and features a unique and innovative world-class curriculum.

Paul Zeitz's answers are taken from an interview conducted by Josephine Lee and have been edited for clarity.

1) In 2011, you wrote the foreword to *Math Survey*, "The Three Epigraphs," to encourage students to get lost, get obsessed, and get together. From your experience, what are the most common stumbling blocks for young explorers, and what is your recommendation to overcome them?

The most common stumbling blocks are psychological; almost every beginner lacks confidence and gives up too easily. Most beginners do not spend enough time thinking about problems and have been conditioned to think that if they don't get "the answer" within a minute or two, then they never will. In reality, most worthwhile things that one grapples with require large amounts of time, failure, wild goose chases, etc. And a baseball-style average (say, 1 in 3 successes) is very, very good.

So the recommendation to deal with this is:

- Get used to a long time scale (working for an hour, a day, etc.)
- When you fail to solve a problem, don't give up but perhaps set it aside
- Simultaneously, improve your confidence by doing problems that are easy for you, at least some of the time
- Find trusted comrades that you can work with whom you don't feel (too much) competition.

2) What do you remember most about your experience at Stuyvesant? Were there any specific students, teachers, or extracurricular activities of Stuyvesant that have had a significant impact on you?

I loved Stuyvesant. It was safe, and everyone was fun to talk to. I am still close friends after nearly 50 years with a few fellow students. Of course, I have very fond memories of Math Team, which back then was just a couple dozen people, completely run by students (although our faculty advisor was a fine mathematician who was very kind to us, but she let us run the show). The older students mentored the younger students. I had the good fortune to be mentored by Eric Lander, who became a famous scientist and President Biden's cabinet-level science advisor.

My very best friends were not Math Team people. I spent lots of time taking very long walks all over NYC with my friends, playing frisbee, hanging out in parks, listening to concerts for free at Lincoln Center by discovering noise leaks in the parking garage, etc. My favorite teachers were Sylvia Schwartz, who taught math with a tremendous aesthetic sense (she was a serious violinist) and Rlene Dahlberg (not misspelled), who taught an independent study English class that was very stimulating, and Vivianne Gluck, who taught history. She inspired me to read history seriously and indeed I became a history major in college (American History with a minor in East Asian History).

3) In your experience as a mathematician and educator, what is one thing you've changed your mind about in the last decade?

I taught high school in an independent private high school in Colorado after college. I discovered that, contrary to my expectations, one of my favorite classes was a terminal algebra class for "weak" math students. The students were pretty uniform in their skill level (which was low), and they considered themselves to be "dumb." Their expectations were quite low, so it turned out to be fun to fool them into doing interesting and hard math through puzzles. They were a very cohesive group that helped one another, and they (and I) had tons of fun.

At the other end of the spectrum, I also loved training the IMO teams in the mid-1990s, including the 1994 "Dream Team" that got a perfect score (no team matched this achievement until China in 2022). These students were super-skilled and super talented, and I had to work like a maniac to keep up with them. So I never got much sleep, but we had tons of fun. One thing that I discovered is that no matter how "smart" you are, you often feel "stupid," because there are always problems that one cannot solve. A large part of my job was helping these students to learn to accept their humanity and relax when something was (perhaps temporarily) elusive to them.

4) Is there an important truth that you find many/most others disagree with? If so, what is it?

Many people, especially at places like Stuyvesant, confuse "intelligence" with virtue. In reality, intelligence is just an attribute. Being "smart" has, sadly, nothing to do with being "good." A "good" student is not necessarily a "smart" or a "accomplished" student.

5) Is there anything else you would like to communicate to current Stuyvesant students?

Every year, it seems to get harder to get into college. It is easy to get overwhelmed and anxious, especially with nervous parents hovering over you. Please relax. First of all, even if you get into your "dream" college, it is just the minor step in a very long path. And if you don't get in, your life is NOT over. They made a mistake, you'll get over it, and you will have a happy and successful life if you let yourself have one. You are the one in control. Strive for situations that give you as much autonomy as possible. This is an undervalued quality that is, in my opinion, the key to a good life.

Stuyvesant Faculty Perspectives: Mr. Patrick Honner: Mathematics is Like a Great Book that Never Ends

Rin Fukuoka

Mr. Patrick Honner has been teaching mathematics for more than 20 years, and he has spent the past 17 years at New York City's specialized high schools, including Stuyvesant since 2019. He holds a B.S. in mathematics and a B.A. in philosophy from Wayne State University, an M.A. in mathematics from the University of Wisconsin, and an M.S.Ed. in mathematics education from the City College of New York.

Honner has received several teaching honors, including the Presidential Award for Excellence in Mathematics and Science Teaching, the Muller Prize for Professional Influence in Education, the University of Chicago's Outstanding Educator Award, and the Math for America's Master Teacher Fellowship (four times). He also won the Sloan Award for Excellence in Teaching Science and Mathematics and was runner-up for the inaugural Rosenthal Prize for Innovation in Math Teaching. Moreover, Honner is a National STEM Ambassador and represents New York City in the New York State Master Teacher Program.

1 Math Without Limits

So, where did it start? Honner says that mathematics has always been part of who he is, a part of how he sees and understands the world. It has always made sense to him. Although he never expected to become a math teacher, once he started learning with, and from, so many brilliant, fun, and creative people, he realized the classroom was where he belonged.

Honner describes math as "a great book that never ends." He goes on to explain that when you're reading a book, it is inevitable for the story to be over, and you'll have to say goodbye. But there is no end to math, since one keeps discovering new ideas, rediscovering old ones, being challenged, and overcoming those challenges. He remarks that there's so much enlightenment and satisfaction to be found in math, and there's no limit to how much it can give.

Honner firmly believes that the key to studying mathematics is patience. Some ideas, especially big and important ones, take a long time to truly comprehend. He details that you must learn to live with that frustration, with that sense of incompleteness. Ask any mathematician, and they'll tell you that they spend a lot of their time in a state of uncertainty and confusion. But keep at it, he says, and keep revisiting those big ideas over and over again. In time, things connect and open up.

2 Trying to Do More for Students

Honner finds that few things are as fun and as exciting as learning, and as a teacher, he gets to be around learning every day. Not only is he present when eyes and minds light up with the discovery of new ideas and the mastery of complex techniques, but he also gets to constantly learn alongside everyone else. He feels very fortunate to be able to organize his life around such enjoyable and fulfilling pursuits.

"One of the most challenging aspects of teaching is that you know you could always be doing more for your students," he says. He explains that this sense is especially true at a place like Stuyvesant, where students are so capable and driven. As a teacher, he knows he could always be pushing his top students a little more, or spending more time with those who need a little extra help, or revising and improving his lessons and activities. He knows that whatever time and energy he puts in will have an impact, but there's only so much time and energy to give.

Honner asserts that great students make great schools. He appreciates the Stuyvesant community, made up of students who are brilliant, creative, fun, curious, driven, and more — as great students everywhere are. But the one thing that really stands out to him is how the students at Stuy truly feel empowered and compelled to lead in the school community. He sees these characteristics in the student organizations and clubs, but also in the hallways and classrooms.

3 Beyond the Classroom

Alongside teaching mathematics, Honner is an active writer and presenter. His work has appeared in *Wired* and the *New York Times* Learning Network, and he writes a column for *Quanta Magazine* where he connects active math research to classrooms and curricula. Honner has authored *Painless Statistics* (2022) and co-authored *SHSAT: New York City Specialized High Schools Admissions Test* (2019). He explains, "I always have a different writing project in the works."

Honner has hosted events like TEDxNYED, delivered keynotes, and spoken before members of Congress and the Senate. He has given talks on a wide variety of topics, including pure mathematics, teaching and technology, mathematics and creativity, teacher leadership and collaboration, and mathematics and writing. Additionally, Honner conducts workshops for teachers and students, presents at local, national, and international conferences, and participates in mathematical outreach.

4 Looking into the Future

Honner is excited about the formalization movement in mathematics. He provides this example: Mathematicians and computer scientists are essentially developing programming languages that can take a mathematical proof as input, and decide its validity by checking it against the axioms. It is already making headway in a variety of fields, and the possibilities seem pretty incredible. Could an AI math engine someday discover entirely new proofs, or even new theorems? Can you imagine a computer conceiving of and proving a mathematical theorem that humans can't understand? We certainly aren't there yet, but it's exciting to think about, he says.

Honner advises passionate math students to keep taking mathematics courses for as long as they can and keep trying new things. You never know where or when some idea will be useful: linear algebra turns out to be important in machine learning, topology in data science, group theory in chemistry, differential equations in finance, and the list goes on and on and on. He remarks that if you keep learning new things, you'll inevitably see them pop up in whatever it is you're trying to understand. "And surround yourself with good people," he concludes. "And if you aren't surrounded by good people, find good people, and go where they are."

Stuyvesant Math and Computer Science Achievements

2021-2022

American Regions Mathematics League (ARML) Competition
NYC Math Team's "Tin Man" team finished in 2nd place overall for the offsite-ARML. Rishabh Das finished in 7th place on the individual round.

Harvard-MIT Mathematics Tournament (HMMT)
NYC Math Team's "Scarecrow" team placed 10th overall, and "Tin Man" placed 5th in the guts round. Rishabh Das finished in 7th place on the combinatorics round, algebra and number theory round, and overall.

Carnegie Mellon Informatics and Mathematics Competition (CMIMC)
NYC Math Team's "Tin Man" team placed 5th overall. Rishabh Das finished in 4th place on the geometry round.

New York State Mathematics League (NYSML)
NYC Math Team's teams, "Tin Man" and "Scarecrow," won 1st place and 2nd place overall.

13th annual Romanian Master of Mathematics (RMM)
Rishabh Das qualified to represent the U.S. in this international math competition held in Bucharest, Romania. The U.S. team won 4th place, and Rishabh won 12th place individually, earning a silver medal.

National Cyber Scholarship Foundation (NCSF) awards
Macy Jiang was honored as a Finalist for scoring among the highest of all the students who completed the NCSF Scholarship Application.

USA Junior Mathematical Olympiad (USAJMO)/USA Mathematical Olympiad (USAMO)
Rishabh Das won a USAMO Gold Award and was named a USAMO Winner by scoring in the top 10 nationwide. In addition, Paul Gutkovich won a USAMO Bronze Award.

USAMO Qualifiers
Rishabh Das (12th)
John Gupta-She (11th)
Paul Gutkovich (11th)

USAJMO Qualifiers
Ian Buchanan (9th)
Andrew Li (10th)

American Mathematics Competition (AMC) 12A
Rishabh Das and Carol Chen were School Winners with scores of 132.

American Mathematics Competition (AMC) 12B
Rishabh Das earned a perfect score of 150 and was School Winner.

American Mathematics Competition (AMC) 10A
Mikayla Lin was School Winner with a score of 117.

American Mathematics Competition (AMC) 10B
Ian Buchanan was School Winner with his score of 118.5.

American Mathematics Competition (AMC) 12A Distinguished Honor Roll
Carol Chen, Rishabh Das, Joseph Othman

American Mathematics Competition (AMC) 12B Distinguished Honor Roll
Rishabh Das, Maxwell Zen

American Mathematics Competition (AMC) 10A Distinguished Honor Roll
Mikayla Lin

80 American Invitational Mathematics Examination (AIME) Qualifiers
Ian Buchanan, Ian Chen Adamczyk, Carol Chen, Corina Chen, Alex Cho, Rishabh Das, Maya Doron-Repa, Lee Eisenberg, Lingjie Fang, Akihito Gohdo, John Gupta-She, Paul Gutkovich, Caleb Hahn, Zhengxi He, Charles Hua, Aaron Hui, Joseph Jeon, Henry Ji, Lawrence Joa, Andrew Juang, Mohammad Khan, Joseph Kim, Jacob Kirmayer, Julia Kozak, Jeremy Ku-Benjet, Keerthan Kumarappan, Anastasia Lee, Josephine Lee, Ryan Lee, Alvin Li, Andrew Li, Justin Li, Arthur Liang, Jerry Liang, Samuel Liao, Curt Lin, Ethan Lin, Mikayla Lin, Ethan Liu, Katherine Liu, Steven Lou, Dylan Ma, Haokai Ma, Nora Miller, Josiah Moltz, Brandon Oconnor, Joseph Othman, Aditya Pahuja, Jacob Paltrowitz, Yuhao Pan, Mark Park, Wonjong Park, Noam Pasman, Daniel Potievsky, Daphne Qin, Leo Schneiderman, Shadman Shaharia, Chenkai Shen, Jennifer Sun, Charles Tang, Ruiwen Tang, Nicholas Tarsis, Benny Wang, Leonna Wang, Maya Wassercug, Ivan Wei, Chun Yeung Wong, Edward Wu, Yujia Xue, Diana Yang, Connor Yau, Eileen Ye, Max Yenlee, Maxwell Zen, Brandon Zhang, Bryan Zhang, Calvin Zhang, Kathleen Zhang, Raymond Zhang, Elie Zheng, Alice Zhu

Mathematics and Computer Science Department Senior Awards
The Dr. Richard Rothenberg Memorial Award for Highest Honors in Mathematics: Rishabh Das

The Richard B. Geller Memorial Scholarship for Mathematics: Jerry Liang, Jennifer Sun

Henrietta O. Midonick Award: Carol Chen

David J. Pichler Memorial Scholarship: Daniela Maksin

The Okean Family Memorial Award for Scholarship in Pure or Applied Mathematics: Alice Zhu, Maxwell Zen

Irene Finkel Memorial Award for Excellence in Mathematics: Arthur Liang, Edward Wu

Highest Honors in Computer Science: Yusuf Elsharawy

High Honors in Computer Science: Reng Geng Zheng, Chistopher Liu, Maxwell Zen, Kenny Lau

Honors in Computer Science: Han Zhang, Raymond Yeung, Wen Hao Dong, Grace Chen, Ishraq Mahid, Josephine Lee, Alvin Li

Golden Ducky for Service to the Computer Science Community: Maxwell Zen, Yaying Liang Li, Rickey Dong, Kirsten Szeto

The 2021 Mathematical Association of America (MAA) Edyth May Sliffe Awards for Distinguished Mathematics Teaching in Middle School and High School
Mr. Brian Sterr was one of the 22 teachers nationwide honored for outstanding work to motivate students in mathematics by participating in one of the MAA American Mathematics Competitions (AMC) competitions.

University of Chicago Outstanding Educator Awards
Mr. Stan Kats and Mr. Topher Mykolyk were among the seven Stuyvesant teachers to be nominated by students accepted to the Class of 2025 for their guidance along the path toward intellectual growth.

Researchers Identify ‘Master Problem’ Underlying All Cryptography

Distinguished Guest Contributor
Erica Klarreich, award-winning journalist

Cryptography is a risky game; a cipher can only be deemed secure if it is impossible to crack, which is extremely difficult to prove given our ever-advancing technology. So how can we ensure the existence of secure cryptography? In this article, Dr. Erica Klarreich, award-winning mathematics and science journalist, explores Rafael Pass and Yanyi Liu's recent discovery of the dependence of cryptography on a problem in computational complexity.

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In 1868, the mathematician Charles Dodgson (better known as Lewis Carroll) proclaimed that an encryption scheme called the Vigenère cipher was “unbreakable.” He had no proof, but he had compelling reasons for his belief, since mathematicians had been trying unsuccessfully to break the cipher for more than three centuries.

There was just one small problem: A German infantry officer named Friedrich Kasiski had, in fact, broken it five years earlier, in a book that garnered little notice at the time.

Cryptographers have been playing this game of cat and mouse, creating and breaking ciphers, for as long as people have been sending secret information. “For thousands of years, people [have been] trying to figure out, ‘Can we break the cycle?’” said Rafael Pass, a cryptographer at Cornell Tech and Cornell University.

Five decades ago, cryptographers took a huge step in this direction. They showed that it's possible to create provably secure ciphers if you have access to a single ingredient: a “one-way function,” something that's easy to carry out but hard to reverse. Since then, researchers have devised a wide array of candidate one-way functions, from simple operations based on multiplication to more complicated geometric or logarithmic procedures.

Today, the internet protocols for tasks like transmitting credit card numbers and digital signatures depend on these functions. “Most of the crypto that is used in the real world is something that can be based on one-way functions,” said Yuval Ishai, a cryptographer at the Technion in Haifa, Israel.

But this advance has not ended the cat-and-mouse game — it has only sharpened its focus. Now, instead of having to worry about the security of every aspect of an encryption scheme, cryptographers need only concern themselves with the function at its core. But none of the functions currently in use have ever been definitively proved to be one-way functions — we don't even know for sure that true one-way functions exist. If they do not, cryptographers have shown, then secure cryptography is impossible.

In the absence of proofs, cryptographers simply hope that the functions that have survived attacks really are secure. Researchers don't have a unified approach to studying the security of these functions because each function "comes from a different domain, from a different set of experts," Ishai said.

Cryptographers have long wondered whether there is a less ad hoc approach. "Does there exist some problem, just one master problem, that tells us whether cryptography is possible?" Pass asked.

Now he and Yanyi Liu, a graduate student at Cornell, have shown that the answer is yes. The existence of true one-way functions, they proved, depends on one of the oldest and most central problems in another area of computer science called complexity theory, or computational complexity. This problem, known as Kolmogorov complexity, concerns how hard it is to tell the difference between random strings of numbers and strings that contain some information.

Rafael Pass, a cryptographer at Cornell Tech and Cornell University, helped find a question that could reveal whether one-way functions — and thus modern cryptography — truly exist.
Hugh Gordon

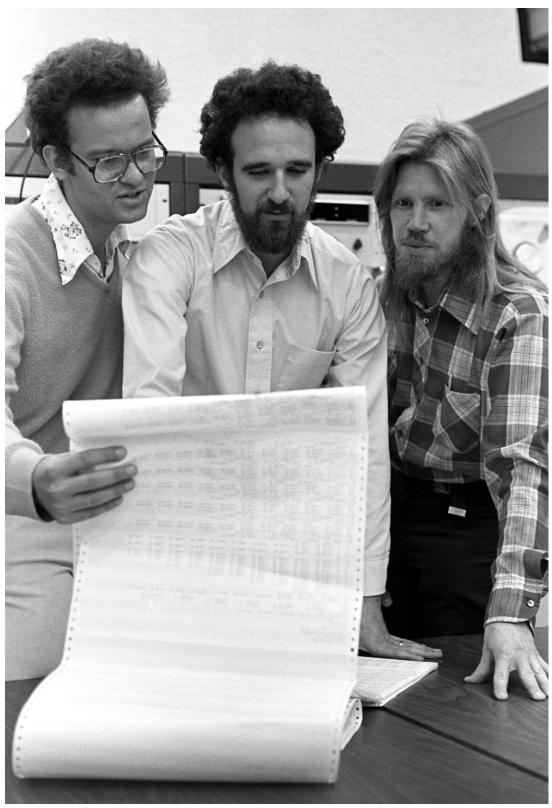
Liu and Pass proved that if a certain version of Kolmogorov complexity is hard to compute, in a specific sense, then true one-way functions do exist, and there's a clear-cut way to build one. Conversely, if this version of Kolmogorov complexity is easy to compute, then one-way functions cannot exist. "This problem, [which] came before people introduced one-way functions, actually turns out to fully characterize it," Pass said.

The finding suggests that instead of looking far and wide for candidate one-way functions, cryptographers could just concentrate their efforts on understanding Kolmogorov complexity. “It all hinges on this problem,” Ishai said. The proof is “breakthrough work on the foundations of cryptography.”

The paper has prompted cryptographers and complexity theorists to work together more closely, spurring a burst of activity uniting their approaches. “Multiple research groups are working to get to the bottom of things,” said Ryan Williams, a computer scientist at the Massachusetts Institute of Technology.

Leveraging Hardness

Usually, a hard problem is an obstacle. But in cryptography, where you can deploy it against your adversaries, it’s a boon. In 1976, Whitfield Diffie and Martin Hellman wrote a groundbreaking paper in which they argued that the particular hardness of one-way functions was precisely what cryptographers needed to meet the demands of the dawning computer age. “We stand today on the brink of a revolution in cryptography,” they wrote.



Martin Hellman (center) and Whitfield Diffie (right), shown here in a 1977 photo with Ralph Merkle, wrote a seminal paper that linked one-way functions to cryptography.
Chuck Painter / Stanford News Service

In the decades that followed, researchers figured out how to build a wide variety of cryptographic tools out of one-way functions, including private key encryption, digital signatures, pseudorandom number generators and zero-knowledge proofs (in which one person can convince another that a statement is true without revealing the proof). Diffie and Hellman's paper was "almost like a prophecy," Pass said. From the single building block of one-way functions, cryptographers managed to build "these super-complex and beautiful creatures," he said.

To get a feel for how one-way functions work, imagine someone asked you to multiply two large prime numbers, say 6,547 and 7,079. Arriving at the answer of 46,346,213 might take some work, but it is eminently doable. However, if someone instead handed you the number 46,346,213 and asked for its prime factors, you might be at a loss. In fact, for numbers whose prime factors are all large, there is no efficient way (that we know of) to find those factors. This makes multiplication a promising candidate for a one-way function: As long as you start with large enough prime numbers, the process seems easy to do, but hard to undo. But we don't know for sure that this is the case. Someone could find a fast way to factor numbers at any moment.

Cryptographers have gleaned an assortment of potential one-way functions from different areas of mathematics, but no single function has a higher claim than another. If, say, multiplication were toppled as a one-way function tomorrow, that wouldn't say anything about the validity of the other candidate one-way functions. Cryptographers have long asked whether there is some quintessential one-way function — one which, if broken, would pull all the other candidates down with it.

In 1985, Leonid Levin, a computer scientist at Boston University, answered this question in a formal sense, demonstrating a "universal" one-way function that is guaranteed to be a one-way function if anything is. But his construction was "very artificial," said Eric Allender, a computer scientist at Rutgers University. It is "not something anybody would have studied for any reason other than to get a result like that."

What cryptographers were really after was a universal one-way function that stemmed from some natural problem — one that would give real insight into whether one-way functions exist. Researchers long had a particular problem in mind: Kolmogorov complexity, a measure of randomness that originated in the 1960s. But its connection with one-way functions was subtle and elusive.

Pass became fascinated with that connection as a graduate student in 2004. Over the years he toyed with the problem, without much success. But he felt sure there was something there, and a burst of activity in Kolmogorov complexity over the past five years only heightened his interest.

Pass tried to persuade several graduate students to explore the question with him, but none were willing to take on what might turn out to be a fruitless project. Then Yanyi Liu started graduate school at Cornell. "Yanyi was fearless," Pass wrote in an email. Together, they plunged in.

What Is Random?

The concept of randomness is, by its nature, tricky to pin down. There's a Dilbert comic strip in which an office tour guide shows Dilbert the accounting department's "random number generator" — which turns out to be a monster who just keeps repeating the number 9. "Are you sure that's random?" Dilbert asks. "That's the problem with randomness," his guide answers, "you can never be sure."

Yanyi Liu worked with Pass to meaningfully connect one-way functions and Kolmogorov complexity.
Changzhi Xie

If someone shows you the number strings 99999999999999999999 and 03729563829603547134 and says they were chosen randomly, you can't exactly debunk that claim: Both strings have the same probability of being created when you pick digits randomly. Yet the second string certainly feels more random.

"We think that we know what we mean when we say, 'That thing is random,'" Allender said. "But it wasn't really until the notion of Kolmogorov complexity was defined that that was shown to have a mathematically meaningful definition."

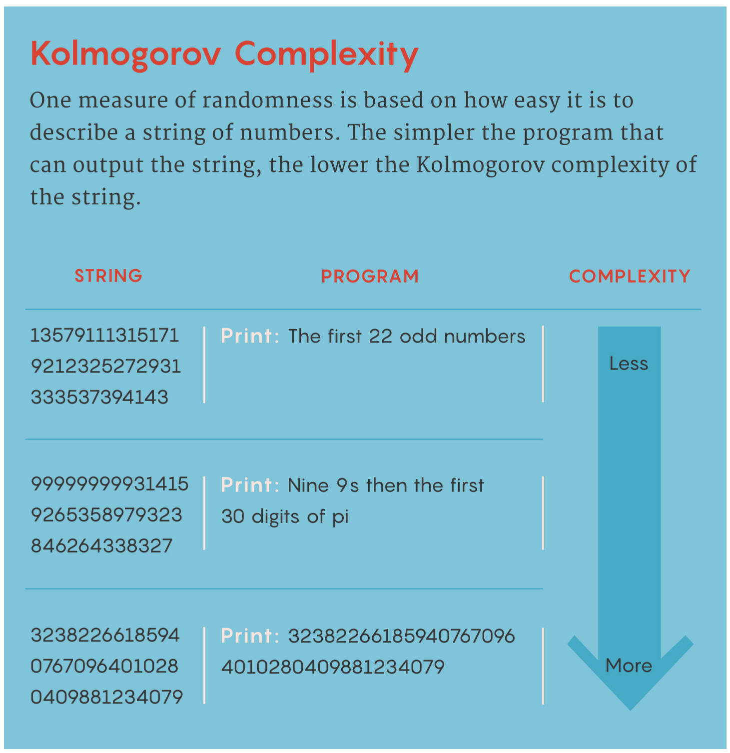
To get at the notion of a random string of numbers, Andrey Kolmogorov decided in the 1960s to focus not on the process by which the string was generated, but on the ease with which it can be described. The string 99999999999999999999 can be concisely described as "20 9s," but the string 03729563829603547134 might not have any description shorter than the string itself.

Kolmogorov defined the complexity of a string as the length of the shortest possible program that produces the string as an output. If we're dealing with, say, thousand-digit strings, some have very short programs, such as "print a thousand 9s" or "print the number 23319" or "print the first thousand digits of π using the following formula. . . ." Other strings are impossible to describe succinctly and have no program shorter than one that writes out the entire string and just tells the computer to print it. And some strings have programs whose length falls somewhere in the middle.

Kolmogorov complexity quickly became one of the core concepts of computer science. The notion is so fundamental that it was independently discovered multiple times in the 1960s. It's "a deep problem, not just about randomness [and] mathematics, but really about science in general," Pass said.

There's just one drawback to Kolmogorov complexity: It's incomputable, meaning that there is no program that can calculate the complexity of every possible string. We know this because if there were such a program, we'd end up with a contradiction.

To see this, imagine we have a program that can compute Kolmogorov complexity for any string. Let's call the program K. Now, let's search for the smallest string of numbers — call it S — whose Kolmogorov complexity is double the length of K. To be concrete, we could imagine that K has 1 million characters, so we're looking for a string S whose Kolmogorov complexity is 2 million (meaning that the shortest program that outputs S has 2 million characters).



Merrill Sherman / Quanta Magazine

With program K in our toolbox, calculating S is easy (though not necessarily quick): We can write a new program that we'll call P. The program P essentially says, "Go through all strings in order, using program K to compute their Kolmogorov complexity, until you find the first one whose Kolmogorov complexity is 2 million." We'll need to use program K when building P, so altogether P will have slightly more than 1 million characters. But this program outputs S, and we defined S as a string whose shortest program has 2 million characters. There's the contradiction.

But this contradiction evaporates if, instead of looking for the shortest program that outputs a string, we look for the shortest reasonably efficient program that outputs the string (where we get to specify what "reasonable" means). After all, the program P takes an enormous amount of time to run, since it has to check so many strings. If we forbid such slow programs, we end up with a notion called "time-bounded" Kolmogorov complexity. This version of Kolmogorov complexity is computable — we can calculate the time-bounded Kolmogorov complexity for every possible string, at least in principle. And in some ways, it is as natural a concept as the original Kolmogorov complexity. After all, Pass said, what we really care about is, "Can you actually generate the string while we live on Earth, or while the universe still exists?"

Since time-bounded Kolmogorov complexity is computable, a natural next question is how hard it is to compute. And this is the question that Liu and Pass proved holds the key to whether one-way functions exist. "It's a lovely insight," Allender said.

More specifically, suppose you've set your sights on a less lofty goal than calculating the exact time-bounded Kolmogorov complexity of every possible string — suppose you're content to calculate it approximately, and just for most strings. If there's an efficient way to do this, Liu and Pass showed, then true one-way functions cannot exist. In that case, all our candidate one-way functions would be instantly breakable, not just in theory but in practice. "Bye-bye to cryptography," Pass said.

Conversely, if calculating the approximate time-bounded Kolmogorov complexity is too hard to solve efficiently for many strings, then Liu and Pass showed that true one-way functions must exist. If that's the case, their paper even provides a specific way to make one. The one-way function that they describe in their paper is too complicated to use in real-world applications, but in cryptography, practical constructions often quickly follow a theoretical breakthrough, Ishai said. The impracticality of Liu and Pass' one-way function, he said, is "not a fundamental limitation."

And if their function can be made practical, it should be used in preference to the candidate one-way functions based on multiplication and other mathematical operations. For if anything is a one-way function, this one is. "If we can break a scheme like that, then all other schemes out there can also be broken," Pass said.

A Richer Theory

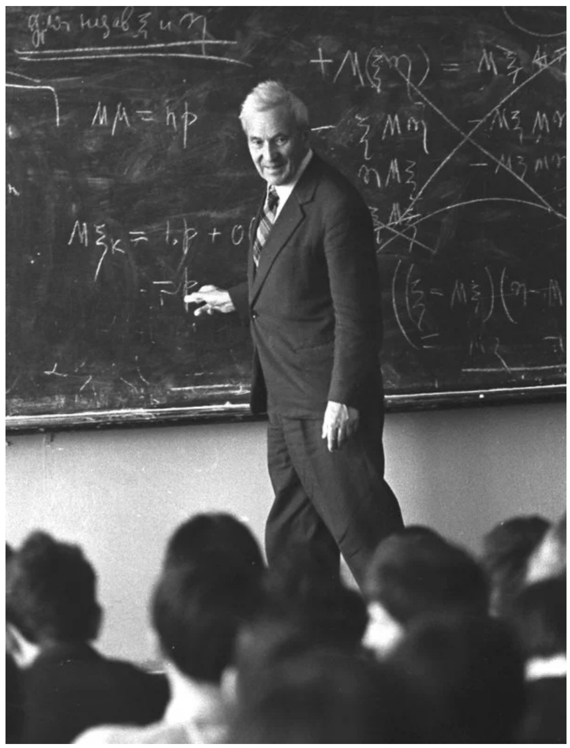
The paper has set off a cascade of new research at the interface of cryptography and complexity theory. While both disciplines investigate how hard computational problems are, they come at the question from different mindsets, said Rahul Santhanam, a complexity theorist at the University of Oxford. Cryptography, he said, is fast-moving, pragmatic and optimistic, while complexity theory is slow-moving and conservative. In the latter field, "there are these long-standing open questions, and once in every dozen years, something happens," he said. But "the questions are very deep and difficult."

Now cryptography and complexity have a shared goal, and each field offers the other a fresh perspective: Cryptographers have powerful reasons to think that one-way functions exist, and complexity theorists have different powerful reasons to think that time-bounded Kolmogorov complexity is hard. Because of the new results, the two hypotheses bolster each other.

"If you believe this [Kolmogorov complexity] problem is difficult . . . then you believe in one-way functions," Williams said. And "if you believe in crypto at all, then you've kind of got to believe that this version of time-bounded Kolmogorov complexity must be hard."

Cryptographers are now faced with the task of trying to make Liu and Pass' one-way function more practical. They are also starting to explore whether any other "master problems" besides time-bounded Kolmogorov complexity might also govern the existence of one-way functions, or of more sophisticated cryptographic tools. Complexity theorists, meanwhile, are starting to dig deeper into understanding the hardness of Kolmogorov complexity.

All of this suggests that the discovery's true legacy might be still to come. "[It's] a seed of something that is likely to develop into a much richer theory," Ishai said.



Andrey Kolmogorov came up with a way to measure randomness based on how easy it was to describe the generation of a string of numbers.
Alexander Makarov / RIA Novosti

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